# Online Supplement
## Observationally Equivalent Algorithms

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Sections S1–S7 reproduce Sections 2–8 of the longer core version of this paper (12 November 2026): every result in full, with the remarks, machine-check notes and the results that the note omits. The remaining sections give the proofs, further results, machine checks and notation. Theorem numbers in the supplement are those of these sections, not of the note. The note's results correspond to them as follows: Theorem 3.1 to Theorem S2.1 and Proposition S2.2; Theorem 3.2 to Theorem S2.3(i); Theorem 4.1 to Theorems S4.1, S4.2 and S4.4; Proposition 5.1 to Propositions S5.1(i) and S12.4; Theorem 5.2 to Theorem S5.3; Proposition 4.2 to Proposition S4.5; Proposition 5.3 to Proposition S5.4; Theorem 6.1 to Theorems S6.1, S6.2 and S6.4.

# S1 Model and motivating examples

Time is discrete. Let $p_t$ be the price, $p^*$ the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + \varepsilon_t, \tag{2}$$

where $v_t$ is an executed probe quantity and $\varepsilon_t$ a price innovation (news). Group $i$ holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{3}$$

The ordering is *price first*: the price responds to the probe, then groups update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{4}$$

and the transfer function from probes to prices, $H(z) = \sum_{j \geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z - a) D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{5}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \sum_j h_j = \kappa/(1 - S)$, where $S = a + \kappa \sum_i q_i$ is the feedback budget. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because groups update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{6}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{7}$$

In the *positive economy* ($q_i \geq 0$), $J$ is entrywise nonnegative, its price row sums to $S$, and the economy is Schur stable iff $S < 1$.

**Lemma S1.1** (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{8}$$

*(ii)* Gradient memory. *Online gradient descent with $L_2$ regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate $\eta$ at a fixed state has the same form; general reinforcement-learning policies need not.*

**Why exponential memory, and what is assumed.** Exponential memory is the oldest model of expectations in economics: adaptive expectations (Cagan, 1956; Nerlove, 1958) and geometric distributed lags (Koyck, 1954), and Muth (1960) showed that exponentially weighted forecasts are optimal for a random walk observed with noise, which is Lemma S1.1(i). Lemma S1.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same reduced-form state. A third example is a desk that each period trades the amount that maximizes $\hat{v}_t u - \frac{1}{2}(\kappa + \lambda)u^2$, a one-period trade-off between its trend forecast $\hat{v}_t$ and quadratic impact and holding costs. Its trade is $u_t = \hat{v}_t/(\kappa + \lambda)$. If the forecast is updated exponentially, $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta x_t$, the trade follows the same recursion, $u_{t+1} = (1-\eta)u_t + w x_t$ with $w = \eta/(\kappa + \lambda)$. This is a static first-order condition, not a dynamic optimum. In these families the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio, its regularization or its cost parameters. That is the private information a supervisor would like to recover, although real algorithms need not belong to any of these families. Applied to the pricing error, a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). Nothing in the model says such trading is profitable.

The results use six modelling choices: linearity; price-first timing within a period; exponential memory (needed for the identification result, Theorem S2.3, and the cost results of Section S4, but not for sufficiency); algorithms that do not change their rules when observed (Proposition S5.4 relaxes this for one group); known primitives $(a, \kappa)$; and stated observation contracts. Unknown primitives can only enlarge the set of candidate populations, so the lower bounds remain valid; Corollary S8.2 in the supplement shows that estimating them leaves the upper-bound rate unchanged. Section S3 replaces the primitive $(a, \kappa)$ by an equilibrium.

**The observer.** The cost results are stated for one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. It knows $(a, \kappa)$, the law of the observation noise and, for the $K$-group results, $K$. It can *wait*, recording prices driven by natural innovations; *probe*, trading a sequence $v_t$ against the groups from a reset state and recording the response; or *acquire disclosure*, receiving each class's flow. Independent resets are a benchmark that no outside participant can run in a live market. We use them because they make the information content of the data exact. Reset probing is a benchmark for what can be learned, not a proposed policy. Section S4 shows separately that passive observation is no cheaper (Theorem S4.4).

# S2 The kernel is a sufficient statistic

This section states the paper's main result. In a market with an insider, noise traders, a dealer and feedback traders, all following arbitrary causal rules, the feedback traders affect every outcome only through their kernel. The notation for the insider follows Kyle (1985), so in this section $x_t$ is the insider's order, not the pricing error.

**Theorem S2.1** (The kernel is sufficient)**.** *Consider a market with the following participants and rules.*

- *A value $v$ and noise trades $z_t$.*
- *An insider who submits $x_t = X_t(v, \mathcal{H}_{t-1})$.*
- *A dealer who sets $p_t = P_t(\mathcal{H}_{t-1}, y_t)$ from the public history $\mathcal{H}_{t-1}$ and the total order flow $y_t = x_t + z_t + f_t$.*
- *Feedback groups whose memory states update on a public price statistic $s_t = S_t(p_0, \dots, p_t)$ by $m_{i,t+1} = r_i m_{i,t} + s_t$ from $m_{i,0} = 0$, with $r_i = 1 - \eta_i$ and aggregate flow $f_t = \sum_i w_i m_{i,t}$, $w_i = q_i \eta_i$.*

*The rules $X$, $P$ and $S$ are arbitrary causal maps. This covers Kyle, inventory-averse, learning and misspecified dealers, and any insider strategy.*

*(i)* $f_t = \sum_{m<t} k_m \, s_{t-1-m}$, *with* $k_m = \sum_i w_i r_i^m$, *the moments of the merged measure $M$.*

*(ii) Two group configurations produce the same aggregate flow on dates $t \le T$ for every signal sequence $(s_t)$ if and only if $k_m = k'_m$ for all $m < T$. On an unbounded horizon this holds if and only if their merged measures coincide. For a fixed signal rule $S$, the 'if' direction holds for every price path. The 'only if' direction needs a rule that can produce an impulse signal. It fails, for example, for $S \equiv 0$, under which every configuration has zero flow.*

*(iii) If $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of $(v, z)$ the following coincide on $t \le T$: the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory. The two games therefore have the same strategy sets and the same payoff for every profile. They have the same equilibria, with the same outcomes, under any equilibrium concept defined from these payoffs and the public history.*

*(iv) Group-level quantities are not invariant. A Gauss and a Radau two-point configuration agree on $k_0, k_1, k_2$, so they are outcome-equivalent for $T = 3$, while their largest groups carry weights $\frac{1}{2}$ and $\frac{3}{4}$.*

**Beyond exponential memory.** The proof of part (iii) uses only one property of the feedback traders: their aggregate flow is a fixed linear function of past public signals. So the result does not depend on exponential memory.

**Proposition S2.2** (Any linear population)**.** *In the market of Theorem S2.1, replace the groups by any finite population of feedback traders $j = 1, \dots, J$ whose flows are causal linear functions of the public signal,*

$$f_{j,t} = \sum_{m<t} \kappa_m^{(j)} \, s_{t-1-m}, \qquad \kappa^{(j)} = (\kappa_0^{(j)}, \kappa_1^{(j)}, \dots) \text{ arbitrary real sequences,}$$

*and write $k = \sum_j \kappa^{(j)}$ for the population's kernel.*

*(i) Two populations produce the same aggregate flow on dates $t \le T$ for every signal sequence if and only if their kernels agree for $m < T$.*

*(ii) If their kernels agree for $m < T$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on $t \le T$, and the two games have the same equilibria under any equilibrium concept defined from payoffs and the public history.*

*(iii) Any decomposition of a kernel into individual kernels is consistent with the same outcomes. Prices, flows and all equilibrium outcomes therefore carry no information about the decomposition beyond $k$.*

*Proof.* (i) Aggregate flow is $f_t = \sum_{m<t} k_m s_{t-1-m}$. If the kernels agree for $m < T$, the flows agree on $t \le T$ for every signal. Conversely, the impulse signal $s_j = \mathbf{1}\{j = 0\}$ gives $f_{m+1} = k_m$. (ii) The strong induction in the proof of Theorem S2.1(iii) (Supplement S9) uses only that $f_t$ is the same function of the signals before $t$ in both markets, which is part (i). (iii) follows from (ii), since outcomes depend on the population only through $k$. □

## S2.1 What prices identify

Theorem S2.1 says that populations with the same kernel cannot be told apart by anything that price-based trading determines. The converse is that prices determine the kernel. In the probing experiment of Section S1, the response of prices to trades is a fixed function of $(a, \kappa, k)$, and it determines the merged measure.

**Theorem S2.3** (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state $x_t$.*

*(i)* Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{9}$$

*Write $a' = a + \kappa M(\{0\})$, so that groups with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0, 1)|$. A group with $\eta = 1$ merges into the anchor $a$ without adding a pole.*

*(ii)* Separation complexity under average power. *Consider testing a single group with rate $\bar{\eta}$ against two groups with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with all rates in $[\epsilon, 1 - \epsilon]$ for some $\epsilon > 0$. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2 (1 - 2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{10}$$

*where $C_{\mathcal{F}}$ depends only on the family, through $\epsilon$, the bound on $|H|$ implied by uniform stability, and the total weight $q$.*

The memory rates are poles of $G$ and *zeros* of $H$, not closed-loop poles. This has a practical consequence verified in the full working paper: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

*Remark* S2.4 (Scope). The result concerns the stated model and observation contract. Groups sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The $\delta^{-4}$ statement in (ii) is a lower bound on a declared family, with family-dependent constants. A group may choose its rate strategically. The observer then still sees only the merged measure of the rules in use; Proposition S5.4 draws the consequence.

# S3 Sufficiency and identification in equilibrium

The identification results of Section S2 take the price impact $\kappa$ and the persistence $a$ of pricing errors as given. This section derives both from dealers who dislike holding inventory. It shows that the equilibrium price is the closed loop of Section S1 with effective parameters and unchanged memory rates. So the kernel stays the sufficient statistic in equilibrium, and the cost of going beyond it (Section S4) keeps its order.

**An inventory equilibrium.**

- *Fundamental and price.* The fundamental $F_t$ is a public random walk with innovation variance $\sigma_F^2$, and the price is $p_t = F_t + x_t$.
- *Groups.* They trade $u_{i,t+1} = r_i u_{i,t} + w_i x_t$, the paper's exponential memory with $w_i = q_i \eta_i$ and $r_i = 1 - \eta_i$.
- *Exogenous flow.* $\xi_t \sim N(0, \sigma_\xi^2)$, i.i.d., stands for noise trades or the observer's probe. Customer flow is $y_t = \sum_i u_{i,t} + \xi_t$.
- *Dealers.* Competitive dealers absorb the flow and lay off a share $1 - \bar{a}$ of their inventory each period, so $H_t = \bar{a} H_{t-1} - y_t$. Each overlapping generation of dealers has CARA risk aversion $A$ and holds $H_t$ to the next date: $H_t = (E_t p_{t+1} - p_t)/(AV)$, with $V = \mathrm{Var}_t(p_{t+1})$.

We study stationary equilibria in which $x_t$ is linear in the dealers' inventory and the groups' next trades. Write $G(D) = \sum_i w_i/(D - r_i) = \int dM(r)/(D - r)$ for the Stieltjes transform of the merged measure.

**Theorem S3.1** (Price impact and persistence in an inventory equilibrium)**.** *Consider stationary linear equilibria of the model above.*

*(i)* Characterization. *The price law*

$$x_t = -\kappa H_t + \kappa \sum_i \frac{u_{i,t+1}}{D - r_i}$$

*with $D \neq 0$ and $D \neq r_i$, is an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa\, G(D), \qquad \kappa(D - \bar{a}) = A\, V\, D, \qquad V = \sigma_F^2 + \frac{\kappa^2 \sigma_\xi^2}{D^2}. \tag{11}$$

*An unexpected unit of flow moves the price by $\kappa/D$.*

*(ii)* No feedback. *Then $D = 1$ and the price law is exactly the paper's $x_t = \bar{a}\, x_{t-1} + \kappa\, y_t$. The persistence of pricing errors is the dealers' carry share, $a = \bar{a}$, and the impact $\kappa$ solves $\kappa(1-\bar{a}) = A(\sigma_F^2 + \kappa^2\sigma_\xi^2)$. An equilibrium exists if and only if $\Delta = (1-\bar{a})^2 - 4A^2\sigma_F^2\sigma_\xi^2 \geq 0$. When $\Delta > 0$ there are two, $\kappa_\pm = \big((1-\bar{a}) \pm \sqrt{\Delta}\big)/(2A\sigma_\xi^2)$; the low-impact one $\kappa_- \approx A\sigma_F^2/(1-\bar{a})$ is the price of dealer inventory risk.*

*(iii)* Invariance. *The coefficients $(\kappa, D, V)$, and hence the price process, depend on the groups only through $G$, and so only through the merged measure $M$. This is Theorem S2.1 in equilibrium form.*

*(iv)* Small feedback. *Let $\Delta > 0$, fix $\delta > 0$ with $\max_i r_i < 1 - \delta$, and let $\epsilon = \sup_{|D-1|\leq\delta}(|G(D)| + |G'(D)|)$ be small. Then there is a unique equilibrium near $(\kappa_-, 1)$. With $G(1) = \sum_i q_i$, the aggregate feedback intensity,*

$$D = 1 - \kappa_- G(1) + o(\epsilon), \qquad \kappa = \kappa_- + \frac{\kappa_-^2(\bar{a} + 2A\kappa_-\sigma_\xi^2)}{\sqrt{\Delta}}\, G(1) + o(\epsilon).$$

*Positive feedback raises the impact $\kappa$, and it raises the impact of an unexpected flow shock, $\kappa/D$, by more.*

**Machine check.** See Supplement S27.

**The identification results hold in equilibrium.** With feedback, dealers price the groups' predictable trades in advance, so the response of prices to flow is no longer the reduced form of Section S1 with the primitive $(a, \kappa)$. The next theorem shows that it is still that closed loop, exactly and path by path, with new parameters and the same memory rates.

**Theorem S3.2** (The equilibrium is the paper's closed loop)**.** *Take an equilibrium of Theorem S3.1 with $D \neq \bar{a}$, and set*

$$a^* = \frac{\bar{a}}{D}, \qquad \kappa^* = \frac{\kappa}{D}, \qquad \tilde{w}_i = \frac{D - \bar{a}}{D - r_i}\, w_i.$$

*(i)* Reduced form, pathwise. *Let $\tilde{u}_{i,t} = \frac{D-\bar{a}}{D-r_i} u_{i,t}$. For every realization, every initial state and every $t$,*

$$x_t = a^* x_{t-1} + \kappa^*\Big(\xi_t + \sum_i \tilde{u}_{i,t}\Big), \qquad \tilde{u}_{i,t+1} = r_i \tilde{u}_{i,t} + \tilde{w}_i x_t.$$

*So the price follows the closed loop (2)–(3) with persistence $a^*$, impact $\kappa^*$, exogenous flow $\xi$ in place of the probe, and groups with the same memory rates and merged weights $\tilde{w}_i$. Its transfer function from flow to price is (5) with $(a^*, \kappa^*, \tilde{w})$. The equilibrium is stable if and only if this closed loop is.*

*(ii)* Zeros. *For $\bar{a} > 0$, distinct rates in $(0, 1)$ and nonzero weights, the memory rates are the transmission zeros of the equilibrium price response, and none cancels (Theorem S4.3(i) with the effective parameters).*

*(iii)* Boundary. *The effective merged measure is $\tilde{M} = h\, M$, where $h(r) = (D - \bar{a})/(D - r)$ is positive. The effective parameters determine the equilibrium ones: $D = 1 - \kappa^*\tilde{M}([0, 1])/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. So the merged measure remains exactly what equilibrium prices identify (Theorem S2.3).*

*(iv)* The rates survive. *Let $(a^*, \kappa^*, W, r_c, \Delta\eta)$ satisfy the hypotheses of Theorem S4.1 with $(a, \kappa) = (a^*, \kappa^*)$. Then the two $K$-group economies of that theorem are the price responses of two inventory equilibria with identical primitives $(A, \sigma_F^2, \sigma_\xi^2, \bar{a})$ and identical $(\kappa, D)$:*

- *$D = 1 - \kappa^* W/(1 - a^*)$, which exceeds every rate;*
- *$\bar{a} = a^* D$ and $\kappa = \kappa^* D$;*
- *any $\sigma_F^2, \sigma_\xi^2 > 0$, with $A = \kappa^* D(1 - a^*)/(\sigma_F^2 + \kappa^{*2}\sigma_\xi^2)$;*
- *group weights $w_i = \tilde{w}_i(D - r_i)/(D - \bar{a}) > 0$.*

*Hence an observer of equilibrium prices who knows the primitives needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments. Theorem S4.2 applies to the effective parameters, and the true weights follow from the effective ones by a Lipschitz map. So $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ in equilibrium.*

**In words.** Dealers' anticipation does not hide the groups and does not move them. It replaces the persistence and impact by $\bar{a}/D$ and $\kappa/D$, and it rescales each group's weight by $(D - \bar{a})/(D - r_i)$. Because that factor rises with $r_i$, slow-memory groups weigh more in equilibrium prices. The memory rates are untouched.

- Every result of Sections S2 and S4 that concerns the map from order flow to prices therefore holds in equilibrium: identification, sample complexity, passive observation, and probing costs computed from the price path. Its constants are evaluated at the effective parameters.
- The lower-bound pair is not an artefact of the reduced form. Two economies with the same dealers, the same risk aversion and the same impact differ only in their groups, and they are as hard to tell apart as Theorem S4.1 says.
- Part (i), the realization in part (iv) and the effective group recursion are machine-checked (`Paper1EquilibriumReducedForm.lean`).
- One assumption is new: the observer's probe is part of the exogenous flow that dealers price as noise. Dealers who recognized a probe would price it differently.

**Further equilibrium results.** Four results in Supplement S9 extend this section. None is needed for the main argument.

- *A strategic insider* with short-lived information (Theorem S9.3). The equilibrium is characterized in closed form and nests Kyle's one-period values. Its coefficients depend on the groups only through $G$, and the observed price is the closed loop driven through a known invertible moving-average filter. So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present. Positive feedback raises total impact, and the insider then trades less and prices reveal less of her news.
- *Fragility* (Theorem S9.2). With positive feedback, a positive-impact equilibrium exists only if dealer risk aversion lies below a threshold. For strong feedback the threshold is inversely proportional to the feedback intensity $G(1)$. With several groups the bound depends on the population only through $G(1)$, again an aggregate.
- *Rents* (Theorem S9.4). A dealer that misjudges the aggregate response by $\varepsilon$ hands an insider with a committed schedule a rent of exact order $\varepsilon$, or charges it one. Public data reveal the aggregate response at the ordinary rate. So the information that protects price formation is cheap to learn.

- *A targeted halt in equilibrium* (Theorem S9.5). Two markets that agree in equilibrium to within the cube of the memory spacing can differ at order one after their largest group is halted.

Disclosure keeps its value in equilibrium too (Corollary S9.1).

# S4 The price of composition

Prices identify the kernel, and with exponential memory the kernel determines the merged measure. A finite record, however, pins down only finitely many moments of $M$, each with noise. This section asks how much data it takes to recover the groups. The answer is the known rate for finite mixtures. What the market adds is that the memories must first be extracted from a feedback loop, and that waiting is no cheaper than trading.

## S4.1 Many unknown groups

**Theorem S4.1** ($K$-group two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes the noisy taps $h_0, \dots, h_{T-1}$ with $T \geq 2K + 1$ ($T = \infty$ allowed), so that the first tap where the two economies below differ, $h_{2K}$, is observed; the bound below uses the full response norm, which dominates any finite prefix. There exist two $K$-group economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \dots, 2K - 2$ coincide while the moment of order $2K - 1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from $N$ independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

*The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.*

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem S4.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which the full working paper matches with an upper bound.

The construction uses the $K$-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have $K$ atoms and integrate polynomials of degree $2K - 2$ exactly. For $K = 2$ it reproduces the order of Theorem S11.1. Numerically, their low moments agree to machine precision and their moment of order $2K - 1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with $K$ on top of the exponent. The two rules also differ in their largest weight for every $K$ we checked: for $K = 2, \dots, 200$ the largest Gauss–Radau weight exceeds the largest Gauss weight, by 0.25 at $K = 2$ and by about $0.79/K^2$ for large $K$ (`equilibrium/gauss_radau_maxweight.py`). So the argument given for $K = 2$ in Proposition S5.1 applies to each such $K$. A proof of this comparison for every $K$ is open. The supervisory result needs less: Lemma S12.3 proves that, for every $K$, some two rules of this kind have different largest weights.

**What counts as an experiment.** In Theorem S4.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance $\sigma^2$ per tap. The exponent is the same for any probe design of bounded energy. For any probe $v$ the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy $E$ each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and the account-level analysis of the full working paper prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section S5).

**Theorem S4.2** (Matching upper bound for every $K$)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j} |s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of $K$-group economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and $N$ independent reset experiments, each observing at least the taps $h_0, \ldots, h_{2K}$ with Gaussian noise of variance $\sigma^2$. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

*with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem S4.1 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.*

**Smaller classes and constants.** On smaller classes the upper bound need not be tight: if the rates are known, for example, two weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments. For $K$ known clustered rates the requirement is at least of order $\sigma^2(\Delta\eta)^{-2(K-1)}$. A weight perturbation along the $(K-1)$-th finite difference preserves the moments of order $0, \ldots, K-2$ and changes the moment of order $K-1$ by order $(\Delta\eta)^{K-1}$. For $K = 3$, the direction $(1, -2, 1)$ on rates $r_c - \Delta\eta, r_c, r_c + \Delta\eta$ changes the second moment by $2(\Delta\eta)^2$. This is still far below the unknown-rate order $(\Delta\eta)^{-(4K-2)}$. The constants $c_0$ and $C$ grow quickly with $K$, through the conditioning of the moment map; the theorem gives the order in $\Delta\eta$ for fixed $K$, not a usable constant for large $K$ (in the full working paper our double-precision Prony implementation already fails at $K = 4$ on the disclosure grid).

## S4.2 Passive data

**Theorem S4.3** (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

*(i) The zeros of $H$ are the memory rates $r_i$ together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so $H$ is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem S2.3.*

*(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2|H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), $f$ determines $\sigma_\xi^2\kappa^2$ and the normalized response $H/h_0$ (recall $h_0 = \kappa$); hence it determines the memory*

*rates. If in addition $\kappa$ (or $\sigma_\xi^2$) and $a$ are known, it determines $H$ and, by Theorem S2.3, the merged measure.*

The result holds for signed weights (mean-reverting groups): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the groups' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

**Theorem S4.4** (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for $\theta$ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

*(i) For a single group at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.*

*(ii) For the moment-matched two-group economies of Theorem S11.1, $D = \Theta(\tau^2(\Delta\eta)^6)$.*

*(iii) For every $K \geq 2$, for the $K$-group Gauss and Gauss–Radau economies of Theorem S4.1, $D = \Theta((\Delta\eta)^{4K-2})$.*

*Hence any test based on $T$ passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of groups. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate $D$ (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation. So the orders are the right ones. We do not claim that this attainment is uniform as the separation shrinks. All constants depend only on $K$, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on $\delta$, $\Delta\eta$ or $T$. In part (ii) $\tau$ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \leq C\,T\,\varepsilon^2$, with $\varepsilon$ the leading moment gap ($\delta^2$, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on $D$ and the fixed-pair error exponents are separate statements.*

**In executed quantity and money.** For a *known* pair (one group at rate $\eta$ against two at $\eta \pm \delta$ with the same total weight), the information needed can be stated in trades. The executed quantity a prober needs is of exact order $\delta^{-2}$: the two price responses first differ at lag two, by $-\kappa^2 q \delta^2$, and a single pulse of size $2\sigma\Phi^{-1}(1-\alpha)/(\kappa^2 q \delta^2)$ attains error $\alpha$. The capital a counterparty must post to fill such probes is of order $\delta^{-4}$, because the adverse mark of a probe grows with the square of its size. Halving the difference between two memories therefore quadruples the trading needed and multiplies the capital by sixteen (Theorems S10.1 and S10.3 and Lemma S10.5, Supplement S10).

**Near instability.** Theorem S4.4 is stated on a family that stays uniformly stable. One might hope that the barrier lifts as the market approaches instability, since the loop then amplifies every difference. It does not. What becomes visible is the distance to instability, not the composition.

**Proposition S4.5** (The barrier does not lift near instability)**.** *Fix $K \geq 2$ and the $K$-group Gauss and Gauss–Radau economies $A$ and $B$ of Theorem S4.1, with positive weights, rates in $(0, 1)$ and $\kappa > 0$. For a margin $\epsilon > 0$, choose persistences $a_A(\epsilon)$ and $a_B(\epsilon)$ so that both economies have*

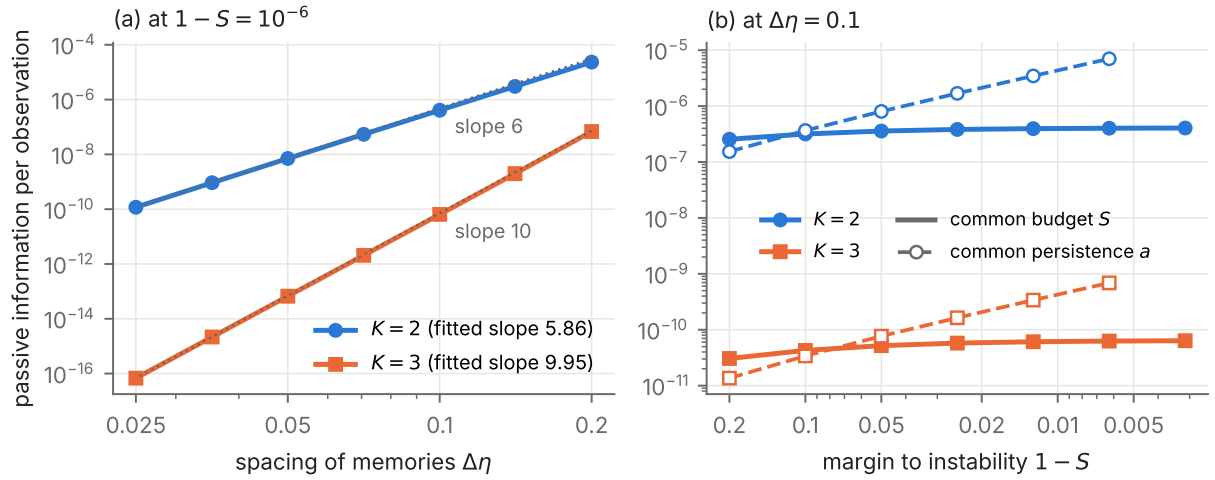



Figure S1: The passive barrier near instability. Both panels show the Gaussian Kullback–Leibler information per observation between the stationary price processes of two populations whose merged measures are the $K$-node Gauss and Gauss–Radau rules on memories spread over $\Delta\eta$ around 0.5 (total weight 0.3, impact $\kappa = 0.5$). (a) At a common feedback budget one millionth below instability, the information falls like $(\Delta\eta)^{4K-2}$ (dotted lines). (b) Held at a common budget (solid), the information stays bounded as the margin $1 - S$ shrinks; held at a common persistence (dashed), it grows without bound, because the budgets then differ. Computed from the closed-form spectra; scripts in `research_extension/fragility_reveals/`.

*feedback budget $S = 1 - \epsilon$, and suppose $a_A(\epsilon), a_B(\epsilon) > 0$ for $\epsilon \le \epsilon_0$. Let $D_\epsilon(B\|A)$ be the Kullback–Leibler divergence rate between the stationary price processes driven by i.i.d. $N(0, \sigma^2)$ innovations. There is $C$, not depending on $\epsilon$ or $\Delta\eta$, such that*

$$D_\epsilon(B\|A) \;\le\; C\,(\Delta\eta)^{4K-2} \qquad \text{for all } 0 < \epsilon \le \epsilon_0 \text{ and all small } \Delta\eta.$$

**In words.** Even as the market approaches a crash, passive prices reveal no more about who is trading than they did in calm conditions, per observation. They reveal more and more about how close the market is to instability: if the two economies share the persistence $a$ instead of the budget, their budgets differ by order $(\Delta\eta)^{2K-1}$, and numerically the divergence rate then grows like $\epsilon^{-1}$. The proof is in Supplement S12.2. Two limits apply. The statement is about the divergence rate per observation; near instability prices are strongly dependent, and a finite-sample bound that is uniform in $\epsilon$ is not proved, although numerically the exact information in $n$ observations stays within 3% below $n$ times the rate down to $\epsilon = 10^{-3}$ (Supplement S12.2). And probing does gain near instability: the squared response difference grows like $\epsilon^{-1}$ (numerical, `research_extension/fragility_reveals`), but probing a market at the edge of instability is not a policy. Figure S1 shows both statements.

# S5 Who needs composition, and what disclosure buys

By Theorem S2.1, composition is worth nothing to anyone trading on public information. It can still matter to a supervisor whose action is aimed at one group. This section gives a decision in which it does and the disclosure that removes the cost of learning it. It ends with a bound on what price-based targeting can achieve when a group can hide.

**A decision in which attribution matters.** A supervisor must keep the feedback budget $S = a + \kappa \sum_i q_i$ below a prudential ceiling $\bar{s}$. It can either cut all activity uniformly or halt the largest verified class. The target is the feedback budget, not the spectral radius of the state matrix: in an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not a restoration of stability.

**Proposition S5.1** (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known $a, \kappa$, and $a < \bar{s} < 1$. Write $q_i = w_i/(1 - r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

*(i) A uniform scale $\theta$ meets the ceiling exactly when $a + \theta(S - a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.*

*(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem S4.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For $N$ independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require*

$$N \ge (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \ge c\sigma^2(\Delta\eta)^{-6}.$$

*The record horizon and the response norm must be the same on both sides.*

*(iii) Tagged-flow estimation under Theorem S5.3, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.*

The proof is in Supplement S12. Proposition S12.4 there extends part (ii) to every $K$, with exponent $4K - 2$.

## S5.1 What class-tagged disclosure buys

**Proposition S5.2** (Tagged flow removes the barrier)**.** *Let the price $x_t$ be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma_\xi^2 \in (0, \infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \ge 0$, $0 \le \lambda_\ell < 1$ the closed-loop modes). Suppose group $i$ has merged weight $w_i > 0$ and the observer receives prices and each group's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance $\sigma_f^2$, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$*

*with instruments* $Z_t = (x_{t-1}, x_t)^\top$ *satisfies* $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,

$$V_i = \Sigma_i^{-1}\Omega_i\Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2[(1+r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)], \tag{12}$$

*where* $A_m = \sum_{s\geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ *and* $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. *The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \;>\; 0. \tag{13}$$

$V_i$ *depends only on group* $i$*'s own* $(w_i, r_i)$, $\sigma_f^2$ *and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy,* $\hat{\beta}_i$ *has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least* $\gamma_{\min} > 0$ *and merged weights at least* $w_{\min} > 0$*, the limiting covariance* $V_i$ *is bounded by a constant that does not depend on* $|r_i - r_j|$*. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family.*

**Machine check.** Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower bound on the determinant of the population instrument matrix that does not depend on the spacing (`p1_tagged_iv_determinant_bound`); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (`p1_iv_boundary_decomposition`). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about $w_i$ does not involve the other groups' rates; the closed form (13) matches the simulated Jacobian (0.1330 against 0.1322 in a two-group economy with $4 \times 10^5$ periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between $5.8 \times 10^{-4}$ and $6.6 \times 10^{-4}$). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem. Proposition S5.2 says that tagging removes the barrier; the next result says by how much.

**Theorem S5.3** (Value of class-tagged disclosure)**.** *Take a rich class of Theorem S4.2: rates* $r_c + \Delta\eta\, s$ *with* $s$ *in a compact set of* $\gamma$*-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem S4.1, and merged weights in a nondegenerate interval* $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. *In each reset experiment the observer applies a unit impulse and records the price response with independent* $N(0, \sigma^2)$ *noise per tap; under tagging it also records each class's flow response* $f_{i,t} = w_i \sum_{s\leq t} r_i^{t-s} h_s$ *with independent* $N(0, \sigma_f^2)$ *noise per tap. Let* $N_p$ *and* $N_t$ *be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants* $0 < c \leq C < \infty$*, depending on* $(\sigma, \sigma_f)$ *and the class but not on* $\Delta\eta$*, such that*

*(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them),* $c \leq N_t \leq C$*, so* $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$*;*

*(ii) to recover the rates to accuracy* $c_0\Delta\eta$*, for a fixed* $c_0 > 0$ *smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target),* $c(\Delta\eta)^{-2} \leq N_t \leq C(\Delta\eta)^{-2}$*, so* $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$*.*

Proposition S5.2 and Theorem S5.3 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

## S5.2 Traders who respond to being watched

The results so far take the groups' rules as given. A group that expects a targeted action has a reason to change its rule. Sufficiency bounds what a supervisor who relies on prices can then achieve.

**Proposition S5.4** (Hiding caps targeting)**.** *A supervisor observes prices, under any probes and for a record of any length, and then targets a given group or not. Being targeted costs the group $P > 0$. The group chooses among available rules. Rule $\rho$ has a private cost $C(\rho) \geq 0$ and leads to targeting with probability $\pi(\rho)$. Suppose one available rule $\rho_0$, with cost $C_0$, gives the population the kernel of a population that the supervisor targets with probability $\alpha$.*

*(i)* $\pi(\rho_0) = \alpha$*, whatever the supervisor's probes, record length and decision rule.*

*(ii) Every rule $\rho^*$ that minimizes the group's expected cost $C(\rho) + P\pi(\rho)$ satisfies*

$$\pi(\rho^*) \;\leq\; \alpha + \frac{C_0 - C(\rho^*)}{P} \;\leq\; \alpha + \frac{C_0}{P}.$$

*(iii) Suppose the group has two rules: its original rule, with cost zero and targeting probability $\pi_T$ on a record of length $T$, and $\rho_0$. It adopts $\rho_0$ whenever $\pi_T > \alpha + C_0/P$. If $\pi_T \to 1$ and $P(1-\alpha) > C_0$, this holds for all large $T$, and the targeting probability is then $\alpha$.*

*Proof.* (i) By Theorem S2.1, populations with the same kernel give the same law of the price record for every probe sequence and every insider and dealer rule. The supervisor's decision is a function of that record and of its own randomization. (ii) $C(\rho^*) + P\pi(\rho^*) \leq C_0 + P\alpha$ by optimality, and $C(\rho^*) \geq 0$. (iii) The original rule costs $P\pi_T$ and $\rho_0$ costs $C_0 + P\alpha$. □

# S6 Beyond the kernel: saturating traders and stress

The kernel is a linear object, and every precedent for it is linear. Real feedback traders have limits: position caps, margin, risk budgets. This section asks what replaces the kernel when trades saturate, and what prices reveal about the part that saturation adds. The answer is a sharper version of the paper's main message. In calm markets, prices reveal the kernel and nothing else. How close traders are to their limits, which is what governs behaviour under stress, is invisible in calm data. It can be learned only by pushing traders toward their limits, or from disclosure.

**Saturating traders.** In the market of Theorem S2.1, let trader $i$ have memory rate $r_i \in [0, 1)$, capacity $c_i > 0$ and leverage $\ell_i \in \mathbb{R}$. Its memory state is $m_{i,t} = \sum_{j<t} r_i^{t-1-j} s_j$, as before, and its flow is

$$f_{i,t} = c_i\, \psi(\ell_i m_{i,t}),$$

for a policy shape $\psi$ common to all traders. A shape is *band-linear* if $\psi(u) = u$ for $|u| \leq 1$. The hard limit $\psi(u) = \max(-1, \min(1, u))$ is the leading case: inside its band trader $i$ trades $c_i \ell_i m_{i,t}$,

and at the band edge its position is capped at $c_i$. Inside the bands the population is the linear one of Section S1, with merged weights $w_i = c_i\ell_i$ and kernel $k_m = \sum_i c_i\ell_i r_i^m$. Write

$$\nu = \sum_i c_i\, \delta_{(r_i,\ell_i)}$$

for the capacity-weighted measure on memory and leverage.

**Theorem S6.1** (What replaces the kernel)**.** *For any shape $\psi$, aggregate feedback flow is $f_t = \sum_i c_i\psi(\ell_i m_{i,t}) = \int \psi(\ell\, m_t(r))\, \nu(dr, d\ell)$, with $m_t(r) = \sum_{j<t} r^{t-1-j} s_j$, so it depends on the population only through $\nu$. If two populations have the same $\nu$, that is $\sum_i c_i g(r_i, \ell_i) = \sum_j c'_j g(r'_j, \ell'_j)$ for every function $g$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and profit and the dealer's inventory coincide at every date. The two games have the same equilibria under any equilibrium concept defined from payoffs and the public history. The kernel is the first leverage moment of $\nu$ at each memory, $k_m = \int r^m \ell\, d\nu$, and in general it does not determine $\nu$.*

The proofs of Theorems S6.1–S6.2 and Proposition S6.3 are in Supplement S13.

**Machine check.** `Paper1SaturatingSufficiency.lean` proves the pathwise statement for every shape $\psi$ and every causal insider, dealer and signal rule (`p1_sat_sufficiency`, through the generic `p1_flow_market_invariance`). The passage to equilibria is on paper, as in Theorem S2.1.

**Theorem S6.2** (No information without stress)**.** *Let $\psi$ and $\psi'$ be band-linear. Take two populations $P$ and $P'$ with the same kernel, $\sum_i c_i\ell_i r_i^m = \sum_j c'_j\ell'_j r'^m_j$ for every $m$, but possibly different capacities and leverages.*

*(i)* Pathwise. *Run both markets with the same strategy profile and the same realization. Let $\tau$ be the first date at which some trader of $P$ or of $P'$, with its state computed from the signals of the $P$-market, leaves its band ($|\ell m_t| > 1$). Then the two markets have identical price and order-flow paths on every date $t < \tau$. On those dates both also coincide with the* linear *market with kernel $k$, so $\tau$ is the first exceedance date of the linear market.*

*(ii)* Records. *Let $O_T$ be any record that is a function of the paths on dates $t < T$, plus noise independent of them. The laws of $O_T$ under $P$ and $P'$ satisfy*

$$\mathrm{TV}(\mathcal{L}_P(O_T), \mathcal{L}_{P'}(O_T)) \le \Pr_{\mathrm{lin}}(\tau < T),$$

*the probability that some trader of $P \cup P'$ leaves its band before $T$ in the linear market.*

*(iii)* Calm passive data. *In the market of Section S1 with innovations $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ i.i.d., no probe, a start from rest, and a linear market with $S_{\mathrm{abs}} = a + \kappa \sum_m |k_m| < 1$, let $s_r^2 = \sup_t \mathrm{Var}_{\mathrm{lin}}(m_t(r))$ and $\vartheta = \max_g |\ell_g| s_{r_g}$ over the $n + n'$ traders of $P \cup P'$. Then*

$$\mathrm{TV} \le T\,(n + n')\, e^{-1/(2\vartheta^2)}, \qquad s_r \le \frac{\sigma_\varepsilon}{(1 - r)(1 - S_{\mathrm{abs}})},$$

*and any test that tells $P$ from $P'$ with both error probabilities at most $\alpha < 1/2$ needs $T \ge (1 - 2\alpha)\, e^{1/(2\vartheta^2)}/(n + n')$ observations.*

*(iv)* Probes. *Without process noise, if the linear market's response to a deterministic probe keeps every trader of $P \cup P'$ inside its band, the two records have the same law, for any number*

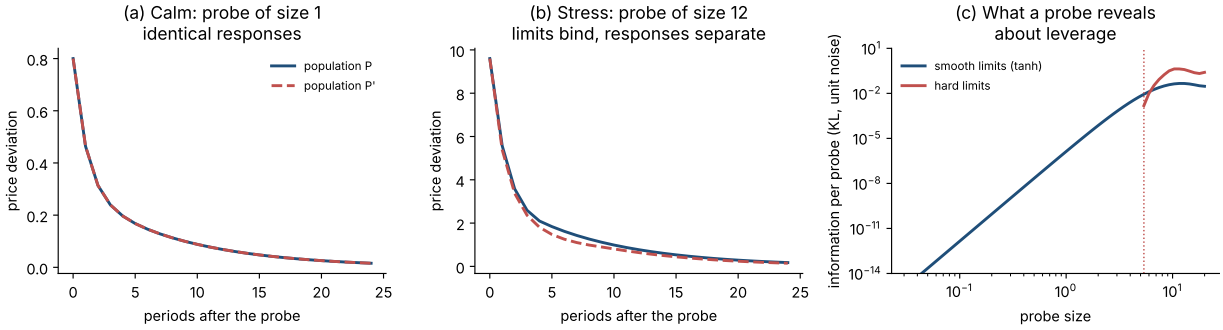



Figure S2: The two layers. Populations $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$ at memory 0.7 have the same kernel ($a = 0.5$, $\kappa = 0.8$). (a) Under hard limits, a calm probe gives identical responses. (b) A stress probe pushes the most levered trader to its limit and the responses separate. (c) Information per probe about the difference (Kullback–Leibler divergence over 30 taps, unit noise): under smooth limits it grows like the sixth power of the probe size (fitted slope 6.00); under hard limits it is exactly zero below the size at which the first trader reaches its limit (5.38, dotted).

*of repetitions and any observation noise. For a single pulse of size $A$, no information is obtained below*

$$A^* = \min_g \Big( |\ell_g| \sup_t \Big| \sum_{j<t} r_g^{t-1-j} h_j \Big| \Big)^{-1},$$

*with $h$ the linear impulse response.*

**Machine check.** The pathwise part (i), including the statement that the market follows the linear market until the first exceedance, is machine-checked for every pair of band-linear shapes and every causal insider, dealer and signal rule (`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`). Parts (ii)–(iv) are on paper.

**An example, and what differs at stress.** At memory $r = 0.7$, take $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$. Both have merged weight 0.1, so they have the same kernel and are equivalent until stress. Under a hard limit, their total capacities, the largest aggregate flow they can deliver, are 1 and 1.25 (`p1_sat_capacity_differs`). In the market of Section S1 with $a = 0.5$ and $\kappa = 0.8$, single pulses up to size 4 give exactly identical responses. Larger pulses push the most levered trader of $P'$ to its limit, and the responses separate by 0.072 at size 6 and 0.364 at size 12. Under Gaussian innovations, the simulated probability that some trader leaves its band within 1,000 dates lies below the bound of part (iii) at every noise level tested. It is zero up to $\sigma_\varepsilon = 0.3$, and 0.0175, 0.0875 and 0.63 at $\sigma_\varepsilon = 0.35$, 0.4 and 0.5. The scripts are `research_extension/nonlinear_scoping/sim_wiener.py` and `sim_passive_clip.py`. Figure S2 shows the two layers.

**Proposition S6.3** (Smooth saturation: information grows like the sixth power of the probe)**.** *Let $\psi = \tanh$. In the probing experiment of Section S1, with reset state, no process noise, a single pulse of size $A$ and taps observed with independent $N(0, \sigma^2)$ noise, compare two populations $P$ and $P'$ with the same kernel and with $\rho_P = a + \kappa \sum_i c_i |\ell_i| / (1 - r_i) < 1$, and likewise $\rho_{P'} < 1$. Let $\hat{S} = \max(\rho_P, \rho_{P'})$, $\Lambda_P = \frac{1}{3} \sum_i c_i |\ell_i|^3 / (1 - r_i)^3$, and $\Delta M_3 = \sum_i c_i \ell_i^3 - \sum_j c'_j \ell'^3_j$.*

*(i) The two price responses satisfy* $\|x^P - x^{P'}\|_2 \le C_D|A|^3$ *with* $C_D = \kappa^4(\Lambda_P + \Lambda_{P'})/(1-\hat{S})^2$. *Hence* $N$ *experiments that tell* $P$ *from* $P'$ *with both errors at most* $\alpha < 1/2$ *need*

$$N \ge \frac{4\sigma^2(1-2\alpha)^2}{C_D^2\, A^6}.$$

*(ii) The first tap after the pulse differs by* $x_1^P - x_1^{P'} = -\frac{1}{3}\kappa^4 \Delta M_3 A^3 + O(A^5)$. *If* $\Delta M_3 \neq 0$, *then for all small enough* $|A|$ *the averaged first tap tells* $P$ *from* $P'$ *with error at most* $\alpha$ *once* $N \ge \left(12\,\sigma\,\Phi^{-1}(1-\alpha)\right)^2/(\kappa^8 \Delta M_3^2 A^6)$. *So the order* $A^{-6}$ *is attained. The cubic coefficient of the first tap identifies* $\sum_i c_i \ell_i^3$, *the third leverage moment of* $\nu$.

## S6.1 What stress reveals

Theorem S6.2 says what calm data cannot show. The next result says what stress does show: everything that matters. Write

$$\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i, |\ell_i|)}$$

for the signed capacity at each memory and each leverage level. A trend follower and a contrarian with the same memory and the same leverage magnitude offset exactly, since $c\,\psi(-\ell m) = -c\,\psi(\ell m)$ for odd $\psi$.

The result holds for any shape with one property. An odd shape $\psi$ has *independent dilations* if rescaled copies of it at distinct scales are linearly independent: for any distinct $s_1, \dots, s_n > 0$ and any weights $w_1, \dots, w_n$, $\sum_k w_k \psi(s_k u) = 0$ for every $u$ implies $w_1 = \cdots = w_n = 0$.

**Theorem S6.4** (Stress identifies the leverage distribution)**.** *Let* $\psi$ *be odd with independent dilations, and let every trader have* $r_i \ge 0$ *and* $\ell_i \neq 0$. *In the probing experiment of Section S1, with reset state and exactly observed responses, two populations give identical price responses to every probe sequence if and only if* $\tilde{\mu}_P = \tilde{\mu}_{P'}$. *Probe responses therefore identify, within each memory class, the net capacity at every leverage level. The kernel* $k_m = \int \lambda r^m\, \tilde{\mu}(dr, d\lambda)$ *is one linear functional of this measure.*

*The property holds or fails as follows.*

*(i) The hard limit has independent dilations. For it the result needs probes of all sizes: probes that keep every memory state inside every band identify only the kernel (Theorem S6.2(iv)).*

*(ii)* tanh *has independent dilations. Moreover, identical responses to all probes in an arbitrarily small neighbourhood of zero already imply* $\tilde{\mu}_P = \tilde{\mu}_{P'}$.

*(iii) A cubic shape* $\psi(u) = u - \gamma u^3$ *does not: the scales* $1, 2, 3$ *with weights* $5, -4, 1$ *give* $\sum_k w_k \psi(s_k u) = 0$ *for every* $u$. *For such a shape, probes identify only the first and third leverage moments within each memory.*

**Machine check.** The general statement of Theorem S6.4 is machine-checked for any number of traders (`p1_thm_stress_identification` in `Paper1StressIdentificationGeneral.lean`). So are the first sentence of part (i) (`p1_clip_indep_dilations`), the first sentence of part (ii) (`p1_tanh_indep_dilations`) and the counterexample in part (iii) (`p1_cubic_not_indep_dilations`). The theorem for the two shapes, with no hypothesis left on $\psi$, is `p1_thm_stress_identification_hard_limit` and `p1_thm_stress_identification_tanh`. Two things are proved on paper only: the small-probe statement in part (ii), and the last sentence

of part (iii). In the formal statement, "identical responses" reads: every pair of a probe and a price path of the first market is such a pair of the second.

The proof (Supplement S13) uses three facts. First, probes can steer the price along any path, so identical responses mean identical maps from past prices to flow. Second, two dates of prices turn that map into a sum of scaled copies of $\psi$, one per memory–leverage pair, with distinct scales for a suitable direction. Third, independent dilations separate the copies. For the hard limit, each copy adds a kink where its pair saturates, and the copy with the largest scale has the first kink. For tanh, each copy approaches its limit at its own exponential rate, and the copy with the smallest scale has the slowest tail. The kink argument is constructive. In the example of Figure S2, read along one direction, the flow map has kinks at 3.702 and 14.813, and their slope jumps return net capacities 0.24989 and 0.99988, against true values 0.25 and 1. Locating the second group needs a memory state of almost 15, about four times the stress level at which the first trader saturates. Knowing the kernel and the first group fixes only the second group's total $c\ell$, not its leverage.

The theorem assumes exactly observed responses; with observation noise, a minimum-distance estimator recovers $\tilde{\mu}$ consistently, without a rate (Supplement Proposition S12.5). Theorem S6.4 turns Section S6 into a two-layer theory. Prices in calm markets identify the kernel. Prices under stress identify the full signed measure $\tilde{\mu}$, which by Theorem S6.1 is all that any price-based outcome depends on. Under hard limits, no information beyond the kernel arrives until the first group saturates. Under smooth limits, the order-$n$ term of the response scales with the probe size to the power $2n+1$. We expect its information to scale with the power $2(2n+1)$, but only the case $n=1$ is proved (Proposition S6.3).

## S6.2 Deleveraging traders and dealers who anticipate

**Proposition S6.5** (Deleveraging traders)**.** *Let each trader follow any causal rule, for example deleveraging gradually, liquidating, or switching strategy after a loss, provided it trades linearly, $w_i m_{i,t}$, as long as its state has stayed inside its band on every date up to $t$. Then Theorem S6.2 holds verbatim for two such populations with the same kernel. Up to the first date at which some trader leaves its band, the paths coincide and equal the linear market's path, and parts (ii)–(iv) follow.*

The proof of the pathwise part is the induction of Theorem S6.2(i). It is machine-checked for arbitrary causal rules outside the bands (`p1_delever_stress_invariance`). Deleveraging that starts before a hard limit only moves the band edge inward. Calm data remain silent until the first trader starts to cut its position.

**After the first breach: an open question.** What happens after a breach is not covered by these results. A natural conjecture is a threshold. If traders who breach their limits liquidate, their sales move the price, and the price move pushes other traders' states toward their own limits. One expects a cascade when the liquidation triggered per unit of price move, summed over the traders close to their limits, exceeds the market's capacity to absorb it, and no cascade otherwise. We have no proof of such a threshold and do not state one.

**Proposition S6.6** (Who sets prices matters)**.** *(i) Theorems S6.2 and S6.4 and Proposition S6.5 hold for any causal dealer rule that is the same in the two markets. This covers every rule that depends on the population only through its kernel, for example a dealer who knows only the kernel, or one who prices current flow.*

*(ii) A dealer who knows the population and quotes on the expected next-period feedback flow uses more than the kernel. Suppose the next memory states are Gaussian, $m_g^+ \sim N(\bar{m}_g, s_g^2)$, with every trader inside its band in expectation, $|\ell_g \bar{m}_g| < 1$, and hard limits. Then the expected next-period flows under two populations with the same kernel differ by at most*

$$\sum_{g \in P \cup P'} c_g \sqrt{2/\pi}\, \sigma_g\, e^{-\delta_g^2/(2\sigma_g^2)}, \qquad \sigma_g = |\ell_g| s_g, \quad \delta_g = 1 - |\ell_g \bar{m}_g|.$$

The proof is in Supplement S13.

**What the supervisor can and cannot learn.** The two results split the population into a part prices reveal in calm markets, the kernel, and a part they reveal only under stress: the distribution of leverage within each memory, weighted by capacity. The second part decides how the market behaves when traders hit their limits, through the total capacity and through how many traders are near their limits. Calm-market price data are therefore the wrong instrument for it, however long the record and however sophisticated the estimator. The options are stress probes, whose cost and risk Proposition S6.3 prices, actual stress episodes, or disclosure of capacities: position limits, margins and risk budgets. This is a different disclosure from the class-tagged flow of Section S5. Tagged flow reveals each group's memory and weight. Capacity disclosure reveals how far each group is from its limit.

## S6.3 Any population

The results above describe two families of rules: linear rules, summarized by the kernel, and saturating rules, summarized by the measure $\nu$. The next theorem drops the family. A population is now represented only by its aggregate feedback flow map $F$, which sends a price path to the flow at each date and may be any causal function: linear, saturating, deleveraging, or switching between strategies after observing past profits.

**Theorem S6.7** (Any population)**.** *In the probing experiment of Section S1, with reset state, exactly observed responses and $\kappa \neq 0$, let two populations have aggregate flow maps $F$ and $F'$.*

*(i)* Sufficiency. *If $F = F'$ on every price path, the two markets give the same prices, order flows, insider profits and dealer inventories for every strategy profile and realization, and the same equilibria.*

*(ii)* Identification. *If every pair of a probe sequence and a price path produced by the first population is produced by the second, then $F = F'$ on every price path. So the two populations give identical responses to every probe sequence if and only if $F = F'$, and a group-level quantity, such as the weight of the largest group, is identified from probe responses if and only if it is a function of $F$.*

*(iii)* Cost. *Let $\mathcal{C}$ be any class of populations that contains the two linear $K$-group economies of Theorem S4.1(i). Then any procedure that decides a group-level quantity separating them, over $\mathcal{C}$, from $N$ reset impulse-response records needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$, and from passive Gaussian records needs at least the number of observations of Theorem S4.4. Every band-linear class whose capacities can be large enough that the impulse responses keep every trader inside its band contains the two economies.*

*Proof.* Part (i) is the induction of Theorem S2.1, which uses only that the flow is a fixed causal function of past public signals. For part (ii), given any price path $x^*$, the probe $v_t = (x_t^* - a x_{t-1}^*)/\kappa -$

$F(x^*)_t$ makes the first market follow $x^*$, because the flow at date $t$ depends only on prices before $t$. By hypothesis the second market also follows $x^*$ under this probe, and subtracting the two price equations gives $F(x^*)_t = F'(x^*)_t$ at every date. The converse is part (i) applied to the probing recursion. A quantity $g$ is identified exactly when $g$ agrees on populations with identical responses, that is, on populations with the same $F$. For part (iii), a procedure that works over $\mathcal{C}$ works in particular on the two economies, so the lower bounds of Theorems S4.1 and S4.4 apply. For a band-linear class, take capacities $c_i$ and leverages $\ell_i = w_i/c_i$ with $c_i$ large enough that $|\ell_i m_{i,t}| \le 1$ along the impulse response, which is bounded; inside the bands each trader trades $c_i \ell_i m_{i,t} = w_i m_{i,t}$, so the population reproduces the linear economy's responses (Theorem S6.2(i)). □

**What this changes, and what it does not.** Part (ii) is the general form of Theorems S2.3 and S6.4: prices identify the aggregate flow map and nothing finer, for any population. For linear rules the map is the kernel, which determines the merged measure; for odd saturating shapes with independent dilations it determines the signed capacity measure; for a cubic shape it determines only two leverage moments. Which group-level quantities prices reveal is therefore a property of the family of rules, and it can be read off from how the family maps to flow maps. Part (iii) says that allowing richer rules cannot make composition cheaper to learn than in the linear case, because the hard linear pair remains in the class. What is not proved is an upper bound for general classes: for some families composition is not identified at all (the cubic shape), and for others the attainable rate is open.

**Machine check.** Parts (i) and (ii) are machine-checked for an arbitrary flow map: `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`); `p1_general_identification`, `p1_general_identification_iff`, `p1_responses_of_flow_eq` and `p1_identified_functional` (`Paper1GeneralIdentification.lean`). The band-linear embedding in part (iii) uses `p1_sat_follows_linear`. The transfer of the lower bounds is on paper.

**What is open.** Three questions remain open. The first is rates for the higher orders of Theorem S6.4: Proposition S6.3 gives the first nonlinear moment, and the general order is not proved. The second is a full equilibrium with forward-looking dealers and saturating or deleveraging traders, including what happens after the first breach, in a fire sale. The third is learning agents whose rules change continuously, not only at a band edge. Related ideas appear in the literature on funding constraints and liquidity spirals (Brunnermeier and Pedersen, 2009) and in evidence that crowding predicts tail risk (Barroso et al., 2022). The identification of sums of nonlinear branches is a known problem in system identification (Dreesen et al., 2017). What is new here is the sufficient measure inside a strategic market, the exact absence of calm-market information about it, and the reduction of its recovery under stress to a known uniqueness problem.

# S7 Scope and limits

**Linearity.** Sufficiency (Theorem S2.1, Proposition S2.2) needs aggregate feedback flow to be a linear function of past public signals, plus independent idiosyncratic shocks if there are any. With group-specific Gaussian demand shocks, a second aggregate enters, the spectral density of the aggregate shock process, and the equivalence class shrinks accordingly (Supplement S9). The passive results of Section S4 are stated for Gaussian innovations. With non-Gaussian innovations, higher-order moments of prices could carry further information; we do not analyse that case. Saturating

rules are treated in Section S6. Other nonlinear rules, and traders who change their rules as they learn, are outside the theorems. The full working paper reports simulations of stylized nonlinear rules and of online learners trained in the closed loop. In those simulations, the costs of Section S4 hold when the competing populations' feedback weights match. Populations whose learned weights differ are told apart by those weights instead. Hard thresholds hide groups from small probes altogether. These are simulations, not theorems.

**Exponential memory.** The identification and cost results need memory to be a finite mixture of exponentials. For general completely monotone memory, the full working paper proves that the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$. Halving the error then squares the data needed, instead of multiplying it by a constant.

**The observer.** The lower bounds hold for an observer who knows $(a, \kappa)$ and the noise law, so they hold a fortiori without that knowledge. The upper bounds assume reset experiments or a known initial state. Strategic responses to being observed are modelled only in Proposition S5.4, and there for one group. Everywhere else the groups do not change their rules when probed.

**Evidence.** The full working paper reports pre-specified tests on two sets of trader-class reports: the CFTC's weekly Commitments of Traders in twenty futures markets and Binance's class ratios in 30 perpetual markets. Of 23 recorded outcomes, 11 pass. The class reports constrain the persistence of class positions far more tightly than prices do. They do not identify learning memory, because that persistence can also reflect turnover, reporting aggregation or inventory retention. A pre-registered out-of-sample test of the feedback premise on Taiwan's tagged exchange flows (spec P1T, holdout 2019–2023; the full working paper reports every diagnostic) gives the following. $D = 0.0426$ $[0.0206, 0.0643]$, $p = 0.000$, meets the registered supportive criterion. The pre-specified reverse-causality check gives $D = -0.0065$ $[-0.0316, 0.0165]$, which does not meet it. The registered verdict is 'supportive'. Read with its pre-specified diagnostic, the test does not provide evidence of feedback beyond prices revealing past flow. We do not claim that tagged flow in Taiwan responds to price deviations. The registered test is supportive, but its pre-specified diagnostic attributes the gain to reverse causality.

A second test, of the two-layer prediction of Section S6, is registered on Binance perpetual futures (spec P7S, amendment A1). Its primary sample is retrospective (July 2024 to October 2026), specified in November 2026 after the sample period, and a prospective window from November 2026 serves as a replication. The primary test was run once on 11 November 2026. A pre-specified test on Binance perpetual futures was inconclusive (calm difference $-0.002$, 90% interval $[-0.007, 0.004]$; difference-in-differences 0.041, $p = 0.553$). It neither supports nor contradicts the prediction of Section S6. The prospective replication covers 16 November 2026 to 14 February 2027; its symbols and thresholds were fixed on its own development data (August to October 2026) before the window opened, and its result is not yet available.

**Open problems.** Four questions remain open: a long-lived strategic insider who times her trades against the groups' memory; rates for the higher orders of the identification of saturating traders, and a full equilibrium with forward-looking dealers and deleveraging traders after a breach; sufficiency for learning agents whose rules change; and a supervisory loss function estimated from real interventions, which would turn the information costs here into a value of disclosure.

# S8 Proofs of the main results

### A lemma used in the proofs.

**Lemma S8.1** (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

*It is completely monotone and nonincreasing; its peak is the direct impact $\kappa$, even though $J$ is non-normal. In price/memory coordinates the price-only gain is* $\sup_t \|e_x^\top J^t\|_2 = 1$.

**Lemma S8.1.** Omit zero-weight groups. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots), \qquad u = (\sqrt{a}, \sqrt{b_1 \eta_1}, \ldots)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to $\kappa u$ and $J'e_x = \sqrt{a}\, u$, so $h_j = (\kappa/a) e_x^\top (J')^{j+1} e_x$. The spectral decomposition of $J'$ gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0, 1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since $J$ is nonnegative with row sums at most one, every price row of $J^t$ has $\ell_1$, hence $\ell_2$, norm at most one, attained at $t = 0$.

**Lemma S1.1.** (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance $P$ solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2 \sigma_\nu^2 = 0$, whose positive root is $P = (\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2 \sigma_\nu^2})/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0, 1)$, which simplifies to (8), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2} H(x_t - \theta)^2 = -H(x_t - \theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

**Theorem S2.3.** With zero initial states, $y_{i,t} = \sum_{s \leq t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s \leq t} k_{t-s} x_s$ and every response is determined by $(a, \kappa, k)$, hence by $M$. Conversely, equal responses for all probes imply equal $H$, hence equal $G$, and partial fractions of $G$ identify the distinct poles with nonzero residues $w_i$, i.e. $M$. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod (z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \neq i} (z - r_j)$, as in (5); at a retained $r_i$ the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j \neq i} (r_i - r_j) \neq 0$ (memory rates are positive because $\eta_i < 1$), so no root of $D$ cancels. The numerator $\kappa z D(z)$ has no roots other than 0 and the $r_i$, so these are the only possible cancellations. The remaining numerator factor $z$ cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\operatorname{supp}(M) \cap (0, 1)|$ for $a' \neq 0$ and $|\operatorname{supp}(M) \cap (0, 1)|$ for $a' = 0$ (an atom at rate 0, a group with $\eta = 1$, contributes the constant $M(\{0\})$ to $G$ and no pole, so it shifts $a$ to $a' = a + \kappa M(\{0\})$ and the argument above applies with $a'$) (numerically, the rank of the Hankel matrix of $(h_j)_{j \geq 1}$ is $K + 1$ and $K$ respectively for $K = 1, 2, 3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z - 1 + \bar{\eta})\big((z - 1 + \bar{\eta})^2 - \delta^2\big)}.$$

The denominator is the product of the three linear factors $z - r$ with $r \in \{1 - \bar{\eta},\, 1 - \bar{\eta} \pm \delta\}$, all in $[\epsilon, 1 - \epsilon]$ by assumption; on $|z| = 1$ each satisfies $|z - r| \geq 1 - r \geq \epsilon$, so $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ there. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1} H_\delta H_0 (G_\delta - G_0)$, whose factor $z^{-1}$ has modulus one on the circle, gives an $\ell_2$ gain at most $C_{\mathcal{F}} \delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal{F}}^2 \delta^4 PT$; the equal-noise KL is at most $C_{\mathcal{F}}^2 \delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (10).

**Theorem S4.1.** Let $(s^A, \omega^A)$ and $(s^B, \omega^B)$ be the $K$-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. To see that $g_K \neq 0$, let the Radau nodes be $-1, s_2, \ldots, s_K$ and put $P(s) = (s + 1) \prod_{j \geq 2} (s - s_j)^2$. This is monic of degree $2K - 1$ and vanishes at every Radau node, so the Radau rule gives it 0. Gauss is exact to degree $2K - 1$, so it gives $\int P \, d\mu > 0$, since $P \geq 0$ on $[-1, 1]$ and vanishes only at the nodes. The two rules agree on every lower-degree term, so they differ on $s^{2K-1}$. Put $r = r_c + \Delta\eta \, s$ and weights $W\omega$, and let $\mu_j^X = W \sum_i \omega_i^X (s_i^X)^j$ for $X \in \{A, B\}$. Then $|\mu_j^X| \leq W$, and $\mu_j^A = \mu_j^B$ for $j \leq 2K - 2$.

*Kernel gap.* By the binomial theorem, $k_m^X = \sum_j \binom{m}{j} r_c^{m-j} (\Delta\eta)^j \mu_j^X$. So, with $n = 2K - 1$, $\Delta k_m = k_m^A - k_m^B = \sum_{j \geq n} \binom{m}{j} r_c^{m-j} (\Delta\eta)^j (\mu_j^A - \mu_j^B)$. Since $\binom{m}{j} \leq \binom{m}{n}\binom{m-n}{j-n}$ for $j \geq n$, this gives $|\Delta k_m| \leq 2W \binom{m}{n} (\Delta\eta)^n (r_c + \Delta\eta)^{m-n}$. Summing with $\sum_m \binom{m}{n} x^{m-n} = (1 - x)^{-(n+1)}$ gives $\|\Delta k\|_1 \leq 2W(\Delta\eta)^{2K-1}/(1 - r_c - \Delta\eta)^{2K}$ over all lags.

*Response gap.* Let $\varsigma = a + \kappa W/(1 - r_c - \Delta\eta) < 1$, the theorem's hypothesis. Each kernel has $\|k^X\|_1 \leq W/(1 - r_c - \Delta\eta)$. From $h_0 = \kappa$ and $h_{t+1} = a h_t + \kappa \sum_j k_j h_{t-j}$, every partial sum of $|h^B|$ is at most $\kappa/(1 - \varsigma)$. The difference $\Delta h = h^A - h^B$ has $\Delta h_0 = 0$ and $\Delta h_{t+1} = a \Delta h_t + \kappa \sum_j (k_j^A \Delta h_{t-j} + \Delta k_j h_{t-j}^B)$, so every partial sum of $|\Delta h|$ is at most $\kappa \|\Delta k\|_1 \|h^B\|_1/(1 - \varsigma)$. Hence, for every horizon,

$$\|h^A - h^B\|_2 \leq \|h^A - h^B\|_1 \leq C_U (\Delta\eta)^{2K-1}, \qquad C_U = \frac{2\kappa^2 W}{(1 - r_c - \Delta\eta)^{2K} (1 - \varsigma)^2}.$$

In the other direction, $\Delta k_j = 0$ for $j < 2K - 1$, so by induction $\Delta h_t = 0$ for $t \leq 2K - 1$, and $\Delta h_{2K} = \kappa \, \Delta k_{2K-1} h_0^B = \kappa^2 W g_K (\Delta\eta)^{2K-1}$. When the taps $h_0, \ldots, h_{T-1}$ are observed with $T \geq 2K + 1$, this gives $\|h^A - h^B\|_2 \geq \kappa^2 W |g_K| (\Delta\eta)^{2K-1}$. With $T \leq 2K$ the observed responses coincide. The impossibility bound holds for every $T$, since it uses only the upper order.

*Testing.* $N$ reset experiments have divergence $N\|h^A - h^B\|_2^2/(2\sigma^2)$. By Pinsker's inequality, a test whose two error probabilities are at most $\alpha$ needs divergence at least $2(1 - 2\alpha)^2$. So $N \geq 4(1 - 2\alpha)^2 \sigma^2 (\Delta\eta)^{-(4K-2)}/C_U^2$, and the constant does not depend on $T$. For the $W_1$ version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta \, s$ scaled to mass $W$, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

**Theorem S4.4.** For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi} \int_{-\pi}^{\pi} (\varrho - 1 - \log \varrho) \, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1} H_1 \Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1} H_1 \Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with $w_k$ the group weights (any factor of $\eta_k$ is absorbed in $\phi$), where $\phi$ is analytic in $\eta$ on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding $\phi$ around

$\bar\eta$, $\Delta G = \sum_{j\ge0} \frac{\partial_\eta^j \phi(z,\bar\eta)}{j!}\,\Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k(\eta_k - \bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+(\Delta\eta)^3$ by (21); for the $K$-group Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j \sum_k(\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \ne 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of $\phi$ in $\eta$ on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j \phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}(z^{-1}H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar\eta))/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function $c$ does not vanish identically: $f = z^{-1}H_0\partial_\eta^j\phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \le \frac12$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-$T$ bound that is uniform in the separation, which we obtain directly. The first $T$ observations are $N(0,\Sigma_j)$ with $\Sigma_j$ the $T\times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2|H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem S4.3), so $0 < \underline f \le f_j \le \bar f < \infty$ with constants that do not depend on the separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \ge \underline f$ and $\|\Sigma_1 - \Sigma_0\| \le \sup_\omega |f_1 - f_0| = \sup_\omega f_0|u| \le \bar f \sup_\omega |u|$. With $A = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}$, whose eigenvalues lie in $[\underline f/\bar f, \bar f/\underline f]$, the inequality $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T\|P_0^T) = \tfrac12\sum_k(\lambda_k - 1 - \log\lambda_k) \le \frac{\bar f}{2\underline f}\,\|A - I\|_F^2 \le \frac{\bar f}{2\underline f}\,T\Big(\frac{\bar f}{\underline f}\Big)^2 \sup_\omega |u|^2 \le C\,T\epsilon^2,$$

with $C$ independent of $T$ and of the separation, because $\sup_\omega|u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on $T$ for every $T$, not only asymptotically. For a fixed pair, $T^{-1}D_{\mathrm{KL}}(P_1^T\|P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

**Theorem S4.3.** In the delay variable $\lambda = z^{-1}$, $H = \kappa\prod_i(1 - r_i\lambda)/[(1-a\lambda)\prod_i(1-r_i\lambda) - \kappa\lambda\sum_i w_i\prod_{j\ne i}(1-r_j\lambda)]$, which is (5) divided through by $z^{K+1}$; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem S4.3. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z\prod_i(z - r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is $H$; Theorem S2.3 then gives the merged measure.

**Proposition S5.2.** With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only group $i$, so $\tilde u_{i,t} = r_i\tilde u_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and, by the ergodic theorem for the stationary linear price process, the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde u_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i\sum_{s\ge0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives $\Sigma_i$. The price is a nonnegative mixture of geometric responses to its innovation (Lemma S8.1), so $\gamma_x(s) = \sum_\ell c_\ell\lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell\sum_m\omega_m/(1-\lambda_\ell\lambda_m) \ge 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i\sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} =$

$\sum_{t=1}^{T}(Z_t - r_i Z_{t+1})\varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance $\sigma_f^2$ and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1})\varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s\in\mathbb{Z}}, \varepsilon_{i,s} : s \le t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (12). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \le \bar{\lambda} < 1$, so with $w_i \ge w_{\min} > 0$ we get $\det \Sigma_i \ge w_{\min}(1 - \bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

**Theorem S4.2.**

*Proof.* The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus $\kappa$ at $t = 0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_1, \dots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each $k_m$ is a polynomial in $h_1, \dots, h_{m+1}$ with coefficients depending on $(a, \kappa)$ (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is $L_h$-Lipschitz, with $L_h$ depending on the class but not on $\Delta\eta$. Let $E$ be the event $\max_{1\le t\le 2K} |\bar{h}_t - h_t| \le \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the $N$ responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \ge 1 - \beta$. On $E$, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m |\hat{k}_m - k_m| \le L_h \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$. Centring at $r_c$ is a fixed triangular linear map, $\mu_m = \sum_{j\le m} \binom{m}{j}(-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j}(s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, $\gamma$-separated nodes $F$ is injective, because a measure with $K$ distinct atoms is determined by its first $2K$ moments. A $C^1$ map that is injective on a compact set $Q$ and has a nonsingular Jacobian there is injective on an open neighbourhood $U$ of $Q$: otherwise there are $x_n \ne y_n$ with $F(x_n) = F(y_n)$ and distance to $Q$ tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at $x$. Applying the quantitative inverse function theorem uniformly on $Q$, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius $\rho_0$ around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is $L$-Lipschitz on each such ball. An error $e$ in $\mu_m$ is an error $e(\Delta\eta)^{-m}$ in $t_m$, largest at $m = 2K-1$. Since $\Delta\eta < 1$ and the centring map is fixed, on $E$ we get $\|\hat{t} - t\|_\infty \le C_1\sqrt{\log(4K/\beta)}\,\epsilon_N$ with $C_1$ equal to $\sqrt{2}L_h$ times the norm of the centring map; and because $\sigma N^{-1/2}\sqrt{2\log(4K/\beta)} \le \sqrt{2}\,\epsilon_N\sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \le 1/\sqrt{2}$. When this is below $\rho_0$, $\hat{t}$ lies in the ball where $F^{-1}$ is $L$-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has $K$ distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly $x$: the Prony polynomial is determined uniquely by the Hankel system, and since the $K$ real nodes of $x$ reproduce $\hat{t}$ it equals $\prod_i(z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where $F^{-1}$ is $L$-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of $L$ (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem S4.1 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight $W$); for any such class the orders match, with constants that depend on the class. □

**Theorem S5.3.**

*Proof.* $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems S4.1 and S4.2. *Upper bounds.* Average the $N$ price responses to estimate $h$ with error of order $\sigma N^{-1/2}$ per tap. Given $h$, class $i$'s mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t-s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \geq \omega_{\min}$, the rates lie in a compact subset of $(0,1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class $i$'s flows with the plug-in $h$ therefore estimates $(w_i, r_i)$ with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class $i$ is $w_i\kappa$, so the mean of the $N$ first-period records gives $P(|\hat{w}_i - w_i| \geq \epsilon) \leq \sigma_f^2/(N\kappa^2\epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by $\epsilon$, or its rate by $\epsilon\Delta\eta$; for small $\epsilon$ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order $\epsilon$, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of $N$ experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

**Unknown primitives.**

**Corollary S8.2** (Unknown primitives cost nothing in the rate)**.** *In the setting of Theorem S4.2, suppose in addition that every memory rate is at least $\varepsilon > 0$ (that is, $r_c - \Delta\eta \geq \varepsilon$), and that the fundamental value $p^*$, the impact $\kappa \in [\underline{\kappa}, \overline{\kappa}] \subset (0, \infty)$ and the persistence $a \in [0, \bar{a}] \subset [0, 1)$ are unknown. Put $\tilde{k}_0 = k_0 + a/\kappa$ and $\tilde{k}_m = k_m$ for $m \geq 1$.*

*(i) The response depends on $(a, \kappa, M)$ only through $\kappa$ and $\tilde{k}$: $h_0 = \kappa$ and $h_{t+1} = \kappa \sum_{m=0}^{t} \tilde{k}_m h_{t-m}$. The persistence $a$ acts exactly like an extra group with memory rate zero and merged weight $a/\kappa$.*

*(ii) The taps $h_0, \ldots, h_{2K+1}$ identify $(\kappa, a, M)$: $\kappa = h_0$; the recursion gives $\tilde{k}_0, \ldots, \tilde{k}_{2K}$; the values $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu' = \sum_i w_i r_i \, \delta_{r_i}$; and $a = \kappa(\tilde{k}_0 - \sum_i w_i)$.*

*(iii) Subtract the pre-probe price to remove $p^*$ (doubling the noise variance), estimate $\hat{\kappa} = \hat{h}_0$, deconvolve with $\hat{\kappa}$ alone, apply Prony's method to the estimated moments of $\mu'$, and set $\hat{w}_i = \widehat{w_i r_i}/\hat{r}_i$ and $\hat{a} = \hat{\kappa}(\hat{\tilde{k}}_0 - \sum_i \hat{w}_i)$. There are constants $c_0, C > 0$ depending on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, \varepsilon, \underline{\kappa}, \overline{\kappa}, \bar{a}, r_c)$ but not on $\Delta\eta$ such that, whenever $\epsilon_N \leq c_0$,*

$$\max_i |\hat{w}_i - w_i| + |\hat{a} - a| + |\hat{\kappa} - \kappa| \;\leq\; C\sqrt{\log(8K/\beta)}\,\epsilon_N$$

*with probability at least $1 - \beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice, the same order as with known primitives.*

*Proof of Corollary S8.2.* (i) Substitute $k_0 = \tilde{k}_0 - a/\kappa$ in $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$: the term $a h_t$ cancels against $\kappa(a/\kappa)h_t$. (ii) Solving the recursion of (i) for $\tilde{k}_t$ gives $\tilde{k}_t = (h_{t+1} - \kappa \sum_{m<t} \tilde{k}_m h_{t-m})/\kappa^2$, a polynomial in $h_1, \ldots, h_{t+1}$ divided by a power of $h_0 = \kappa$. For $m \geq 1$, $\tilde{k}_m = \sum_i w_i r_i^m = \sum_i (w_i r_i) r_i^{m-1}$, so $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu'$, which has $K$ distinct atoms with positive weights; these moments determine $\mu'$ (Theorem S2.3). Since $r_i \geq \varepsilon > 0$, $w_i = (w_i r_i)/r_i$, and then $a = \kappa(\tilde{k}_0 - \sum_i w_i)$. (iii) On the event that every estimated tap is within $\sigma N^{-1/2}\sqrt{2\log(8K/\beta)}$ of its mean (probability at least $1 - \beta/2$ by the Gaussian

tail and a union bound over $2K+2$ taps, with the doubled variance absorbed into the constant), $\hat{\kappa} \geq \underline{\kappa}/2$ once $\epsilon_N \leq c_0$, so the map from taps to $\hat{\tilde{k}}$ is Lipschitz with a constant that depends on $(K, \underline{\kappa}, \overline{\kappa}, \bar{a}, W)$ and not on $\Delta\eta$. The weights of $\mu'$ lie in $[\varepsilon\omega_{\min}, \omega_{\max}]$ and its rates are those of the class of Theorem S4.2; the proof of that theorem uses only a $\Delta\eta$-free Lipschitz deconvolution, the rescaling by $(\Delta\eta)^{-m}$ and weights bounded below, so it applies verbatim to $\mu'$ and gives $\max_i |\widehat{w_i r_i} - w_i r_i| \leq C_1\sqrt{\log(8K/\beta)}\,\epsilon_N$ and rate errors smaller by a factor $\Delta\eta$, with probability at least $1-\beta/2$. The maps $(v, r) \mapsto v/r$ on $r \geq \varepsilon/2$ and $(\kappa, \tilde{k}_0, w) \mapsto \kappa(\tilde{k}_0 - \sum_i w_i)$ are Lipschitz on the bounded set reached, and $|\hat{\kappa} - \kappa|$ and $|\hat{\tilde{k}}_0 - \tilde{k}_0|$ are of order $\sigma N^{-1/2} \leq \epsilon_N$. Combining gives the bound. □

**Proposition S2.2.** The proof is given after the statement in the main text.

# S9 Equilibrium: further results and proofs

**Corollary S9.1** (Disclosure in equilibrium)**.** *Take a positive-impact equilibrium of Theorem S3.1 with positive weights and $D > \max_i r_i$.*

*(i) The groups' flows obey $u_{i,t+1} = r_i u_{i,t} + w_i x_t$ with their* true *weights. The price is the stable closed loop of Theorem S3.2, driven by $\xi$, with positive effective weights. So Proposition S5.2 applies: tagged flow identifies each group's $(r_i, w_i)$, with a limiting covariance that has no singularity as rates collide.*

*(ii) On classes in which both the true and the effective weights lie in compact positive intervals, Theorem S5.3 holds in equilibrium. Tagging cuts the experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ for weights and $(\Delta\eta)^{-(4K-4)}$ for rates.*

**In words.** What makes disclosure valuable survives the dealers' optimization. Prices still hide the groups at the same polynomial cost, and each group's own flow still reveals that group's memory and weight directly.

**Theorem S9.2** (Fragility: where liquidity disappears)**.** *Let there be one feedback group with weight $w > 0$ and memory rate $r \in (0,1)$, with $\bar{a} \in [0,1)$ and $\sigma_F, \sigma_\xi > 0$. Call an equilibrium of Theorem S3.1 with $\kappa > 0$ and $D > 0$ a positive-impact equilibrium. For $D \in (m,1)$ with $m = \max(\bar{a}, r)$, let*

$$\kappa_w(D) = \frac{(1-D)(D-r)}{w}, \qquad \Psi_w(D) = \frac{\kappa_w(D)\,(D-\bar{a})\,D}{\sigma_F^2 D^2 + \kappa_w(D)^2\sigma_\xi^2}, \qquad A^*(w) = \max_{D\in(m,1)} \Psi_w(D).$$

*(i)* Threshold.

- *The positive-impact equilibria are exactly the pairs $(\kappa_w(D), D)$ with $D \in (m,1)$ and $\Psi_w(D) = A$, and every one of them is stable.*
- *A positive-impact equilibrium exists if and only if $A \leq A^*(w)$.*
- *For $A < A^*(w)$ there are at least two; for $A > A^*(w)$ there are none.*

*(ii)* Monotonicity. *$A^*(w)$ is strictly decreasing in $w$. As $w \downarrow 0$ it tends to $A_0^* = (1-\bar{a})/(2\sigma_F\sigma_\xi)$, the threshold of Theorem S3.1(ii).*

*(iii)* Bounds and rate. *For every $w > 0$,*

$$A^*(w) < \min\Big\{A_0^*,\ \frac{\Phi^*}{\sigma_F^2\, w}\Big\},$$

*where*

$$\Phi^* = \max_{D\in[m,1]} \frac{(1-D)(D-r)(D-\bar{a})}{D} \le \frac{(1-\bar{a})(1-r)^2}{4},$$

*and $w\,A^*(w) \to \Phi^*/\sigma_F^2$ as $w \to \infty$. In terms of the feedback intensity $G(1) = w/(1-r)$, strong feedback gives $A^* \approx \Phi^*/(\sigma_F^2(1-r)G(1))$, and always $A^* < (1-\bar{a})(1-r)/(4\sigma_F^2 G(1))$.*

*(iv)* Several groups. *Let all weights be positive.*

- *The positive-impact equilibria with $D > \max_i r_i$ are exactly the pairs $((1-D)/G(D), D)$ with $D \in (\max(\bar{a}, \max_i r_i), 1)$ at which the analogue of $\Psi_w$, with $\kappa = (1-D)/G(D)$, equals $A$.*
- *Every such equilibrium has*

$$A < \min\Big\{A_0^*,\ \frac{(1-\sqrt{\bar{a}})^2}{\sigma_F^2\, G(1)}\Big\},$$

  *whatever the number of groups and their rates.*
- *Every such equilibrium is stable.*

**In words.** With positive feedback, dealers can make a market only if their dislike of risk is below a threshold, and the threshold falls as feedback grows: for strong feedback it is inversely proportional to the feedback intensity. Past the threshold, no price impact is consistent with dealers' optimization. Below it, every equilibrium has stable prices. Liquidity does not erode through growing instability; it disappears when the two equilibria merge, at a point that stable prices give no warning of. At the calibration of Supplement S9 ($\bar{a} = 0.6$, $r = 0.7$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$), $A^*$ is 0.40 without feedback, 0.25 at $G(1) = \frac{1}{15}$ and 0.034 at $G(1) = \frac{2}{3}$, and $w\,A^*$ approaches $\Phi^* = 0.00688$. The necessary conditions behind parts (i) and (iii) are machine-checked (`Paper1EquilibriumReducedForm.lean`). With several groups, part (iv) gives a threshold that depends on the population only through its total feedback intensity $G(1)$. At the same calibration, numerically, the exact thresholds for two and three groups lie between 28% and 86% of that bound, and the equilibria on the threshold curve are stable. Every such equilibrium is stable (part (iv)), as the scan also shows.

**Theorem S9.3** (A strategic insider in the inventory equilibrium)**.** *Add three things to the model of Theorem S3.1:*

- News. *The fundamental moves by $F_{t+1} = F_t + \theta_t + \varepsilon_{t+1}$, with $\theta_t \sim N(0, \sigma_\theta^2)$ and $\varepsilon_t$ i.i.d. of variance $\sigma_\varepsilon^2$.*
- An insider. *At date $t$ a short-lived insider privately observes $\theta_t$, and nothing else, submits $d_t$, and values the position at $F_{t+1}$.*
- What groups and dealers see. *The groups trade on the observed deviation $s_t = p_t - F_t$. The dealers price from total flow.*

*Let $\nu_t = \xi_t + d_t$ be the unexpected flow.*

*(i)* Characterization. *The strategy $d_t = b\theta_t$ and the price $p_t = F_t + \lambda\nu_t + x_t$, with $x_t$ the price law of Theorem S3.1, form an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa G(D), \quad \kappa(D - \bar{a}) = AVD, \quad \lambda = \frac{b\sigma_\theta^2}{\sigma_\nu^2}, \quad b = \frac{D}{2(\kappa + \lambda)},$$

$$\sigma_\nu^2 = \sigma_\xi^2 + b^2\sigma_\theta^2, \qquad V = \sigma_\varepsilon^2 + \frac{\sigma_\theta^2\sigma_\xi^2}{\sigma_\nu^2} + \Big(\frac{\kappa + \lambda}{D}\Big)^2\sigma_\nu^2.$$

*One unit of unexpected flow moves the price by $(\kappa + \lambda)/D$, and the insider's expected profit per date is $b\sigma_\theta^2/2$.*

*(ii)* Kyle limit. *Without inventory risk ($A = 0$), $\kappa = 0$, $D = 1$, $\lambda = \sigma_\theta/(2\sigma_\xi)$ and $b = \sigma_\xi/\sigma_\theta$. These are Kyle's one-period values at every date, and the feedback groups do not move prices.*

*(iii)* Invariance. *The coefficients depend on the groups only through $G$, as Theorem S2.1 requires.*

*(iv)* Observer. *The observed deviation follows*

$$s_t = a^* s_{t-1} + \kappa^* \sum_i \tilde{u}_{i,t} + c_0\nu_t - c_1\nu_{t-1}, \qquad c_0 = \frac{\kappa + \lambda}{D}, \quad c_1 = \frac{\bar{a}\lambda}{D},$$

*with $(a^*, \kappa^*, \tilde{u})$ as in Theorem S3.2. This is the closed loop of Section S1 driven through a known moving-average filter $c_0 - c_1 z^{-1}$, which is invertible because $c_1 < c_0$.*

- *The memory rates remain the transmission zeros of the feedback part.*
- *The lower-bound pair of Theorem S4.1 is realized by two such equilibria with identical primitives.*
- *So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present.*

**In words.** A short-lived insider who knows the next piece of news trades against noise traders and against the feedback traders' predictable flow. Dealers net the predictable flow out. So the insider's trades move prices through two channels: an information part $\lambda$ and an inventory part, both amplified by feedback through $D$. Positive feedback raises the total impact. The insider then trades less and earns less, and prices reveal less of her news. Negative feedback does the opposite.

- *An example from Supplement S9.* One positive-feedback group lowers the share of the insider's information impounded in prices from 0.372 to 0.366. One negative-feedback group raises it to 0.415.
- *What prices cannot identify.* It is still worth nothing to anyone trading on public information.
- *Machine check.* `Paper1EquilibriumInsider.lean` checks the observer's pathwise reduced form, the insider's best response, the Kyle limit and the posterior variance.
- *Scope.* The insider's information is short-lived, and she conditions only on her news. Two cases are not solved: an insider who also saw the public state, who would trade against the predictable mispricing; and a long-lived Kyle insider, who would also time her trades against the groups' memory.

**A misjudged dealer and a targeted halt.** The next two results use the simplest linear market inside Theorem S2.1. They are summarized in Section S3.

- Dates are $t = 0, \ldots, T - 1$ and the price change is $s_t = p_t - p_{t-1}$, with $p_{-1} = 0$.

- The dealer moves the price by $s_t = \lambda(y_t - \hat{f}_t)$, with $\lambda > 0$, where $\hat{f}_t = \sum_{m<t} \hat{k}_m s_{t-1-m}$ is the feedback flow it predicts from its kernel $\hat{k}$.
- The insider, with value $v$ of mean zero and variance $\Sigma_0$ independent of the noise, commits to a schedule $x = bv$, $b \in \mathbb{R}^T$. Schedules may change sign, so pump-and-dump patterns are allowed.
- The kernel error is $\Delta k = k - \hat{k}$, with $\varepsilon = \sum_{m \le T-2} |\Delta k_m|$.

Let $C$ be the lower-triangular matrix of ones and $L$ the strictly lower Toeplitz matrix with $L_{t,t-1-m} = \Delta k_m$.

**Theorem S9.4** (Rents from a misjudged aggregate response)**.** *(i) The insider's expected profit is $J(b) = \Sigma_0(\mathbf{1}^\top b - b^\top A b)$, with $A = \lambda C(I - \lambda L)^{-1}$. If $Q = \frac{1}{2}(A + A^\top) \succ 0$, the optimal schedule is $b^* = \frac{1}{2}Q^{-1}\mathbf{1}$ and the value is $V(\Delta k) = \frac{\Sigma_0}{4}\mathbf{1}^\top Q^{-1}\mathbf{1}$.*

*(ii) If $\lambda T\varepsilon \le \frac{1}{4}(1 - \lambda\varepsilon)$, then $Q \succ 0$ and*

$$|V(\Delta k) - V(0)| \le \frac{2\Sigma_0 T^2 \varepsilon}{1 - \lambda\varepsilon}.$$

*(iii) To first order, $V(\Delta k) = V(0) - \Sigma_0 \sum_m \omega_m \Delta k_m + O(\varepsilon^2)$, with $\omega_m = \frac{(T-m)(T-m-1)}{2(T+1)^2} > 0$ and $V(0) = \frac{\Sigma_0 T}{2\lambda(T+1)}$. The rent is therefore of exact order $\varepsilon$. The insider gains when the dealer overstates the feedback ($\Delta k < 0$) and loses when it understates it.*

*(iv) If the dealer estimates the aggregate kernel from $n$ observations with error $\varepsilon_n = O_p(n^{-1/2})$, the rent is $O_p(n^{-1/2})$. Separating the groups behind the same kernel still needs order $(\Delta\eta)^{-(4K-2)}$ experiments (Theorem S4.1).*

**Theorem S9.5** (Targeted intervention in equilibrium)**.** *Take the Gauss and Radau pairs of Theorem S2.1(iv), with total weight $W$, on rates $c + ds$ with $\rho = |c| + |d| < 1$. Place them in the market above, with the same dealer rule $(\lambda, \hat{k})$, the same insider schedule and the same draws.*

*(i) The kernels agree for $m \le 2$, and $|k^G_m - k^R_m| \le 2W\binom{m}{3}|d|^3\rho^{m-3}$ for $m \ge 3$. Hence $\|k^G - k^R\|_1 \le 2W|d|^3/(1-\rho)^4$.*

*(ii) Before any intervention, the price paths differ by at most $\lambda^2 T\|k^G - k^R\|_1\|x + z\|$ divided by $(1 - \lambda\|k^G - \hat{k}\|_1)(1 - \lambda\|k^R - \hat{k}\|_1)$, which is $O(|d|^3)$. The insider's values differ by $O(|d|^3)$.*

*(iii) Halting the largest group leaves kernels $\frac{W}{2}(c - d\sigma)^m$ (the Gauss weights tie; we remove the group at $c + d\sigma$) and $\frac{W}{4}(c-d)^m$, which differ by $\frac{W}{4}$ at $m = 0$ for every spacing. The post-halt responses therefore differ at order one, and so do the post-halt feedback budgets $\sum_m k_m$.*

*Proof of Theorem S2.1. Part (i).* By induction, $m_{i,t} = \sum_{m<t} r_i^m s_{t-1-m}$. Multiplying by $w_i$ and summing over $i$ gives $f_t = \sum_{m<t} k_m s_{t-1-m}$.

*Part (ii).* If $k_m = k'_m$ for $m < T$, then by (i) the two aggregate flows agree on $t \le T$ for every signal. Conversely, the impulse signal sequence $s_j = \mathbf{1}\{j = 0\}$ gives $f_{m+1} = k_m$, so equal flows on $t \le T$ for every signal sequence force $k_m = k'_m$ for $m < T$. This direction quantifies over signal sequences, not over price paths under a fixed rule $S$: a rule such as $S \equiv 0$ never produces the impulse.

On an unbounded horizon, equality of all $k_m = \int r^m \, dM(r)$ is equality of all moments of two finite signed measures on $[0, 1]$. Polynomials are dense in $C[0, 1]$, so the measures coincide. In terms of $\eta$ this is the condition (9) of Theorem S2.3.

*Part (iii).* Fix a strategy profile and a realization. We argue by strong induction on $t \le T$. If prices and flows agree before $t$:

- the signals agree before $t$, because $S_j$ depends on prices up to $j$;
- the aggregate flows at $t$ agree, by part (i) and the equal kernels, since $f_t$ uses signals before $t$ only;
- the insider's orders at $t$ agree, by causality;
- hence the total flows $y_t$ agree, and then the prices $p_t$ agree, by causality of the dealer's rule.

The insider's profit $\sum_{t<T}(v - p_t)x_t$ and the dealer's inventory $-\sum_{t<T} y_t$ are functions of these paths.

Strategy sets are the same in the two games, because the rules are defined on the same histories. Every payoff is an expectation of the same function of $(v, z)$, so best responses coincide and so do the equilibria.

*Part (iv).* The Gauss rule has weights $\frac{1}{2}, \frac{1}{2}$ and nodes $\pm\sigma$ with $\sigma^2 = \frac{1}{3}$. The Radau rule has weights $\frac{1}{4}, \frac{3}{4}$ and nodes $-1, \frac{1}{3}$. Both integrate $1, s, s^2$ exactly, so on rates $c + ds$ the kernels agree for $m \le 2$. The largest weights are $\frac{1}{2}$ and $\frac{3}{4}$. □

**Machine check.** `Paper1EquilibriumInvariance.lean` proves the following:

- part (i): `p1_eqAggFlow_kernel`;
- both directions of part (ii) on a finite horizon: `p1_eqAggFlow_of_kernel` and `p1_kernel_of_eqAggFlow`;
- the pathwise statement of part (iii) for arbitrary causal insider, dealer and signal rules: `p1_eq_market_invariance` and `p1_eq_insider_profit_invariance`;
- part (iv): `p1_eq_group_level_differs`.

The unbounded-horizon moment statement and the passage from pathwise equality to equal equilibria are on paper.

**Group-specific demand shocks.** Suppose group $i$ also trades an idiosyncratic shock $\epsilon_{i,t}$ through its memory, so that $f_t$ gains the term $\sum_i w_i \sum_{m<t} r_i^m \epsilon_{i,t-1-m}$. The same induction shows that the equilibrium depends on the groups only through the kernel and the law of this aggregate shock process. For independent Gaussian shocks with variances $\sigma_i^2$, that law is determined by the spectral density $\sum_i w_i^2 \sigma_i^2 |1 - r_i e^{i\omega}|^{-2}$. This is a second merged object, and the invariance class shrinks accordingly. The paper's groups have no idiosyncratic shocks.

**The textbook benchmark.** Take a risk-neutral, competitive dealer that knows the kernel. It can compute $f_t$ from public prices and price only $y_t - f_t$, so predictable feedback flow then has no price impact, and the closed loop of Section S1 is absent in that equilibrium. The theorem holds in that benchmark as well. Feedback reaches prices in equilibrium when the dealer bears inventory risk, or when it does not know the kernel and must learn it. These are the cases studied in the equilibrium extension.

**Numerical illustration.** `equilibrium/e1_demo.py` runs one market. The insider submits $x_t = 0.3(v - p_{t-1})$. The dealer sets $p_t = p_{t-1} + 0.3\, y_t - 0.05\, (I_t - I_{t-1})$. The signal is $s_t = p_t - p_{t-1}$. Two group configurations are compared, the Gauss and Radau pairs of part (iv) on rates $0.6 + 0.2s$, with total weight 0.5, under the same $(v, z)$ draws. Over 200 draws:

- price paths coincide to machine precision on the first four dates ($t \le 3$), with largest gap $2 \times 10^{-16}$;
- from date five, where $k_3$ first differs (by $\frac{2}{9}d^3$ times the total weight), they separate, with largest gap $1.8 \times 10^{-4}$ at date five and $1.9 \times 10^{-2}$ by date twenty;
- the largest group's share of feedback positions stays at medians of 0.59 and 0.79.

The result file is `equilibrium/e1_demo_results.json`.

**Proof of Theorem S9.4.** *Part (i).* Total flow is $y = x + z + f$. The dealer subtracts $\hat{f}$, so $s = \lambda(x + z + Ls)$. The matrix $L$ is strictly lower triangular, so $I - \lambda L$ is invertible and $s = \lambda(I-\lambda L)^{-1}(x+z)$. Prices are $p = Cs$. With $x = bv$, realized profit is $\sum_t (v-p_t)x_t = v^2 \mathbf{1}^\top b - v\, b^\top C s$. Taking expectations with $E[v] = 0$ and $v$ independent of $z$ gives $J(b)$. When $Q \succ 0$, $J$ is strictly concave, and its maximizer and value are as stated.

*Part (ii).* A Toeplitz operator is bounded by the $\ell_1$ norm of its symbol, so $\|L\| \le \varepsilon$. For $\lambda\varepsilon < 1$ this gives $\|(I - \lambda L)^{-1}\| \le 1/(1 - \lambda\varepsilon)$ and $(I - \lambda L)^{-1} - I = \lambda L(I - \lambda L)^{-1}$. Since $\|C\| \le T$, we get $\|Q - Q_0\| \le \|A - A_0\| \le \lambda^2 T\varepsilon/(1 - \lambda\varepsilon)$. Moreover $Q_0 = \frac{\lambda}{2}(I + \mathbf{1}\mathbf{1}^\top) \succeq \frac{\lambda}{2} I$. Under the stated condition $\|Q - Q_0\| \le \lambda/4$, so $Q \succeq \frac{\lambda}{4} I$. Then

$$|V - V_0| = \tfrac{\Sigma_0}{4} |\mathbf{1}^\top Q^{-1}(Q_0 - Q)Q_0^{-1}\mathbf{1}| \le \tfrac{\Sigma_0}{4}\, T \cdot \tfrac{4}{\lambda} \cdot \tfrac{2}{\lambda} \cdot \tfrac{\lambda^2 T\varepsilon}{1-\lambda\varepsilon}.$$

*Part (iii).* $Q_0^{-1}\mathbf{1} = \frac{2}{\lambda(T+1)}\mathbf{1}$, which gives $V(0)$. Since $A = A_0 + \lambda^2 CD + O(\varepsilon^2)$, the derivative of $\mathbf{1}^\top Q^{-1}\mathbf{1}$ is $-u^\top(\lambda^2 CD)u$ with $u = Q_0^{-1}\mathbf{1}$. Using $\mathbf{1}^\top C = (T, T-1, \ldots, 1)$ and $(L\mathbf{1})_t = \sum_{m<t} \Delta k_m$, we get $\mathbf{1}^\top CL\mathbf{1} = \sum_m \Delta k_m \frac{(T-m)(T-m-1)}{2}$. Multiplying by $\frac{\Sigma_0}{4} \cdot \frac{4}{\lambda^2(T+1)^2} \cdot \lambda^2$ gives $\omega_m$.

*Part (iv).* This is part (ii) with $\varepsilon = \varepsilon_n$. The rate for $\varepsilon_n$ is an assumption on the dealer's estimator, for example least squares of order flow on lagged price changes in a stationary market. It is not proved here.

**Proof of Theorem S9.5.** *Part (i).* Write $\mu_j^G, \mu_j^R$ for the $j$th moments of the two rules, which are at most one in absolute value. Expanding $(c + ds)^m$ gives

$$k_m^G - k_m^R = W \sum_{j \ge 3} \binom{m}{j} c^{m-j} d^j (\mu_j^G - \mu_j^R).$$

The terms with $j \le 2$ vanish because both rules are exact for those degrees. Since $\binom{m}{j} \le \binom{m}{3}\binom{m-3}{j-3}$, the sum is at most $2W\binom{m}{3}|d|^3\rho^{m-3}$. Finally, $\sum_{m\ge3}\binom{m}{3}\rho^{m-3} = (1-\rho)^{-4}$.

*Part (ii).* With residual matrices $L_G, L_R$ built from $k^G - \hat{k}$ and $k^R - \hat{k}$,

$$\begin{aligned} p^G - p^R &= \lambda C\left[(I - \lambda L_G)^{-1} - (I - \lambda L_R)^{-1}\right](x + z) \\ &= \lambda^2 C(I - \lambda L_G)^{-1}(L_G - L_R)(I - \lambda L_R)^{-1}(x + z). \end{aligned}$$

Bound each factor as in the proof of Theorem S9.4. The value comparison follows from the same perturbation argument, with base point $L_R$ instead of 0.

*Part (iii).* Removing one of the two Gauss groups leaves weight $W/2$. Removing the Radau group with weight $3/4$ leaves $W/4$. At $m = 0$ the kernel is the remaining weight.

**Proof of Theorem S3.1.** *Part (i).* Conjecture $x_t = -\kappa H_t + \sum_i \beta_i u_{i,t+1}$ and set $D = 1 - \sum_i \beta_i w_i$. The next-period expectation follows from three facts: $E_t\xi_{t+1} = 0$, $E_t H_{t+1} = \bar{a} H_t -$

$\sum_i u_{i,t+1}$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Together they give

$$D\,E_t x_{t+1} = -\kappa \bar{a} H_t + \sum_i (\kappa + \beta_i r_i) u_{i,t+1}.$$

The dealers' condition $x_t = E_t x_{t+1} - AVH_t$ must hold for all states. Matching the coefficient on $u_{i,t+1}$ gives $\beta_i = (\kappa + \beta_i r_i)/D$, that is, $\beta_i = \kappa/(D - r_i)$. The definition of $D$ then becomes $D = 1 - \kappa G(D)$, and matching on $H_t$ gives $\kappa(D - \bar{a}) = AVD$.

For the variance, $x_{t+1}$ depends on $\xi_{t+1}$ through $H_{t+1}$ and through its own effect on $u_{i,t+2}$. Its innovation $\delta$ therefore satisfies $\delta = \kappa\xi_{t+1} + \sum_i \beta_i w_i \delta$, so $\delta = \kappa\xi_{t+1}/D$ and $V = \sigma_F^2 + \kappa^2\sigma_\xi^2/D^2$.

Conversely, take a solution of (11). Then the conjectured law satisfies the dealers' condition in every state (`p1_inventory_foc`), and $V$ is consistent with it (`p1_inventory_innovation`). If the induced dynamics are stable, the process is stationary and the solution is an equilibrium.

*Part (ii).* With no groups, $\beta$ is empty and $D = 1$. Then $x_t = -\kappa H_t$ and $H_t = \bar{a} H_{t-1} - y_t$ give $x_t = \bar{a} x_{t-1} + \kappa y_t$. The quadratic $A\sigma_\xi^2\kappa^2 - (1 - \bar{a})\kappa + A\sigma_F^2 = 0$ has discriminant $\Delta$ (`p1_inventory_benchmark`).

*Part (iii).* The system (11) involves the groups only through $G$. The price process is a function of the history of flows, and by Theorem S2.1(i) it depends on the groups only through the kernel $k_m = \int r^m\, dM$. A finite measure on $[0, 1]$ is determined by $G$ off its support, and also by its moments.

*Part (iv).* Write $\Phi(\kappa, D; M) = \big(D + \kappa G(D) - 1,\ \kappa(D - \bar{a}) - A(\sigma_F^2 D + \kappa^2\sigma_\xi^2/D)\big)$. At $M = 0$ the zero is $(\kappa_-, 1)$. There the Jacobian with respect to $(\kappa, D)$ is triangular with determinant $\sqrt{\Delta} \neq 0$: $\partial_D \Phi_1 = 1$, $\partial_\kappa \Phi_1 = G = 0$, and $\partial_\kappa \Phi_2 = (1 - \bar{a}) - 2A\kappa_-\sigma_\xi^2 = \sqrt{\Delta}$. The implicit function theorem, applied in the norm of the statement, gives a unique zero near $(\kappa_-, 1)$ that is continuous in $M$.

Linearizing, $dD = -\kappa_- G(1)$. Differentiating the second equation and using $A(\sigma_F^2 + \kappa_-^2\sigma_\xi^2) = \kappa_-(1 - \bar{a})$ gives $d\kappa = -\kappa_-(\bar{a} + 2A\kappa_-\sigma_\xi^2)\, dD/\sqrt{\Delta}$, and so the stated formula. Finally, $G(1) = \sum_i w_i/(1 - r_i) = \sum_i q_i$.

At $M = 0$ the transition matrix has eigenvalues $\bar{a}$ and the $r_i$, all inside the unit disk, so the dynamics remain stable for small $M$.

**Proof of Theorem S3.2.** *Part (i).* Write the implicit form of the price at $t$, $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t}$, and the price law at $t - 1$, $x_{t-1} = -\kappa H_{t-1} + \sum_i \beta_i u_{i,t}$. Use $H_t = \bar{a} H_{t-1} - \sum_i u_{i,t} - \xi_t$. Then

$$Dx_t - \bar{a} x_{t-1} = \kappa\xi_t + \sum_i (\kappa + \beta_i(r_i - \bar{a})) u_{i,t}.$$

Since $\kappa = \beta_i(D - r_i)$, the coefficient is $\beta_i(D - \bar{a}) = \kappa(D - \bar{a})/(D - r_i)$. Divide by $D$.

The scaled flows inherit the group recursion, because the scaling is constant (`p1_eq_scaled_cohort`). The states $(H_t, u_{\cdot,t+1})$ and $(x_t, \tilde{u}_{\cdot,t+1})$ are related by an invertible linear map, since $\kappa \neq 0$ and every scale factor is nonzero. So the two dynamics are conjugate and have the same eigenvalues.

*Part (ii).* The effective weights are nonzero because $D \neq \bar{a}$. Apply Theorem S4.3(i) to the closed loop of part (i).

*Part (iii).* Since $G(D) = \sum_i w_i/(D - r_i) = \sum_i \tilde{w}_i/(D - \bar{a})$, the equation $D = 1 - \kappa G(D)$ reads $D = 1 - \kappa^* D\, \tilde{M}([0, 1])/(D(1 - a^*))$. Solve for $D$; then $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. Conversely, $(\bar{a}, \kappa, D, M)$ determine the effective objects. So two equilibria give the same price response if and only if their effective objects agree, and then their merged measures agree. This is consistent with Theorem S2.1.

*Part (iv).*

- *$D$ exceeds every rate.* The hypothesis of Theorem S4.1 is $a^* + \kappa^* W/(1 - r_c - \Delta\eta) < 1$. It gives $\kappa^* W/(1 - a^*) < 1 - r_c - \Delta\eta$, so $D > r_c + \Delta\eta$, which is at least every rate.
- *Each economy is an equilibrium.* With the stated choices, $D + \kappa G(D) = 1$ holds (`p1_eq_realize_D`), and so does $\kappa(D - \bar{a}) = AVD$ with $V = \sigma_F^2 + \kappa^{*2}\sigma_\xi^2$ (`p1_eq_realize_dealer`). By part (i) the equilibrium price response is the given closed loop, which is stable. So each economy is an equilibrium by Theorem S3.1(i).
- *The two economies share everything but the groups.* They have the same total effective weight $W$, so they share $D$, $\kappa$ and the primitives, and their price responses are those of Theorem S4.1.
- *Upper bound.* On a class where the effective parameters are known or estimated at the parametric rate (Corollary S8.2), Theorem S4.2 recovers $\tilde{w}$. Then $D$ follows from part (iii), and $w_i = \tilde{w}_i(D - r_i)/(D - \bar{a})$ is Lipschitz on the class.

**Proof of Theorem S9.2.** *Part (i).*

- *$D$ lies in $(m, 1)$.* Let $\kappa > 0$ and $D > 0$. The dealer equation $\kappa(D - \bar{a}) = AVD > 0$ gives $D > \bar{a}$. The group equation is $(D - 1)(D - r) + \kappa w = 0$. If $D < r$, it forces $D > 1$, which is impossible; and $D = r$ is excluded. So $D \in (r, 1)$, and $\kappa = \kappa_w(D)$.
- *The equilibria are the solutions of $\Psi_w(D) = A$.* Multiplying the dealer equation by $D$ gives $\kappa(D - \bar{a})D = A(\sigma_F^2 D^2 + \kappa^2\sigma_\xi^2)$, which is $\Psi_w(D) = A$. Conversely, every such $D$ solves the system.
- *Stability.* By Theorem S3.2(i) the dynamics are those of the closed loop with $c = \kappa^*\tilde{w} = (1 - D)(D - \bar{a})/D \in (0, 1)$. Its characteristic polynomial is $p(\lambda) = \lambda^2 - (a^* + r + c)\lambda + a^* r$. The roots are real and nonnegative, because the discriminant is at least $(a^* - r)^2$ and the coefficients alternate in sign. Next, $p(1) = (1 - a^*)(1 - r) - c = (D - \bar{a})(D - r)/D > 0$, so 1 lies outside the interval between the roots. Their product $a^* r$ is below 1, so both roots are below 1.
- *Existence and count.* $\Psi_w$ is continuous and positive on $(m, 1)$ and tends to 0 at both ends: as $D \to 1$, $\kappa_w \to 0$; as $D \to m$, either $D - \bar{a} \to 0$ or $\kappa_w \to 0$. So its maximum $A^*$ is attained. Every $A \in (0, A^*)$ is taken at least twice, once on each side of a maximizer, and no $A > A^*$ is taken.

*Part (ii).*

- *A parametrization by $\kappa$.* For fixed $\kappa \in (0, (1 - r)^2/(4w)]$, the solutions of $\kappa_w(D) = \kappa$ are $D_\pm(\kappa, w) = \big((1 + r) \pm \sqrt{(1 - r)^2 - 4\kappa w}\big)/2$.
- *The larger root gives the maximum.* $\tilde{\Psi}(\kappa, D) = \kappa(D - \bar{a})D/(\sigma_F^2 D^2 + \kappa^2\sigma_\xi^2)$ is strictly increasing in $D$ for $D > \bar{a}$. The numerator of its $D$-derivative is $\bar{a}\sigma_F^2 D^2 + (2D - \bar{a})\kappa^2\sigma_\xi^2 > 0$. So $A^*(w) = \max_\kappa \tilde{\Psi}(\kappa, D_+(\kappa, w))$.
- *Strict decrease.* Let $w_1 < w_2$, and let $\kappa^\circ$ attain $A^*(w_2)$. Then $\kappa^\circ$ is feasible at $w_1$, and $D_+(\kappa^\circ, w_1) > D_+(\kappa^\circ, w_2)$. So $A^*(w_1) \geq \tilde{\Psi}(\kappa^\circ, D_+(\kappa^\circ, w_1)) > A^*(w_2)$.
- *The limit as $w \downarrow 0$.* The bound $A^* < A_0^*$ of part (iii) is one half. For the other, fix $\kappa_0 = \sigma_F/\sigma_\xi$. As $w \downarrow 0$, $D_+(\kappa_0, w) \to 1$, and $\tilde{\Psi}(\kappa_0, D_+) \to (1 - \bar{a})/(2\sigma_F\sigma_\xi)$.

*Part (iii).*

- *The bound $A_0^*$.* By the arithmetic–geometric mean inequality, $\sigma_F^2 D^2 + \kappa^2\sigma_\xi^2 \geq 2\sigma_F\sigma_\xi\kappa D$. So $\Psi_w(D) \leq (D - \bar{a})/(2\sigma_F\sigma_\xi) < A_0^*$. Machine-checked in squared form as $4A^2\sigma_F^2\sigma_\xi^2 \leq (D - \bar{a})^2$ (`p1_fragility_benchmark_bound`).

- *The bound $\Phi^*/(\sigma_F^2 w)$.* Dropping $\kappa^2\sigma_\xi^2 > 0$ from the denominator gives $\Psi_w(D) < (1-D)(D-r)(D-\bar{a})/(\sigma_F^2 wD) \le \Phi^*/(\sigma_F^2 w)$. Machine-checked as $A\sigma_F^2 wD \le (1-D)(D-r)(D-\bar{a})$ (`p1_fragility_feedback_bound`).
- *The bound on $\Phi^*$.* It uses $(1-D)(D-r) \le (1-r)^2/4$ and $(D-\bar{a})/D \le 1-\bar{a}$.
- *The limit as $w \to \infty$.* We have $\kappa_w = O(1/w)$ uniformly on $[m,1]$. So $w\Psi_w(D) \to (1-D)(D-r)(D-\bar{a})/(\sigma_F^2 D)$ uniformly, and the maxima converge.

*Part (iv).*

- *$\kappa$ is determined by $D$.* Let $D > \max_i r_i$. All weights are positive, so $G(D) > 0$. Then $D = 1-\kappa G(D)$ with $\kappa > 0$ gives $D < 1$ and $\kappa = (1-D)/G(D)$. The dealer equation then reads as in part (i), and $D > \bar{a}$ as there.
- *Comparison with $G(1)$.* For $D \in (\max_i r_i, 1)$, each $D - r_i < 1 - r_i$, so $G(D) > G(1)$.
- *The bound.* The dealer equation gives $A\sigma_F^2 D < \kappa(D-\bar{a}) < (1-D)(D-\bar{a})/G(1)$. Moreover

$$(1-D)(D-\bar{a}) = D(1-\sqrt{\bar{a}})^2 - (D-\sqrt{\bar{a}})^2 \le D(1-\sqrt{\bar{a}})^2.$$

  Machine-checked as `p1_fragility_K_bound`. The bound $A_0^*$ follows as in part (iii).
- *Stability.* By Theorem S3.2(i), the dynamics are those of the closed loop with $(a^*, \kappa^*, \tilde{w})$. Here $a^* \ge 0$, $\kappa^* > 0$, $\tilde{w}_i > 0$ and $r_i \ge 0$, so its transition matrix in the state $(x_t, y_{1,t}, \dots, y_{K,t})$ is nonnegative.
  - *The spectral radius is the root above the rates.* By Perron–Frobenius, the spectral radius $\rho$ is an eigenvalue. Every eigenvalue $z \notin \{0, r_i\}$ solves $F(z) = 1$, where $F(z) = a^*/z + \kappa^* \sum_i \tilde{w}_i/(z - r_i)$. No $r_i$ is an eigenvalue, because $\tilde{w}_i \ne 0$ (Theorem S4.3(i)). $F$ decreases strictly from $+\infty$ to 0 on $(\max_i r_i, \infty)$, so it has exactly one root $z^*$ there, and $\rho \ge z^*$. Being real and at least $z^*$, $\rho$ equals $z^*$.
  - *The criterion.* Hence $\rho < 1$ if and only if $F(1) < 1$, that is, $a^* + \kappa^* \sum_i \tilde{w}_i/(1-r_i) < 1$. With $g_i = w_i/(D-r_i)$ this reads $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_effective_stability_iff`).
  - *It holds.* Each $1 - r_i > 1 - D > 0$ and $\kappa \sum_i g_i = 1 - D$, so $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_stability_K`).

**Proof of Corollary S9.1.** *Part (i).*

- The group recursion is part of the model.
- By Theorem S3.2(i), the price is the closed loop with $a^* \ge 0$, $\kappa^* > 0$ and $\tilde{w}_i > 0$, driven by the i.i.d. flow $\xi$.
- It is stable by Theorem S9.2(iv).
- These are the hypotheses of Proposition S5.2. Its regression involves only group $i$'s own flow and the price, so it delivers the true $(r_i, w_i)$.

*Part (ii).*

- *Price-only side.* By Theorem S3.2, the price-only experiment is the reduced form at the effective parameters. The realization of Theorem S3.2(iv) puts the Gauss and Gauss–Radau pair inside the equilibrium class, so the price-only side of Theorem S5.3 holds with the effective weights.

- *Tagged side.* Each group's flow response to a unit impulse is $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$, where $h$ is the equilibrium response and $h_0 = \kappa^* > 0$. That is the tagged record of Theorem S5.3, and its argument uses only that $h$ is bounded with $h_0 > 0$.
- On the stated classes the true weights are a Lipschitz function of the effective ones, and conversely, so the orders are unchanged.

**Numerical checks of Theorems S3.2 and S9.2.** The script `equilibrium/e4_reduced_form.py` runs three checks.

- *Reduced form.* For four solved equilibria, the equilibrium price path and the reduced-form path with $(a^*, \kappa^*, \tilde{w})$ agree to $4.4 \times 10^{-16}$ over 5,000 simulated periods. The four are: one group with positive feedback, one with negative feedback, two groups, and the Gauss pair. Their impulse responses agree to $10^{-16}$. For one group of weight 0.02 at $r = 0.7$, for example, $a^* = 0.629$, $\kappa^* = 0.621$ and $\tilde{w} = 0.0279$.
- *Realization.* Take the Gauss and Gauss–Radau pair with $K = 2$, $a^* = 0.5$, $\kappa^* = 0.5$, $W = 0.2$, $r_c = 0.6$ and $\Delta\eta = 0.1$, so that the stability index of Theorem S4.1 is 0.833. Both economies are equilibria with $D = 0.8$, $\bar{a} = 0.4$, $\kappa = 0.4$ and $A = 0.188$, and both equilibrium equations hold to machine precision.
  - Their true group weights are $(0.064, 0.036)$ and $(0.0375, 0.0625)$.
  - Their equilibrium impulse responses differ in $\ell_2$ by $1.22\,d^3$, $1.25\,d^3$ and $1.29\,d^3$ at $d = 0.1$, 0.05 and 0.025. That is the order $(\Delta\eta)^{2K-1}$.
- *Fragility.* The maximum of $\Psi_w$ reproduces the bisection thresholds of the solver to $10^{-12}$ ($A^* = 0.2547$, 0.1297, 0.0677, 0.0342 at $w = 0.02$, 0.05, 0.1, 0.2).
  - $A^* = 0.3967$ at $w = 5 \times 10^{-4}$, against the limit 0.4.
  - $w\,A^* = 0.006875$ at $w = 1$ and 0.0068764 at $w = 10$, against $\Phi^* = 0.0068764$.

*Several groups.* `equilibrium/e4_fragility_K.py` scans $D$ for three populations at weights 0.005 to 1 per group. The populations are two groups at rates 0.5 and 0.85, three at 0.3, 0.6 and 0.8, and the Gauss pair.

- The exact threshold $A^*$ lies below the part (iv) bound in all 15 cases, at 28% to 86% of it. For example, for two groups with $G(1) = 1.73$, $A^* = 0.0084$ against a bound of 0.0293.
- Every equilibrium on the threshold curve is stable. At the threshold the spectral radius is between 0.79 and 0.93; it approaches 1 only at the end of the curve where $D \to \max_i r_i$ and $A \to 0$.

The result files are `equilibrium/e4_reduced_form_results.json` and `equilibrium/e4_fragility_K_results.json`.

**Proof of Theorem S9.3.** *Part (i).*

- *Dealers' information.* The groups' flow is predictable. So the dealers' information about $\theta_t$ is $\nu_t$, and $E[\theta_t \mid \nu_t] = \lambda \nu_t$ with $\lambda = \mathrm{Cov}(\theta, \nu)/\mathrm{Var}(\nu) = b\sigma_\theta^2/\sigma_\nu^2$.
- *The price decomposes.* $E_t F_{t+1} = F_t + \lambda \nu_t$ and $E_t \nu_{t+1} = 0$, so $E_t p_{t+1} - p_t = E_t x_{t+1} - x_t$. The dealers' condition is therefore the one of Theorem S3.1. Its state is $(H_t, u_{\cdot,t+1})$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Coefficient matching gives the same equations for $\beta$, $D$ and $\kappa$ (`p1_inventory_necessity`).

- *The price's response to $\nu_t$.* Since the groups trade on $s_t = x_t + \lambda\nu_t$ and $\sum_i \beta_i w_i = 1 - D$, the implicit form is $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t} + (1-D)\lambda\nu_t$. The response of $s_t$ to $\nu_t$ is therefore $(\kappa + (1-D)\lambda)/D + \lambda = (\kappa + \lambda)/D$.
- *The insider.* $F_{t+1} - p_t = \theta_t + \varepsilon_{t+1} - m_t - c\,\xi_t - c\,d_t$, where $c = (\kappa + \lambda)/D$ and $m_t$ is the predictable part of $s_t$. The insider does not observe the public state, and $m_t$ has mean zero and is independent of $\theta_t$. So she maximizes $(\theta - c\,d)\,d$. So $d = \theta/(2c)$ and the profit is $\theta^2/(4c)$ (`p1_insider_best_response`). Taking expectations, $b\sigma_\theta^2/2$.
- *The variance.* The innovation of $p_{t+1}$ given dealers' information is $(\theta_t - \lambda\nu_t) + \varepsilon_{t+1} + c\,\nu_{t+1}$, which gives $V$ (`p1_insider_posterior_variance`).
- *Converse.* A solution of the system satisfies every condition just used.

*Part (ii).* With $A = 0$, $\kappa(D - \bar{a}) = 0$ forces $\kappa = 0$, and then $D = 1$. Then $b = 1/(2\lambda)$ and $\lambda\sigma_\nu^2 = b\sigma_\theta^2$ give $4\lambda^2\sigma_\xi^2 = \sigma_\theta^2$ (`p1_insider_kyle_limit`).

*Part (iii).* The groups enter only through $D = 1 - \kappa G(D)$.

*Part (iv).*

- *The reduced form.* Combine the implicit form at $t$, the price law at $t-1$ and $H_t = \bar{a}H_{t-1} - \sum_i u_{i,t} - \nu_t$, as in Theorem S3.2. This gives $Ds_t - \bar{a}s_{t-1} = \sum_i \beta_i(D - \bar{a})u_{i,t} + (\kappa + \lambda)\nu_t - \bar{a}\lambda\nu_{t-1}$ (`p1_insider_reduced_form`).
- *The filter is invertible.* $c_1 < c_0$ because $\bar{a} < 1$, $\lambda \geq 0$ and $\kappa \geq 0$. So the observer, knowing the primitives, inverts the filter, and is left with the closed loop of Theorem S3.2.
- *Realization.* Proceed as in Theorem S3.2(iv):
  - set $D = 1 - \kappa^* W/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$;
  - solve the insider's pair $(\lambda, b)$ given $(\kappa, D)$;
  - set $A = \kappa(D - \bar{a})/(VD)$.

  The two economies then share every primitive.

**Numerical check of Theorem S9.3.** The script `equilibrium/e6_insider.py` solves the system at $A = 0.05$, $\sigma_\varepsilon^2 = 1$, $\sigma_\theta^2 = \sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$, and simulates $2 \times 10^5$ dates for each solution.

| Groups | $\kappa$ | $D$ | $\lambda$ | $b$ | $(\kappa+\lambda)/D$ | profit | inform. |
|---|---|---|---|---|---|---|---|
| none | 0.1656 | 1 | 0.4835 | 0.7704 | 0.6490 | 0.0963 | 0.372 |
| +0.02 at 0.7 | 0.1692 | 0.9883 | 0.4816 | 0.7592 | 0.6586 | 0.0949 | 0.366 |
| −0.2 at 0.7 | 0.1463 | 1.0775 | 0.4928 | 0.8430 | 0.5931 | 0.1054 | 0.415 |
| two groups, or split† | 0.1679 | 0.9924 | 0.4823 | 0.7632 | 0.6552 | 0.0954 | 0.368 |

Informativeness (inform.) is $b^2\sigma_\theta^2/\sigma_\nu^2$, the share of $\mathrm{Var}(\theta)$ resolved by the price. †Weights $0.005, 0.005$ at rates $0.5, 0.85$, or the same merged measure split into three groups; both give identical coefficients.

- In every case, the dealers' first-order residual is below $5 \times 10^{-17}$, and the observer's reduced form holds to $4.4 \times 10^{-16}$.
- The regression of $\theta$ on $\nu$ reproduces $\lambda$ to the third decimal.
- The simulated profit at $b' = 0.6b, \ldots, 1.4b$ peaks at $b$, as does the analytic profit.
- As $A \downarrow 0$ (from 0.05 to 0.001), $(\lambda, b)$ moves from $(0.482, 0.759)$ to $(0.49999, 0.994)$, toward Kyle's $(0.5, 1)$.

**Numerical solution of Theorem S3.1.** The script `equilibrium/e4_inventory.py` solves the system (11) at $A = 0.2$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$.

*How the solver finds equilibria.*

- For one group with $w > 0$, every equilibrium with $\kappa > 0$ and $D > 0$ has $D \in (\max(\bar{a}, r), 1)$ and $\kappa = (1 - D)(D - r)/w$. A scan over $D$ therefore finds all of them.
- For several groups, the solver follows the branch of $D(\kappa)$ that is continuous with $D = 1$.
- Each solution is checked by simulating $2 \times 10^5$ periods of the closed loop. The check computes the dealers' first-order residual and the innovation variance of the price.

*Benchmark.* Here $\Delta = 0.12$, and the two equilibria are $\kappa_- = 0.536$ and $\kappa_+ = 7.46$, as in part (ii).

*Results.*

| Groups (weights at memories) | $\kappa$ | $D$ | $\kappa/D$ | spectral radius |
|---|---|---|---|---|
| none | 0.5359 | 1 | 0.5359 | 0.600 |
| 0.02 at 0.7 | 0.5916 | 0.9533 | 0.6206 | 0.787 |
| $-0.2$ at 0.7 | 0.4212 | 1.1767 | 0.3579 | 0.597 |
| $0.005, 0.005$ at $0.5, 0.85$ | 0.569147 | 0.970292 | 0.586573 | 0.881 |
| $0.0025, 0.0025, 0.005$ at $0.5, 0.5, 0.85$ | 0.569147 | 0.970292 | 0.586573 | 0.881 |
| Gauss pair, $0.01, 0.01$ | 0.5759 | 0.9649 | 0.5969 | 0.766 |
| Radau pair, $0.005, 0.015$ | 0.5743 | 0.9662 | 0.5945 | 0.738 |
| 0.2 at 0.7; or $0.1, 0.1$ at $0.5, 0.85$ | no equilibrium | | | |

- *The simulation checks hold in every case.* The relative first-order residual is below $4 \times 10^{-16}$. The simulated innovation variance matches $\kappa^2 \sigma_\xi^2 / D^2$ to within 0.3%.
- *Invariance (part (iii)).* The split configuration has the same merged measure as the two-group one, and its coefficients agree to all printed digits.
- *Gauss and Radau pairs.* They agree on the first three kernel moments, and their impacts differ by 0.0016.
- *Negative feedback* lowers impact.

*First-order formula of part (iv)*, for one group at $r = 0.7$:

- at $w = 0.005$, $\kappa = 0.5457$ against the first-order value 0.5449, and $D = 0.9906$ against 0.9911;
- at $w = 0.01$, $\kappa = 0.5574$ against 0.5540.

Doubling $w$ multiplies the error by about 4.5, as a second-order remainder should.

*Existence region.* For one group at $r = 0.7$, an equilibrium exists for dealer risk aversion below a threshold $A^*$. This is computed by bisection, with the full scan over $D$ at each step:

| $w$ | 0 | 0.02 | 0.05 | 0.1 | 0.2 |
|---|---|---|---|---|---|
| $G(1) = w/(1 - r)$ | 0 | 0.067 | 0.167 | 0.333 | 0.667 |
| $A^*$ | 0.400 | 0.255 | 0.130 | 0.068 | 0.034 |

- Without feedback, $A^* = (1 - \bar{a})/(2\sigma_F \sigma_\xi) = 0.4$, which is the closed form of part (ii).
- For larger $G(1)$, $A^*$ falls roughly in proportion to $1/G(1)$.
- At the threshold the two equilibria merge and disappear.

The results file is `equilibrium/e4_inventory_results.json`.

**Numerical checks.**

- `equilibrium/e2_rents.py`, with $T = 10$, $\lambda = 0.5$ and $\Delta k_m = \pm 0.4\,\epsilon\,(0.7)^m$:
  - the exact rent is $-0.322\epsilon$ when the dealer understates feedback and $+0.322\epsilon$ when it overstates it, as $\epsilon \to 0$;
  - this matches the first-order formula $\mp\Sigma_0 \sum_m \omega_m |\Delta k_m|$ to three digits at $\epsilon = 0.0125$, and to within 6% at $\epsilon = 0.4$.
- `equilibrium/e3_halt.py`, with the market of `e1_demo.py`, $c = 0.6$, $W = 0.5$, 200 draws and 40 dates:
  - the largest pre-halt price gap is $3.8 \times 10^{-4}$, $2.9 \times 10^{-3}$ and $2.3 \times 10^{-2}$ for $d = 0.05$, 0.1 and 0.2, a ratio to $d^3$ of 3.0, 2.9 and 2.8;
  - the largest post-halt gap is 0.196, 0.190 and 0.177.

The result files are `e2_rents_results.json` and `e3_halt_results.json`. Parts of both theorems are machine-checked.

- *Theorem S9.4(i):* the symmetrization, the optimal schedule and its value (`Paper1EquilibriumExtras.lean`).
- *Theorem S9.5(i):* the kernel bound $|k_m^G - k_m^R| \le 2W\binom{m}{3}|d|^3(|c| + |d|)^{m-3}$ for every $m \ge 3$ (`Paper1HaltKernelBound.lean`).
- *Theorem S9.5(iii):* the post-halt gap $W/4$.

The remaining parts are proved on paper.

# S10 Execution cost and capital

**Theorem S10.1** (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single group of weight $q$ and rate $\eta$ with two groups of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through $\kappa v_t$, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h_j^\delta - h_j^0)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

*(i) For every probe with $\sum_t |v_t| \le V$,*

$$D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{14}$$

*(ii) The optimal worst-case error over these probes and all classifiers is exactly*

$$e_T^*(V) = \Phi\Big(-\frac{V\|d_T\|_2}{2\sigma}\Big), \tag{15}$$

*attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed $V$. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of $T$.*

*(iii) The first two taps coincide and $d_2 = -\kappa^2 q\delta^2$. A single pulse of size*

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{16}$$

*and the third observation achieve error exactly $\alpha$. The necessary and sufficient quantity order is $\delta^{-2}$.*

*(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is $\delta^{-4}$, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.*

*(v) For a fixed monetary price exposure $\lambda \neq 0$, additive reserve $L$ and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.*

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and $\kappa$ currency per squared quantity.

**Lemma S10.2** (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{17}$$

*In the positive stable economy $T_h$ is positive definite, so no nonzero round trip produces a deterministic gain.*

Positivity here means that the response is a positive mixture of decaying exponentials (Lemma S8.1), whose spectral density is positive. Positive, decreasing taps alone do not suffice: for $h = (1, 0.9, 0.1)$ the $3 \times 3$ matrix $T_h$ has determinant $-0.468$, so some round trip earns money. The conclusion concerns modelled execution cost only.

We verify the ledger identity (17) against an explicit cash account to $5.7 \times 10^{-13}$ on 1,000 random probes, and the smallest eigenvalue of $T_h$ over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

## S10.1 Capital supplied by a counterparty

**Theorem S10.3** (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \ge 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \le E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \le V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{18}$$

*equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \le B$. The cap holds across all positive group representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least $\delta^{-4}$, and a single positive pulse attains that order.*

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure S3 illustrates the experiment.

**Proposition S10.4** (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \le V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

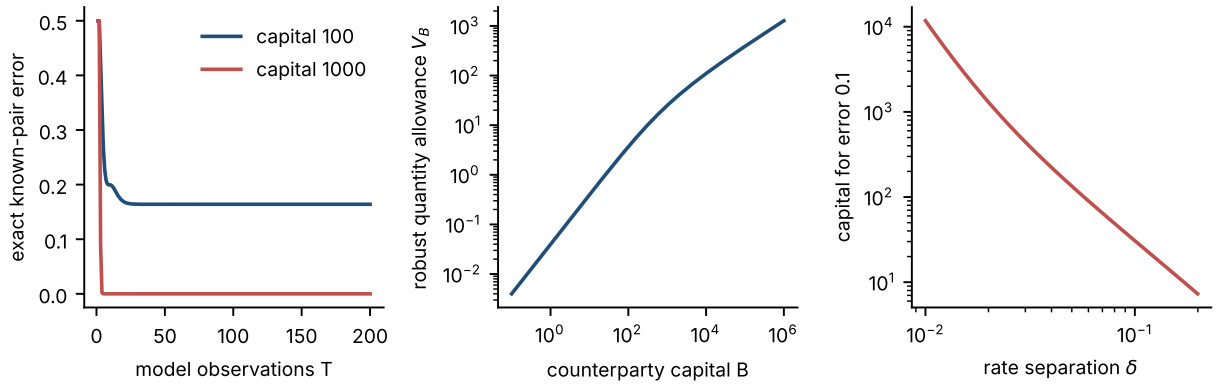



Figure S3: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (18). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (see the disclosure contracts of the full working paper).

## S10.2 Capital paid by the probe trader

**Lemma S10.5** (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence $v$ changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to $P_*$. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

*Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires*

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{19}$$

*without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse $A$ in (16) achieves error $\alpha$ and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*)A$. Necessary and sufficient capital orders are $\delta^{-4}$.*

Lemma S10.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the $\delta^{-4}$ rate before the exact frequency-domain optimization of the account-level analysis of the full working paper. Its explicit constant is very conservative, because it scales as $m/K_*^2 \propto \epsilon^7 (1 - S)^5$: at the calibrated parameters it is $4.9 \times 10^{-4}$ par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as

a lemma; the account-level analysis of the full working paper computes the sharp constant, within 5% of a feasible design at every separation.

**Theorem S10.1.** In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1-\epsilon(1-S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1-S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_*\delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_*\delta^2\|v\|_1$, and Gaussian KL with Pinsker gives (14). For (ii), the mean difference is a lower-triangular convolution matrix times $v$; each column is a shifted copy of a prefix of $d_T$, so $\|\mu_\delta - \mu_0\|_2 \leq V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters $h_2$ with factor $\kappa^2$ while $h_0, h_1$ coincide; a pulse $A$ separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the $\ell_2$ gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \leq |\lambda|\kappa V$, attained by a single pulse.

**Lemma S10.2.** After the uncontrolled update, the pre-probe displacement at step $j$ is $\sum_{i<j} h_{j-i}v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2}v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \leq \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

**Theorem S10.3.** Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \leq x_t \leq \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of $h$ bounds every entry by $\kappa/2$. Hence $B - E_t \leq \kappa U_t^2/2 + cU_t \leq \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (18). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse $A$ funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

**Proposition S10.4.** The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

**Lemma S10.5.** Marking residual inventory at $P_*$ gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum|v_t| \geq 0$. For each mode $\lambda \leq r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1 - 2\lambda\cos\omega + \lambda^2) \geq (1-r)/(1+r)$, so $v^\top T_h v \geq m\|v\|_2^2$ and $\|v\|_2^2 \leq 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (19). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory $A$; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \leq 1$.

# S11 Two unknown groups: pole fitting and the global two-point bound

Because the memory rates are zeros of $H$ (Theorem S2.3), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A

three-step inversion repairs this: (1) compute a minimal realization of $H$ itself from the Hankel matrix of $h_1, h_2, \ldots$ (with $h_0$ as the direct term) by Kung's eigensystem realization; (2) compute its transmission zeros, which are the memory rates and the timing zero at 0 (transforming the sequence first, for example to $h_{t+1} - ah_t$, cancels these zeros); (3) project reconstructed flow onto the recovered memory filters to obtain the weights (full working paper, internet appendix). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-group systems exactly; with noise, per-group weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

## S11.1 A global two-point bound for any estimator

A Cramér–Rao benchmark (full working paper) applies only to estimators that are unbiased near the true parameter. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

**Theorem S11.1** (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem $w_k$ denotes group $k$'s trading intensity $q_k$, not the merged weight $q_k\eta_k$ of Section S1, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar{\eta} - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{20}$$

*(i) Label groups in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k\eta_k^m$ for $m \in \{0, 1, 2\}$, while*

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{21}$$

*Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.*

*(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-group economies with total weight $q$, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,*

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{22}$$

*so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2/\|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. For $\Delta\eta \geq 2\tau q/(q + 2\tau)$ ($\approx 0.018$ at the stated values) the same bound holds without any labelling convention, for the permutation-invariant loss*

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k (|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|),$$

*because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$. For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem S4.1.*

*(iii) Conversely, the matching upper bound $N = O((\Delta\eta)^{-6})$ on rich separated two-group classes is Theorem S4.2 with $K = 2$. On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ for the deconvolved kernel $g_m = \sum_i w_i \eta_i r_i^m$, and perturbation analysis gives the same order.*

Table S1: Two-point sample complexity for two-group weight recovery

| $\Delta\eta$ | $\Delta M_3$ | $\|\Delta h\|_2$ | $N^*_{\text{LeCam}}$ (95%) | $N_{\text{CR}}$ |
|---|---|---|---|---|
| 0.30 | $2.67 \times 10^{-4}$ | $1.98 \times 10^{-3}$ | 7.1 | 47.8 |
| 0.20 | $7.92 \times 10^{-5}$ | $5.60 \times 10^{-4}$ | 88.2 | 597.0 |
| 0.14 | $2.72 \times 10^{-5}$ | $1.89 \times 10^{-4}$ | 775.3 | 5,267.9 |
| 0.10 | $9.90 \times 10^{-6}$ | $6.84 \times 10^{-5}$ | 5,927.6 | 40,374.2 |
| 0.08 | $5.07 \times 10^{-6}$ | $3.49 \times 10^{-5}$ | 22,740.5 | 155,048.9 |
| 0.06 | $2.14 \times 10^{-6}$ | $1.47 \times 10^{-5}$ | 128,333.0 | 875,732.3 |
| 0.05 | $1.24 \times 10^{-6}$ | $8.50 \times 10^{-6}$ | 383,859.1 | 2,620,321.2 |
| 0.04 | $6.34 \times 10^{-7}$ | $4.35 \times 10^{-6}$ | 1,466,361.5 | 10,012,662.4 |

*Worst-case failure at* $\Delta\eta = 0.10$, $N = 50$, *1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Estimators: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least $\tau$.

We recomputed the two-point quantities exactly (Table S1): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table S1).

*Remark* S11.2 (The hard directions carry no risk). The two economies of Theorem S11.1 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1-a-\kappa\sum_i q_i)$ is identical, 2.353 at the calibration of Table S1: the directions in which prices are least informative about the groups are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

# S12 The supervisory decision in detail

This section completes Section S5.

### Proof of Proposition S5.1.

*Proof of Proposition S5.1.* Part (i) follows by scaling the intensities. For (ii), expand $(1 - r_c - \Delta\eta s)^{-1}$: the first three moments agree, so aggregate budgets differ by order $(\Delta\eta)^3$. Each intensity converges to $W\omega_i/(1 - r_c)$, giving the stated dominant-weight gap. This exceeds the aggregate discrepancy at sufficiently small spacing. A ceiling between the two post-halt budgets reduces the action problem to testing two Gaussian mean vectors. Their common-covariance testing error is $\Phi(-\sqrt{N}\|h^G - h^R\|_2/(2\sigma))$; the response bound in Theorem S4.1 gives the lower bound. Part (iii) follows from continuity of $w/(1 - r)$ on the declared compact parameter set and the fixed margin. □

**General-$K$ scope.** The same reduction gives exponent $4K-2$ for a pair whose price responses have that moment-matching order *and* whose dominant intensities have a fixed gap. Moment matching alone does not supply the gap: the symmetric odd/even binomial pair has equal maximal normalized weights. Lemma S12.3 and Proposition S12.4 below supply a pair with a gap for every $K$. The $K=3$ rows below are numerical examples for the Gauss and Gauss–Radau pair.

**Explicit robustness condition.** Smooth best responses alone do not preserve separation: the constant smooth map $\Phi(M)=0$ sends both post-halt budgets to $a$. Instead assume that, over the stated action and holding horizon, $|\widetilde{R}_j - R_j| \le \varepsilon$ for $j \in \{G,R\}$, with $R_R < R_G$. Then

$$R_R + \varepsilon \le \bar{s} < R_G - \varepsilon \quad \Longrightarrow \quad \widetilde{R}_R \le \bar{s} < \widetilde{R}_G. \tag{23}$$

Indeed, $\widetilde{R}_R \le R_R + \varepsilon$ and $\widetilde{R}_G \ge R_G - \varepsilon$. A nonempty robust window exists if $2\varepsilon < R_G - R_R$. This is a maintained response bound, not an equilibrium result. The same allowance must cover the alternative uniform action. At the $K=2$, $\Delta\eta = 0.10$ calibration, $R_R = 0.532$, $R_G = 0.569909$, and $\bar{s} = 0.55$ has margin 0.018; an allowance 0.005 fits within it.

**A complete conditional cost-and-error model.** Let $j$ denote the economy, $\pi_j$ its declared prior, $H$ the holding horizon, $b$ the surplus per active class per period, $C_h$ the implementation cost of halting and $L$ the penalty per period for violating the ceiling. These are assumed accounting primitives, not estimated social welfare. An envelope $S_{\max}$ supports the robust uniform action

$$\theta_\varepsilon = \min\{1, \max\{0, (\bar{s} - \varepsilon - a)/(S_{\max} - a)\}\}, \qquad L_T = HbK(1-\theta_\varepsilon).$$

This action is feasible only if its worst-case budget meets the ceiling (in particular $a + \varepsilon \le \bar{s}$). For a halt,

$$L_{h,j} = Hb + C_h + HL\,\mathbf{1}\{R_j + d_j > \bar{s}\}, \quad \mathcal{L} = F + Nc + \sum_j \pi_j\{p_j L_{h,j} + (1-p_j)L_T\}.$$

Here $F + Nc$ is the information bill, $p_j$ the probability of halting, and $|d_j| \le \varepsilon$ the action response. Require $\Pr_j(\text{halt and violate}) \le \alpha$ separately in each world. Compare this loss with the best feasible no-purchase lottery over the *same two actions*, and allow refusing to buy information. Neither arbitrary alternative policies nor an optimum over sample size are claimed. Active probing additionally incurs its execution loss and capital requirement; these are not automatically the per-record fee $c$ and must not be counted twice.

**Existing runs and the remaining synthetic check.** The archived `R_P1_targeted_intervention.json` supplies six response/decision rows. Recomputing them independently from the stated state recursion agrees to $2.1\times10^{-17}$ in the response discrepancy. Those rows supply $S_j, R_j$ and Gaussian testing distances; they contain no measured disclosure fees, intervention payoffs or strategic response bounds. The existing GPU and historical diagnostics cannot fill these quantities by relabelling their outputs. The new CPU study `studies/supervisory_release_20261105/run_study.py` therefore evaluates only this missing accounting layer, with exact Gaussian probabilities rather than a new large simulation.

For this check alone, assume independent scalar disclosures $Z_n \sim N(R_j, \sigma_t^2)$ and halt when $\bar{Z} + z_{1-\alpha}\sigma_t/\sqrt{N} + \varepsilon \le \bar{s}$. This is an illustrative measurement contract, *not* a claim that class positions or tagged flow directly observe post-halt budgets. If a world is unsafe under the response allowance, this gate falsely halts with probability at most $\alpha$ under the stated unbiased Gaussian

law. The price comparator observes the full 400-tap Gaussian reset record of the same known pair, with a likelihood-ratio rule constrained by the same per-world error budget. Both information channels have the same assumed monetary fee here; neither represents free passive observation.

With $K = 2$, $N = 32$, $\sigma_t = 0.02$, $\varepsilon = 0.005$, $\alpha = 0.05$, $\pi_R = 1/2$, $H = b = 1$, $C_h = 0.02$, $L = 10$, $F = 0.05$, and $c = 0.0005$, the uniform loss is 1.449219, the information bill is 0.066, and disclosure loss is 1.305132. Its conditional net advantage is 0.144087 surplus units. Halt probabilities are approximately zero in G and 0.978928 in R. Buying the price record yields loss 1.515219 at this sample size; its optimal acquisition decision is to decline and retain the no-data action. An informed oracle before information fees has loss 1.234609.

Across 2,304 deterministic configurations varying fees, noise, horizon, prior, sample size and allowance, paid disclosure improves on the no-data benchmark in 1,207, worsens it in 1,079, and ties in 18. These counts have no population or statistical-significance interpretation. Negative cases remain in the output. A downward measurement bias of 0.03 produces false-halt probability 0.418776; a future response of 0.025 despite an allowance of 0.005 makes halting unsafe in R with probability 0.978928. The latter also invalidates the uniform-action guarantee, so payoff comparisons under that misspecification are not presented as a fair safety comparison. The experiment establishes conditional accounting and failure modes, not real-market gains, a validated equilibrium response, or the general-$K$ theorem.

**Proposition S12.1** (A robust scalar disclosure gate)**.** *Let $Z_1, \dots, Z_N$ be independent $N(R, v)$ readings of the fixed post-halt feedback budget, $N > 0$, and let the realized action response $d$ satisfy $d \le \varepsilon$. For $t > 0$, halt only if $\bar{Z} + t + \varepsilon \le \bar{s}$. Then the probability of halting while the realized budget exceeds the ceiling is at most*

$$\Pr\{\bar{Z} + t + \varepsilon \le \bar{s},\ R + d > \bar{s}\} \le \frac{v}{Nt^2}.$$

*Consequently $v \le \alpha N t^2$ gives a per-world error bound $\alpha$.*

*Proof.* On an unsafe-halt event, $\bar{Z} + t + \varepsilon \le \bar{s} < R + d \le R + \varepsilon$, so $t \le |\bar{Z} - R|$. Chebyshev's inequality for the actual Gaussian sample mean gives the bound. Lean checks the product Gaussian law, the sample statistic, this event inclusion and the probability inequality in `p1_supervisory_gaussian_unsafe_halt`; it is not merely a check of a scalar named "probability bound." The response allowance and the interpretation of $Z$ remain explicit assumptions. □

This conservative gate uses $t = \sigma_t/\sqrt{N\alpha}$. It differs from the sharper Gaussian-quantile gate above, whose exact normal probabilities are numerically evaluated but whose quantile identity is not formalized here. Under the same calibration with $N = 128$, the conservative gate has radius 0.007906, information bill 0.114, loss 1.349034 and conditional net advantage 0.100185. Its halt probabilities are approximately zero in G and 0.998023 in R. Across the same 2,304 sensitivity configurations it improves on the fixed-menu no-data benchmark in 781, worsens it in 1,499 and ties in 24. Both gates' rows remain in the result file: 4,608 configurations in total. The welfare and error interpretation remains conditional on the declared observation and response contracts.

**Operational bridge from tagged flows to a halt decision.** The scalar-reading contract above can be replaced, within the reset model, by five observed channels: two first class-flow responses, two second responses, and the first price response. For a unit pulse write $f_{i,0} = \kappa w_i$, $f_{i,1} = w_i(\kappa r_i + h_1)$, and $h_1 = \kappa a + \kappa^2 \sum_i w_i$. Thus

$$w_i = f_{i,0}/\kappa, \qquad r_i = f_{i,1}/f_{i,0} - h_1/\kappa, \qquad q_i = w_i/(1 - r_i).$$

Assume known impact, verified stable class membership, $0 \le r_i \le 0.95$, independent unbiased Gaussian reset readings with per-channel variance $\sigma_f^2$, and $N$ independent records. A simultaneous radius $e = \sigma_f\sqrt{5/(N\alpha)}$ covers the five sample means with probability at least $1-\alpha$ by Chebyshev and a union bound. Interval division, only when the first-flow lower endpoint is positive, gives bounds on $w_i, r_i, q_i$; intersect rate bounds with the declared class. Halt only if the remaining upper budget plus the response allowance meets the ceiling. Otherwise retain the same robust uniform fallback. On simultaneous coverage, every selected halt is safe. This supplies a conditional observation-to-action argument without feeding the unknown $R$, $w$ or $r$ to the rule.

The separate Lean module `Paper1FlowDecision.lean` covers exact flow inversion, ratio and clipped intervals, the intensity interval, selected-halt inequality, actual product-Gaussian joint mean bound and a probability-composition theorem with an explicit rule-soundness premise. Their deterministic composition, clipping implementation and executable simulation still require separate correspondence checks; no claim of a fully formalized executable decision algorithm is made. The pinned build receipt records the checked declarations.

The exploratory script `studies/paper1_uplift_20261105/flow_decision.py` runs 20,000 independent replications per world in each of 60 configurations: $N \in \{32, 128, 512, 2048, 8192\}$, $\sigma_f \in \{0.0005, 0.002, 0.005\}$, $H \in \{1, 10\}$ and $F \in \{0.05, 0.25\}$. Other quantities use the two-group calibration above, a per-channel-vector record fee 0.0005, and the same class-halt or throttle action menu. Fourteen configurations have positive estimated net information value; the others are retained. The largest observed gain is exploratory, selected from the grid: at $N = 128$, $\sigma_f = 0.0005$, $H = 10$ and $F = 0.05$, no-data loss is 14.492188, information-inclusive loss 12.370094 and net gain 2.122094 surplus units. A simultaneous grid-adjusted Hoeffding interval is reported in the result file; zero observed unsafe halts is not a proof of zero probability. Costs, horizons and response bounds are synthetic, and no real-market gain or calibrated welfare conclusion follows. Six additional fixed stress cases retain both smaller biases with no observed unsafe halt and failures: a downward second-flow bias of 0.010 produces unsafe halts in G in all 20,000 replications, while a response 0.030 exceeding the allowance 0.005 produces unsafe halts in R in all replications. Swapped second-flow labels lead to abstention in both worlds. These are synthetic contract falsifications, not guarantees under misspecification.

**Proposition S12.2** (No-data action and information purchase)**.** *For the fixed halt/throttle menu, let $\overline{L}_h = \sum_j \pi_j L_{h,j}$ and let $p_{\max} = \alpha$ if a halt is unsafe in at least one world, otherwise $p_{\max} = 1$. Assuming the fallback meets the ceiling, the best constant no-data halt probability is $p = 0$ when $\overline{L}_h \ge L_T$, and $p = p_{\max}$ when $\overline{L}_h < L_T$. A specified information rule with expected action loss $L_I$ is strictly preferred if and only if*

$$F + Nc < \mathcal{L}_0 - L_I.$$

*Allowing refusal to purchase gives loss* $\min\{\mathcal{L}_0, F + Nc + L_I\} \le \mathcal{L}_0$.

*Proof.* The no-data loss is affine in its feasible halt probability: $L_T + p(\overline{L}_h - L_T)$. Its slope determines the minimizing endpoint. Subtracting the information rule's total loss gives the break-even inequality, and the minimum inequality proves the optional-purchase statement. These algebraic decisions are machine checked in `Paper1SupervisoryDecision.lean`. They do not establish an optimum over other actions, sample sizes, or information contracts. □

## S12.1 The targeted halt for every $K$

Proposition S5.1(ii) is stated for two groups. Its argument needs two populations whose kernels agree to order $(\Delta\eta)^{2K-2}$ and whose largest groups differ at order one. This subsection supplies

such a pair for every $K$.

Let $\mu$ be the uniform probability measure on $[-1,1]$, let $p_0, p_1, \dots$ be its orthonormal polynomials, and put $\lambda_K(s) = 1/\sum_{j<K} p_j(s)^2$. Call a rule *admissible* if it has $K$ distinct nodes in $[-1,1]$ and integrates every polynomial of degree at most $2K-2$ exactly against $\mu$. The Gauss rule and the Gauss–Radau rule of Theorem S4.1 are admissible.

**Lemma S12.3** (Largest weights in the family)**.** *Fix $K \geq 2$.*

*(i) An admissible rule with nodes $t_1, \dots, t_K$ has weights $\omega_i = \lambda_K(t_i)$. In particular its weights are positive.*

*(ii) There is $c_0 > 0$ such that, for every $|c| \leq c_0$, the zeros of $p_K - c\,p_{K-1}$ are the nodes of an admissible rule $R_c$. A point is a node of $R_c$ for at most one $c$.*

*(iii) There are $c_1 \neq c_2$ in $[-c_0, c_0]$ for which $R_{c_1}$ and $R_{c_2}$ have different largest weights. Their moments of degree $2K-1$ differ.*

*Proof.* (i) Let $\ell_i$ be the Lagrange polynomials of the nodes. Each product $\ell_i \ell_j$ has degree $2K-2$, so the rule integrates it exactly: $\int \ell_i \ell_j \, d\mu = \sum_m \omega_m \ell_i(t_m)\ell_j(t_m) = \omega_i \mathbf{1}\{i=j\}$. Taking $i=j$ gives $\omega_i = \int \ell_i^2 d\mu > 0$. So $\ell_i/\sqrt{\omega_i}$, $i = 1, \dots, K$, is an orthonormal basis of the polynomials of degree at most $K-1$, as is $p_0, \dots, p_{K-1}$. The sum of squares of an orthonormal basis at a point does not depend on the basis, so $\sum_{j<K} p_j(s)^2 = \sum_i \ell_i(s)^2/\omega_i$. At $s = t_i$ the right side is $1/\omega_i$.

(ii) The zeros of $p_K$ are real, simple and inside $(-1,1)$. Simple real zeros move continuously with the coefficients, so for $|c| \leq c_0$, with $c_0$ small, $q_c = p_K - c\,p_{K-1}$ has $K$ simple zeros in $(-1,1)$. Give these nodes the interpolatory weights. Let $f$ have degree at most $2K-2$ and write $f = q_c g + \rho$ with $\deg g \leq K-2$ and $\deg \rho \leq K-1$. Both $p_K$ and $p_{K-1}$ are orthogonal to polynomials of degree at most $K-2$, so $\int q_c g \, d\mu = 0$, and the rule integrates $\rho$ exactly and gives $q_c g$ the value zero. Hence the rule is exact for $f$. If $s$ is a zero of $q_c$ and of $q_{c'}$ with $c \neq c'$, then $p_{K-1}(s) = 0$ and so $p_K(s) = 0$. By the three-term recurrence a common zero of $p_K$ and $p_{K-1}$ is a zero of $p_{K-2}$, and so on down to $p_0 = 1$, which has none.

(iii) The reciprocal $1/\lambda_K$ is a polynomial of degree $2K-2 \geq 2$, so every level set of $\lambda_K$ is finite. Suppose all the rules $R_c$, $|c| \leq c_0$, had the same largest weight $v$. By (i) each $R_c$ would have a node in the finite set $\{\lambda_K = v\}$, and by (ii) different values of $c$ would give different nodes. There are infinitely many $c$, so this is impossible. For the last statement, let $\nu$ be the difference of the two rules. It is supported on at most $2K$ points. If the two rules also agreed at degree $2K-1$, $\nu$ would vanish on every polynomial of degree at most $2K-1$, including the one that equals 1 at one support point and 0 at the others. So $\nu$ would be zero, which contradicts (ii) and the positivity of the weights. □

**Proposition S12.4** (Supervisory separation for every $K$)**.** *Fix $K \geq 2$ and let $R_{c_1}$ and $R_{c_2}$ be the rules of Lemma S12.3(iii), with nodes $s_i^j$, weights $\omega_i^j$ and largest weights $v_1 < v_2$. Under the hypotheses of Theorem S4.1, let economy $j \in \{1,2\}$ have $K$ groups with rates $r_c + \Delta\eta\, s_i^j$ and merged weights $W\omega_i^j$, and write $q_i^j = W\omega_i^j/(1 - r_c - \Delta\eta\, s_i^j)$.*

*(i) The response bounds of Theorem S4.1 hold for this pair: $\|h^1 - h^2\|_2 \leq C_U(\Delta\eta)^{2K-1}$ for every horizon, and $\|h^1 - h^2\|_2 \geq \kappa^2 W|g|(\Delta\eta)^{2K-1}$ when at least $2K+1$ taps are observed, where $g \neq 0$ is the difference of the two rules at degree $2K-1$.*

*(ii) The feedback budgets satisfy $|S^1 - S^2| = O((\Delta\eta)^{2K-1})$, and the largest groups satisfy $\max_i q_i^2 - \max_i q_i^1 = W(v_2 - v_1)/(1 - r_c) + O(\Delta\eta)$.*

*(iii) For sufficiently small spacing there is a nonempty interval of ceilings at which halting the largest group meets the ceiling in economy 2 and not in economy 1. For $N$ independent reset records with Gaussian noise $\sigma^2 I$, a rule that makes this decision with both errors at most $1/4$ needs*

$$N \;\geq\; \left(2\Phi^{-1}(0.75)\,\sigma/\|h^1-h^2\|_2\right)^2 \;\geq\; c\,\sigma^2(\Delta\eta)^{-(4K-2)}.$$

*Proof.* (i) The proof of Theorem S4.1 uses four properties of its pair: nodes in $[-1,1]$, positive weights of unit mass, equal moments up to degree $2K-2$, and a nonzero difference at degree $2K-1$. Lemma S12.3 gives all four. (ii) Expanding $(1-r_c-\Delta\eta\, s)^{-1}$ in powers of $\Delta\eta$, the sum $\sum_i q_i^j$ is $W\sum_{m\geq 0}(\Delta\eta)^m\mu_m^j/(1-r_c)^{m+1}$, where $\mu_m^j$ is the moment of degree $m$ of rule $j$. The moments agree for $m\leq 2K-2$, which gives the first statement. Each $q_i^j$ equals $W\omega_i^j/(1-r_c)+O(\Delta\eta)$, and the maximum is continuous, which gives the second. (iii) The post-halt budgets are $R^j=S^j-\kappa\max_i q_i^j$, so $R^1-R^2=\kappa W(v_2-v_1)/(1-r_c)+O(\Delta\eta)>0$ for small spacing, and any ceiling strictly between $R^2$ and $R^1$ separates the two decisions. The decision is then a test between two Gaussian mean vectors, as in the proof of Proposition S5.1, and (i) gives the bound. □

**What is and is not shown.** The pair in Proposition S12.4 is two members of the family $R_c$, not the Gauss and Gauss–Radau pair itself. For that pair the largest weights differ in every case we computed ($K=2,\dots,200$), and a proof of that comparison for every $K$ is open. Lemma S12.3(ii) proves admissibility only for $c$ near zero. Numerically, for $K=2,\dots,14$ the rules $R_c$ are admissible on the whole range between the two Gauss–Radau rules, their weights equal $\lambda_K$ at the nodes to $7\times 10^{-13}$, and on a grid of 41 values of $c$ the largest weight is smallest at the Gauss rule and largest at the Gauss–Radau rules (`equilibrium/every_k_halt_check.py`). The counting step in part (iii) is machine-checked (`p1_largest_weight_not_constant` in `Paper1EveryKHalt.lean`), with parts (i) and (ii) and the finiteness of the level sets as hypotheses. The rest is proved on paper.

## S12.2 Proof of Proposition S4.5

Write $D_X(z)=z-a_X-\kappa G_X(z)$ for economy $X$, so that $H_X=\kappa z/D_X$ and the stationary price spectrum is $f_X(\omega)=\sigma^2\kappa^2/|D_X(e^{i\omega})|^2$. Because both economies have budget $1-\epsilon$, $D_A(1)=D_B(1)=\epsilon$.

*Step 1: the difference vanishes at zero frequency.* $D_B-D_A=-(a_B-a_A)-\kappa(G_B-G_A)$, and the choice of persistences gives $a_B-a_A=-\kappa(G_B(1)-G_A(1))$. With $E=G_B-G_A$ this is $\Delta(z)\equiv D_B(z)-D_A(z)=-\kappa\,(E(z)-E(1))$, which does not depend on $\epsilon$. Write $E(z)=\sum_m e_m z^{-m}$ with $e_m=k_m^B-k_m^A$. Then $|E(e^{i\omega})-E(1)|\leq|\omega|\sum_m m|e_m|$. The rates are $r_c+\Delta\eta\, s$, and expanding $(r_c+\Delta\eta\, s)^m$ in powers of $\Delta\eta$, the moments of order up to $2K-2$ of the two normalized rules cancel, so $|e_m|\leq 2W\sum_{j\geq 2K-1}\binom{m}{j}r_c^{m-j}(\Delta\eta)^j$. Summing, $\sum_m m|e_m|\leq C_1(\Delta\eta)^{2K-1}$, with $C_1$ depending on $K,W,r_c$ and a bound on $\Delta\eta$. Hence $|\Delta(e^{i\omega})|\leq\kappa C_1(\Delta\eta)^{2K-1}|\omega|$.

*Step 2: the denominator is bounded below by $\epsilon+|\omega|$.* Let $a^*=\lim_{\epsilon\to 0}a_A(\epsilon)$ and $D^*=D_A-\epsilon$, the denominator of economy $A$ at budget one; $D_A=D^*+\epsilon$. At budget one the closed-loop matrix $J$ of the price and the groups is entrywise nonnegative with a positive diagonal and positive first row and column (since $a^*>0$ and all weights are positive), so it is primitive. Its Perron root is 1, it is simple, and no other eigenvalue lies on the unit circle. The zeros of $D^*$ in $|z|\geq 1$ are eigenvalues of $J$, so $D^*$ has no zero on the unit circle except $z=1$, and there $D^{*\prime}(1)=1+\kappa\sum_i w_i r_i/(1-r_i)^2>0$. Hence there are $\delta,c_1,C_2,m>0$ with $|\operatorname{Im}D^*(e^{i\omega})|\geq c_1|\omega|$ and $\operatorname{Re}D^*(e^{i\omega})\geq -C_2\omega^2$ for $|\omega|\leq\delta$, and $|D^*(e^{i\omega})|\geq m$ for $\delta\leq|\omega|\leq\pi$. For $|\omega|\leq\delta$: if $\epsilon\geq 2C_2\omega^2$ then $\operatorname{Re}D_A\geq\epsilon/2$ and $|D_A|\geq\max(\epsilon/2,c_1|\omega|)$; if $\epsilon<2C_2\omega^2$ then $|D_A|\geq c_1|\omega|$ and $\epsilon<2C_2\delta|\omega|$. In both

cases $|D_A(e^{i\omega})| \geq c_2(\epsilon + |\omega|)$. For $|\omega| \geq \delta$ and $\epsilon \leq m/2$, $|D_A| \geq m/2$. So $|D_A(e^{i\omega})| \geq c_3(\epsilon + |\omega|)$ on $[-\pi, \pi]$ for all $\epsilon \leq \epsilon_1$.

*Step 3: the bound.* By Steps 1 and 2, $\rho(\omega) = |\Delta/D_A| \leq \kappa C_1(\Delta\eta)^{2K-1}|\omega|/(c_3(\epsilon + |\omega|)) \leq C_3(\Delta\eta)^{2K-1}$, uniformly in $\epsilon$ and $\omega$. The ratio of spectra is $u = f_B/f_A = |D_A/D_B|^2 = |1+\Delta/D_A|^{-2}$, so for $\rho \leq 1/2$, $|u-1| \leq C_4\rho$. The divergence rate of stationary Gaussian processes is $D_\epsilon(B\|A) = \frac{1}{4\pi}\int_{-\pi}^{\pi}(u - 1 - \log u)\, d\omega$, and $u - 1 - \log u \leq (u-1)^2$ for $|u-1| \leq 1/2$. Hence $D_\epsilon(B\|A) \leq \frac{1}{2}C_4^2C_3^2(\Delta\eta)^{4K-2}$ for all $\epsilon \leq \min(\epsilon_0, \epsilon_1)$ and all $\Delta\eta$ small enough that $C_3(\Delta\eta)^{2K-1} \leq 1/2$. For $\epsilon_1 < \epsilon \leq \epsilon_0$ the family is uniformly stable and the bound is that of Theorem S4.4. □

**Numerical check.** For $K = 2, 3$, $\Delta\eta \in \{0.1, 0.05, 0.025\}$ and margins from $10^{-1}$ to $10^{-6}$, the ratio $D_\epsilon(B\|A)/(\Delta\eta)^{4K-2}$ increases to a finite limit as $\epsilon \to 0$ (0.41 to 0.48 for $K = 2$, 0.65 to 0.71 for $K = 3$; `fragility_uniform_check.py`). With a common persistence instead of a common budget, the divergence rate grows like $\epsilon^{-1}$, which is the budget difference becoming visible. This proposition is not machine-checked.

**Finite samples (numerical).** The proposition bounds the information per observation. A finite-sample bound uniform in $\epsilon$ is not proved. Numerically, the exact Kullback–Leibler divergence between the laws of $n$ consecutive stationary observations, computed from the state-space autocovariances, lies between 97.8% and 99.8% of $n$ times the per-observation rate in every case computed: $K = 2$, $\Delta\eta \in \{0.1, 0.2\}$, $\epsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and $n \in \{250, 1000, 4000\}$, at the parameters of Figure S1 (`research_extension/fragility_reveals/finite_sample_check.py`). So, in these cases, data requirements computed from the per-observation rate are not loosened by finite samples.

## S12.3 Stress identification with noisy observations

Theorem S6.4 assumes exactly observed responses. The next result shows that the identification survives observation noise: a minimum-distance estimator recovers the signed capacity measure consistently. It gives no rate.

**Proposition S12.5** (Noisy stress identification)**.** *Let $\psi$ be continuous, odd and with independent dilations. Fix $n \geq 1$, $0 < \ell_{\min} \leq \ell_{\max}$, $0 \leq \bar{r} < 1$ and $\bar{c} > 0$, and let $\Theta$ be the set of populations of at most $n$ traders, written $\theta = (c_i, r_i, \ell_i)_{i \leq n}$ with $c_i \in [0, \bar{c}]$, $r_i \in [0, \bar{r}]$ and $\ell_{\min} \leq |\ell_i| \leq \ell_{\max}$ (a trader with $c_i = 0$ is absent). Let $(v^{(j)})_{j \geq 1}$ enumerate the probe sequences of finite length with rational entries, $v^{(j)}$ of length $T_j \leq j$, and let $h_j(\theta) \in \mathbb{R}^{T_j}$ be the reset response of the price to $v^{(j)}$, without process noise. In round $N$, each probe $j \leq N$ is run $N$ times, every price is observed with independent $N(0, \sigma^2)$ noise, and $\hat{h}_{j,N}$ is the average of the $N$ observed responses. Let $\hat{\theta}_N$ minimize*

$$Q_N(\theta) = \sum_{j \leq N} 2^{-j} \min(1,\ \|\hat{h}_{j,N} - h_j(\theta)\|)$$

*over $\Theta$. If the true population $\theta_0 \in \Theta$, then $\tilde{\mu}(\hat{\theta}_N) \to \tilde{\mu}(\theta_0)$ in probability, in the weak topology on signed measures on $[0, \bar{r}] \times [\ell_{\min}, \ell_{\max}]$.*

*Proof. Continuity.* For a fixed probe, the response at each date is a finite composition of the market recursion, sums, products and $\psi$, so $h_j$ is continuous on $\Theta$, which is compact. The map $\theta \mapsto \tilde{\mu}(\theta) = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i, |\ell_i|)}$ is continuous into the weak topology, because $\operatorname{sgn}(\ell_i)$ is constant on each of the two components $\ell_i \geq \ell_{\min}$ and $\ell_i \leq -\ell_{\min}$; the weak topology on this bounded set of measures is metrized by the bounded-Lipschitz distance $\rho$.

*Identification.* Let $Q(\theta) = \sum_j 2^{-j} \min(1, \|h_j(\theta_0) - h_j(\theta)\|)$. If $Q(\theta) = 0$, the two populations give identical responses to every rational probe sequence. Responses are continuous in the probe, by the same composition argument, and rational sequences of each length are dense, so the responses agree for every probe sequence. Drop the traders with $c_i = 0$ from both populations; the remaining traders satisfy the hypotheses of Theorem S6.4, so $\tilde{\mu}(\theta) = \tilde{\mu}(\theta_0)$. Conversely $\tilde{\mu}(\theta) = \tilde{\mu}(\theta_0)$ gives $Q(\theta) = 0$ by the same theorem.

*Uniform convergence.* Since $\min(1, \|\cdot\|)$ satisfies the triangle inequality,

$$\sup_\theta |Q_N(\theta) - Q(\theta)| \le \sum_{j \le N} 2^{-j} \|\hat{h}_{j,N} - h_j(\theta_0)\| + 2^{-N},$$

and $\mathbb{E}\|\hat{h}_{j,N} - h_j(\theta_0)\| \le \sigma\sqrt{T_j/N} \le \sigma\sqrt{j/N}$. The expectation of the right side is at most $\sigma N^{-1/2} \sum_j 2^{-j}\sqrt{j} + 2^{-N} \to 0$, so $\sup_\theta |Q_N - Q| \to 0$ in probability.

*Consistency.* $Q_N(\hat{\theta}_N) \le Q_N(\theta_0) \le \sup |Q_N - Q|$, since $Q(\theta_0) = 0$; hence $Q(\hat{\theta}_N) \le 2 \sup |Q_N - Q| \to 0$ in probability. Fix $\delta > 0$. The set $F_\delta = \{\theta \in \Theta : \rho(\tilde{\mu}(\theta), \tilde{\mu}(\theta_0)) \ge \delta\}$ is closed in a compact set, hence compact, and $Q$ is continuous and, by identification, strictly positive on it, so $q_\delta = \min_{F_\delta} Q > 0$ (if $F_\delta$ is empty there is nothing to prove). Then $\Pr(\rho(\tilde{\mu}(\hat{\theta}_N), \tilde{\mu}(\theta_0)) \ge \delta) \le \Pr(Q(\hat{\theta}_N) \ge q_\delta) \to 0$. □

The argument is the standard one for extremum estimators: identification, compactness, continuity and uniform convergence. It inherits the hard limit's requirement of probes of all sizes, and it is silent on rates. For tanh, Proposition S6.3 shows the information per probe grows like the sixth power of its size for small probes, so rates for the full measure will depend on how far the design pushes the market.

# S13 Saturating traders: proofs and checks

### Theorem S6.1.

*Proof.* $f_t = \sum_i c_i g_t(r_i, \ell_i)$ with $g_t(r, \ell) = \psi(\ell \sum_{j<t} r^{t-1-j} s_j)$, so equal measures give equal flows for every signal. The rest is the induction of Theorem S2.1(iii), which uses only that the two flow maps agree and depend on signals before $t$. The last sentence follows from the example below Theorem S6.2. □

### Theorem S6.2.

*Proof.* (i) By strong induction on $t < \tau$. If the paths agree before $t$, the signals agree before $t$, so every trader of either population has the same state in both markets. All are inside their bands, so both aggregate flows equal the linear flow $\sum_{j<t} k_{t-1-j} s_j$, which is the same for $P$ and $P'$ because the kernels agree. The insider's order and the dealer's price are causal, so they agree at $t$. The same argument with $P'$ replaced by the linear market (the band-linear shape $\psi'(u) = u$) gives the second statement. (ii) Build both markets, and the record noise, on one probability space. By (i) the records coincide on $\{\tau \ge T\}$, so the total-variation distance is at most $\Pr(\tau < T)$, and $\tau$ is computed on the linear market. (iii) In the linear market from rest, $x = g * \varepsilon$ with $g = h/\kappa$, and $m_t(r) = (\varphi_r * x)_t$ with $\varphi_r(j) = r^j$ shifted by one date. So $m_t(r)$ is centred Gaussian with variance at most $s_r^2 \le \sigma_\varepsilon^2 \|\varphi_r\|_1^2 \|g\|_1^2$. Here $\|\varphi_r\|_1 = 1/(1-r)$. Also $|h_t|$ is dominated by the response of the positive economy with persistence $a$ and kernel $|k_m|$, whose total mass is $\kappa/(1 - S_{\mathrm{abs}})$, so $\|g\|_1 \le 1/(1 - S_{\mathrm{abs}})$. For a standard normal $Z$ and $z \ge 0$, $\Pr(|Z| > z) \le e^{-z^2/2}$. A union bound

over the $T$ dates and $n + n'$ traders gives the bound. A test with both errors at most $\alpha$ needs $\mathrm{TV} \geq 1 - 2\alpha$. (iv) Without process noise, the linear path is deterministic, so $\tau \geq T$ surely and (ii) gives distance zero. For a pulse, the linear state is $m_t(r) = A \sum_{j<t} r^{t-1-j} h_j$. □

**Proposition S6.3.**

*Proof.* For $u \geq 0$, $\phi(u) = u - \tanh u$ has $\phi(0) = 0$ and $\phi'(u) = \tanh^2 u \in [0, u^2]$. So $0 \leq u - \tanh u \leq u^3/3$, and by oddness $|\tanh u| \leq |u|$ and $|\tanh u - u| \leq |u|^3/3$ for all $u$. (i) *Size of each path.* Since $|\tanh u| \leq |u|$, $|x_t| \leq a|x_{t-1}| + \kappa \sum_i c_i|\ell_i| \, (\varphi_{r_i} * |x|)_t$ for $t \geq 1$, with $|x_0| = \kappa|A|$. So $|x|$ is dominated by the pulse response of the positive economy with weights $c_i|\ell_i|$. That response has peak $\kappa|A|$ (Lemma S8.1) and total mass $\kappa|A|/(1 - \rho_P)$. Hence $\|x\|_\infty \leq \kappa|A|$ and $\|x\|_2 \leq \|x\|_1 \leq \kappa|A|/(1 - \rho_P)$, and the same holds for the memory states, with an extra factor $1/(1 - r_i)$. *Difference.* Write $c_i \tanh(\ell_i m_{i,t}) = c_i \ell_i m_{i,t} + R_{i,t}$ with $|R_{i,t}| \leq c_i|\ell_i|^3 |m_{i,t}|^3/3$. The linear parts give $(k * x)_t$ with the common kernel. So $e = x^P - x^{P'}$ solves the linear closed loop driven by $u_t = \sum_i R_{i,t} - \sum_j R'_{j,t}$ in place of the probe, and $e = h * u$. By Young's inequality, $\|e\|_2 \leq \|h\|_1 \|u\|_2$, and $\|h\|_1 \leq \kappa/(1 - S_{\mathrm{abs}}) \leq \kappa/(1 - \hat{S})$ as in Theorem S6.2(iii). Also $\| \, |m_i|^3 \|_2 \leq \|m_i\|_\infty^2 \|m_i\|_2 \leq \kappa^3 |A|^3/((1 - r_i)^3 (1 - \rho_P))$, so $\| \sum_i R_i \|_2 \leq \Lambda_P \kappa^3 |A|^3/(1 - \rho_P)$. Combining gives the bound on $\|e\|_2$. For $N$ experiments, the Kullback–Leibler divergence is $N\|e\|_2^2/(2\sigma^2)$. Pinsker's inequality and $\mathrm{TV} \geq 1 - 2\alpha$ give the requirement. (ii) Both markets have $x_0 = \kappa A$ and $m_{i,1} = x_0$, so $x_1^P - x_1^{P'} = \kappa(\sum_i c_i \tanh(\ell_i \kappa A) - \sum_j c'_j \tanh(\ell'_j \kappa A))$. The linear terms cancel because $k_0$ is common, and $\tanh u = u - u^3/3 + O(u^5)$ gives the expansion. For $|A|$ small, $|x_1^P - x_1^{P'}| \geq \kappa^4 |\Delta M_3| |A|^3/6$. The test that compares the average of $N$ first taps with the midpoint has error $\Phi(-\sqrt{N} \, |x_1^P - x_1^{P'}|/(2\sigma))$, which gives the stated $N$. □

**Theorem S6.4.**

*Proof. Step 1: probes steer the price.* Write $F_t$ for aggregate feedback flow, a function of $x_0, \ldots, x_{t-1}$. Since $x_t = a x_{t-1} + \kappa(F_t + v_t)$ with $\kappa \neq 0$, the probe $v_t = (x_t^* - a x_{t-1}^*)/\kappa - F_t(x_{<t}^*)$ realizes any target path $x^*$, and $F_t = (x_t - a x_{t-1})/\kappa - v_t$ is read from the response. So two populations give identical responses to all probes if and only if their flow maps agree on all paths.

*Step 2: sufficiency of $\tilde{\mu}$.* The shape is odd, so $c\,\psi(\ell m) = c\,\mathrm{sgn}(\ell)\,\psi(|\ell| m)$ and $F_t = \int \psi(\lambda \, m_t(r)) \, \tilde{\mu}(dr, d\lambda)$. Equal $\tilde{\mu}$ give equal flow maps, hence identical responses.

*Step 3: identification.* Let $D = \tilde{\mu}_P - \tilde{\mu}_{P'} = \sum_k d_k \delta_{(r_k, \lambda_k)}$ with distinct atoms and $\lambda_k > 0$. Take the path with $x_0 = \theta u$ and $x_1 = u$. Then $m_2(r) = (r\theta + 1)u$, and equal flow maps at date 2 give $g(u) = \sum_k d_k \psi(s_k u) = 0$ for all $u$, with $s_k = \lambda_k(r_k \theta + 1)$. For two distinct atoms, $\lambda_k(r_k\theta + 1) = \lambda_j(r_j\theta + 1)$ is an affine equation in $\theta$ that is not an identity, so it has at most one root. Fix $\theta > 0$ outside these finitely many roots. Then the $s_k$ are positive, because $r_k \geq 0$, and distinct. Independent dilations give $d_k = 0$ for every $k$, so $D = 0$. □

**Theorem S6.4, parts (i)–(iii).**

*Proof.* (i) Let $g(u) = \sum_k w_k \, \mathrm{clip}(s_k u) = 0$ for all $u$, with distinct $s_k > 0$, and let $s_1$ be the largest scale. At $u = 1/s_1$ every copy is inside its band or at its edge, so $g(1/s_1) = \sum_k w_k s_k/s_1 = 0$. If there is a second-largest scale $s_2$, then at $u = 1/s_2$ the first copy is saturated and the others are not, so $w_1 + \sum_{k \neq 1} w_k s_k/s_2 = 0$. The first equation gives $\sum_{k \neq 1} w_k s_k = -w_1 s_1$, so $w_1(1 - s_1/s_2) = 0$ and $w_1 = 0$. If there is no other scale, the first equation is $w_1 = 0$. Remove $s_1$ and repeat. The statement on probes inside the bands is Theorem S6.2(iv).

(ii) For every $y$, $1 - \tanh y = 2e^{-2y}/(1 + e^{-2y})$. Let $g(u) = \sum_k w_k \tanh(s_k u) = 0$ for all $u$, with distinct $s_k > 0$, and let $s_1$ be the smallest scale. Letting $u \to \infty$ gives $\sum_k w_k = 0$, so

$\sum_k w_k(1-\tanh(s_k u))=0$ for all $u$. Multiply by $e^{2s_1 u}$ and let $u\to\infty$: the first term tends to $2w_1$, and every other term is at most $2|w_k|e^{-2(s_k-s_1)u}$, which tends to 0. So $w_1=0$. Remove $s_1$ and repeat.

*Small probes.* $g$ is real-analytic, so if it vanishes near 0 it vanishes everywhere, and the argument above applies. A direct route: the odd Taylor coefficients of tanh are $\tau_{2n-1}=2^{2n}(2^{2n}-1)B_{2n}/(2n)!$, which are nonzero because the Bernoulli numbers $B_{2n}$ are nonzero. If $g$ vanishes near 0, then $\sum_k(w_k s_k)(s_k^2)^{n-1}=0$ for $n=1,\dots,K$, a Vandermonde system in the distinct numbers $s_k^2$, so every $w_k=0$. In Step 1 of the theorem, $|F_t|\le\sum_i c_i|\ell_i||m_{i,t}|$, so small target paths need only small probes.

(iii) $\sum_k w_k s_k=5-8+3=0$ and $\sum_k w_k s_k^3=5-32+27=0$, so $\sum_k w_k\psi(s_k u)=u\sum_k w_k s_k-\gamma u^3\sum_k w_k s_k^3=0$. For a cubic shape the date-$t$ flow map is $\int\lambda\, m_t(r)\,d\tilde\mu-\gamma\int\lambda^3 m_t(r)^3\,d\tilde\mu$, which depends on $\tilde\mu$ only through the measures $\lambda\,\tilde\mu$ and $\lambda^3\tilde\mu$ in $r$. □

**Proposition S6.6.**

*Proof.* (i) The proofs of these results use the dealer's rule only through causality and through its being the same map in both markets. (ii) The linear parts of the two expected flows agree because the kernels agree, so the merged weight at each memory is the same. For each trader, $|\mathbb{E}[c\,\mathrm{clip}(Y)]-c\,\mathbb{E}Y|\le c\,\mathbb{E}[(|Y|-1)_+]$ with $Y=\ell m^+\sim N(\ell\bar m,\sigma^2)$. Since $|Y|\le|\ell\bar m|+\sigma|Z|$, this is at most $c\,\mathbb{E}[(\sigma|Z|-\delta)_+]=c\big(2\sigma\phi(\delta/\sigma)-2\delta(1-\Phi(\delta/\sigma))\big)\le c\sqrt{2/\pi}\,\sigma e^{-\delta^2/(2\sigma^2)}$. □

**Numerical checks.** The scripts are `research_extension/nonlinear_scoping/identification_and_figure.py`, `sim_wiener.py`, `sim_passive_clip.py` and `dealer_anticipation.py`. Two populations with the same $\tilde\mu$ (a trend follower of capacity 2 and a contrarian of capacity 1, against one trend follower of capacity 1, all at $r=0.7$ and $|\ell|=0.1$) give identical responses to 50 random probe sequences of standard deviation 10, with gap exactly 0, under both shapes. The constructive recovery in the main text locates the kinks at 3.702 and 14.813 (true 3.704 and 14.815) and returns net capacities 0.24989 and 0.99988.

# S14 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m=\int_{[0,\rho]} r^m\,d\mu(r)$ with $\rho<1$, where $\mu$ is any nonnegative measure of total mass $W$: a finite or infinite mixture of exponential memories with merged weight $W$. Write $\mathcal{M}(\rho,W)$ for this class, assume the closed loop is uniformly stable, $s\equiv a+\kappa W/(1-\rho)<1$, and measure accuracy by

$$d(\mu,\mu')=\sup\Big\{\int f\,d(\mu-\mu')\,:\ f\ \text{1-Lipschitz on } [0,\rho],\ f(0)=0\Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly $W$. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree $L$, and let $c_*=\inf_{L\ge1} L\,E_L(|x|)$. Bernstein (1914) showed that $L\,E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_*>0$; numerically $c_*=1/4$, attained at $L=2$ (Table S2).

**Lemma S14.1** (Moment-matched group pairs)**.** *For every $n\ge2$ there are two probability measures on $[0,\rho]$, each with exactly $n$ atoms, whose moments of orders $0,\dots,2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\,E_{2n-2}(|x|)\ge c_*\rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r-\rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Supplement S8). Each measure is an $n$-atom merged measure in the class; one atom may sit at rate 0 (a group with $\eta=1$), which the class allows.

**Theorem S14.2** (Minimax rate for general memory)**.** *Let $N$ independent reset experiments observe the impulse response with Gaussian noise of variance $\sigma^2$ per tap, and let $B(\rho)=(1+\sqrt{2})(1+2/\rho)$ and $C_h=\kappa^2/(1-s)^2$.*

*(i)* Lower bound. *Let $\Lambda_N=\log(N/\sigma^2)+\log(2C_h^2W^2/((1-\rho)\rho)^2)$. Whenever $\Lambda_N>4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu)\geq \frac{c_*\,\rho W\log(1/\rho)}{\Lambda_N}\Big)\;\geq\;\frac{1}{4e},$$

*and the bound already holds over $n$-group economies, for a finite $n$ that grows with $N$.*

*(ii)* Upper bound. *With $a$ and $\kappa$ known, invert the averaged response into kernel values $\hat{k}_0,\dots,\hat{k}_L$, with $L+1=\lfloor\log(N/\sigma^2)/(4\log B(\rho))\rfloor$, and let $\hat{\mu}$ be any measure on $[0,\rho]$ minimizing $\max_{j\leq L}|\int r^j d\hat{\mu}-\hat{k}_j|$ (a linear program). For every $\beta\in(0,1)$ there is $N_0$ depending only on $(\rho,W,a,\kappa,\beta)$ such that for $N/\sigma^2\geq N_0$,*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu)>\frac{3\pi\rho W\log B(\rho)}{\log(N/\sigma^2)}\Big)\;\leq\;\beta.$$

*Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$, and reaching accuracy $w$ requires and suffices $N/\sigma^2=\exp(\Theta(1/w))$.*

*Proof sketch.* (i) Put the measures of Lemma S14.1, scaled to mass $W$, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int(1-rz)^{-1}d(\mu_A-\mu_B)=z^{2n-1}\int r^{2n-1}(1-rz)^{-1}d(\mu_A-\mu_B)$; on $|z|=1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A-k^B\|_2$ by the same quantity. The response difference is $H_AH_B$ times the shifted kernel difference, so $\|h^A-h^B\|_2\leq C_h\|k^A-k^B\|_2$. Choosing the smallest $n$ for which the Kullback–Leibler divergence of the $N$-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz $f$ with $f(0)=0$, Jackson's theorem gives a polynomial $q$ of degree $L$ with $\sup|f-q|\leq\pi\rho/(4(L+1))$, and expanding $q$ in Chebyshev polynomials of $2r/\rho-1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu},\mu)\leq\pi\rho(W+\delta)/(2(L+1))+2\delta\rho B(\rho)^{L+1}$, where $\delta$ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z)=(1-az-\kappa/H(z))/(\kappa z)$, whose error is at most $2((1+s)/\kappa)^2(L+2)$ times the largest response error, so $\delta=O_P(\sigma N^{-1/2}L\sqrt{\log L})$. The choice of $L$ makes $\delta B(\rho)^{L+1}=O_P((\sigma^2/N)^{1/4}L\sqrt{\log L})$, negligible against the first term. Full proofs are in Supplement S8. □

**Machine check.** Lean verifies part (i), for $N$ reset experiments of any length, using only Lean's standard axioms. The formal proof replaces the pair of Lemma S14.1 by an elementary one: the binomial weights $W\binom{2L+1}{k}/2^{2L}$ on the rates $\rho k/(2L+1)$, odd $k$ against even $k$, which have $n=L+1$ atoms each and equal moments of orders $0,\dots,2n-2$. The 1-Lipschitz function $f(r)=(h/\pi)(1-\cos(\pi r/h))$, $h=\rho/(2L+1)$, vanishes at 0 and at even nodes and equals $2h/\pi$ at odd nodes, so the

pair is at distance at least $2\rho W/(\pi(2n-1))$, more than the $c_*\rho W/(2n-2)$ of the lemma, without Bernstein's theorem. Because the raw moments agree, the kernels agree below lag $2n-1$ and differ by at most $2W\rho^{2n-1}/(1-\rho)$ in $\ell_1$ over all lags; the closed-loop $\ell_1$ bounds, the Gaussian divergence and the testing lemma then show that, with $C = 2\kappa^2 W/((1-\rho)(1-s)^2)$, $\Lambda = \log(2NC^2/\sigma^2) > 0$ and $L = \lceil \Lambda/(4\log(1/\rho)) \rceil$, no estimator reaches accuracy $2\rho W \log(1/\rho)/(\pi(\Lambda + 6\log(1/\rho)))$ with probability at least $3/4$ under both economies. This is the order of part (i), with success probability $3/4$ in place of $1/(4e)$ and constants of its own. An estimator $\hat{\mu}$ enters through its integrals $\int f\, d\hat{\mu}$ of 1-Lipschitz functions with $f(0) = 0$, which is how $d(\hat{\mu}, \mu)$ is defined. Lean also verifies part (ii). Mathlib has no Jackson theorem, so the formal proof builds one: on $M = n+2$ equally spaced nodes the Fejér–Korovkin kernel $K(u) = |\sum_{k\le n} a_k e^{iku}|^2 / \sum_k a_k^2$, $a_k = \sin(\pi(k+1)/(n+2))$, is nonnegative, has node mean one and node mean of $K(u)(1-\cos u)$ equal to $1 - \cos(\pi/(n+2))$ (from the recurrence of the sines and the orthogonality of roots of unity), so the positive operator it defines maps an $L$-Lipschitz function on $[-1,1]$ to a polynomial of degree at most $n$ within $L\pi/(n+2)$, by Cauchy–Schwarz on the nodes. Written in Chebyshev polynomials and rescaled to $[0,\rho]$, its coefficients sum in absolute value to at most $\rho(n+1)B(\rho)^n$. With an explicit bound for the deconvolution, $\alpha e(1+\beta')^t$ for kernel value $t$, and the sub-Gaussian union bound, any measure on $[0,\rho]$ that fits the first $L+1$ deconvolved kernel values at least as well as the truth, the linear-programming estimate among them, is within a finite-sample bound of $\mu$ with probability at least $1-\beta$; taking $L = \lfloor \log(N/\sigma^2)/(8\log D) \rfloor$, $D = B(\rho)(1+\beta')$, gives $16\pi\rho W \log D/\log(N/\sigma^2)$ for all $N/\sigma^2$ beyond a threshold depending only on $(a, \kappa, \rho, s, W, \beta)$. The formal constants are larger than the theorem's $3\pi$ and $B(\rho)$, because the formal deconvolution bound grows geometrically in $L$; the order is the same.

Theorem S14.2 replaces a polynomial barrier with a logarithmic one. With exponential memory and $K$ known groups, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of groups is also known (Theorem S15.1). The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly ($\rho^{2n-1}$ decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0,\rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table S2 checks the construction and the estimator. The pairs of Lemma S14.1 have exactly $n$ atoms, agree in all moments below order $2n-1$ to $10^{-61}$, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho = 0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for $N$ from $10^4$ to $10^{24}$: a $10^{20}$-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

**Full proofs.**

**Lemma S14.1.** Let $L = 2n-2$ and let $p^*$ be the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree $L$, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin,

Table S2: General memory: moment-matched group pairs and the moment estimator

| | | | | | | |
|---|---|---|---|---|---|---|
| *A. Best approximation of* $\lvert x\rvert$ | | | | | | |
| $L$ | 2 | 4 | 6 | 10 | 20 | 30 |
| $L\,E_L(\lvert x\rvert)$ | 0.2500 | 0.2705 | 0.2756 | 0.2785 | 0.2797 | 0.2800 |
| *B. Pairs of Lemma S14.1* | | | | | | |
| Atoms per measure $n$ | 2 | 4 | 6 | 8 | 10 | 12 |
| $W_1(\mu_A,\mu_B)$, $\rho=0.9$ | 0.2250 | 0.1037 | 0.0669 | 0.0494 | 0.0391 | 0.0324 |
| $n\,W_1$, $\rho=0.9$ | 0.450 | 0.415 | 0.402 | 0.395 | 0.391 | 0.388 |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.9$ | $4.41\times10^{-2}$ | $1.03\times10^{-3}$ | $4.08\times10^{-5}$ | $1.96\times10^{-6}$ | $1.05\times10^{-7}$ | $5.95\times10^{-9}$ |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.99$ | $7.41\times10^{-2}$ | $5.84\times10^{-3}$ | $1.07\times10^{-3}$ | $2.76\times10^{-4}$ | $8.33\times10^{-5}$ | $2.76\times10^{-5}$ |
| *C. Linear-programming estimator, continuous memory distribution,* $\rho=0.9$ | | | | | | |
| $\log_{10}N$ | 4 | 8 | 12 | 16 | 20 | 24 |
| Moments used $(L+1)$ | 2 | 3 | 5 | 6 | 7 | 8 |
| Error $d(\hat\mu,\mu)$ | 0.00878 | 0.00759 | 0.00432 | 0.00445 | 0.00382 | 0.00399 |
| Error $\times\log(N/\sigma^2)$ | 0.202 | 0.245 | 0.179 | 0.226 | 0.229 | 0.276 |

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to $10^{-61}$ (60-digit arithmetic); every $W_1$ exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; $L$ from the rule in Theorem S14.2(ii); measure fitted on a 401-point grid; 20 replications.

1981) there are points $t_0<\cdots<t_{L+1}$ at which $\lvert x\rvert-p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i=1/\prod_{j\neq i}(t_i-t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i)=0$ for every polynomial $p$ of degree at most $L$; multiplying by $-1$ if necessary, $w_i$ has the sign of $\lvert t_i\rvert-p^*(t_i)$ for every $i$. Exactly $n$ weights are positive and $n$ negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x\mapsto\rho(x+1)/2$ gives $n$-atom probability measures $\pi_A$ (positive part) and $\pi_B$ (negative part) on $[0,\rho]$ with equal moments of orders $0,\dots,L$. For the 1-Lipschitz function $f(r)=\lvert r-\rho/2\rvert=(\rho/2)\lvert x\rvert$,

$$\int f\,d(\pi_A-\pi_B)=\frac{\rho}{2}\int\big(\lvert x\rvert-p^*(x)\big)\,d(\pi_A-\pi_B)=\frac{\rho}{2}(E+E)=\rho E,$$

because $p^*$ integrates to the same value under both measures, $\lvert x\rvert-p^*=E$ on the support of $\pi_A$ and $-E$ on that of $\pi_B$. By Kantorovich–Rubinstein duality $W_1(\pi_A,\pi_B)\geq\rho E\geq c_*\rho/L$.

**Theorem S14.2.** Write $K(z)=\sum_m k_m z^m$, $D=1-az-\kappa zK$ and $H=\kappa/D$, which is the generating function of the recursion $h_t=ah_{t-1}+\kappa\sum_m k_m h_{t-1-m}$ with $h_0=\kappa$. Since $k_m\geq0$ and $\sum_m k_m=\int(1-r)^{-1}d\mu\leq W/(1-\rho)$, the $\ell_1$ norm of $az+\kappa zK$ is at most $s<1$, so $\lVert H\rVert_1\leq\kappa/(1-s)$ and $\lVert 1/H\rVert_1=\lVert D\rVert_1/\kappa\leq(1+s)/\kappa$.

(i) Let $\mu_A=W\pi_A$ and $\mu_B=W\pi_B$ with $\pi_A,\pi_B$ from Lemma S14.1; both lie in $\mathcal{M}(\rho,W)$. With $\nu=\mu_A-\mu_B$, $\sum_m(k^A_m-k^B_m)z^m=\int(1-rz)^{-1}d\nu(r)=z^{2n-1}\int r^{2n-1}(1-rz)^{-1}d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $\lvert z\rvert=1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\lVert k^A-k^B\rVert_2\leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A-H_B=z(K_A-K_B)H_AH_B$, so by Young's inequality $\lVert h^A-h^B\rVert_2\leq\lVert H_A\rVert_1\lVert H_B\rVert_1\lVert k^A-k^B\rVert_2\leq C_h\lVert k^A-k^B\rVert_2$. The Kullback–Leibler divergence between the $N$-sample Gaussian experiments is $N\lVert h^A-h^B\rVert_2^2/(2\sigma^2)\leq(N/\sigma^2)\,2C_h^2W^2\rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho)\geq\Lambda_N-2\log(1/\rho)$. Take $n$ the smallest integer with this property and $n\geq2$; then $2n-2<\Lambda_N/(2\log(1/\rho))$ when $\Lambda_N>4\log(1/\rho)$. Since $d$ is a norm on signed measures and $d(\mu_A,\mu_B)=W\,W_1(\pi_A,\pi_B)\geq Wc_*\rho/(2n-2)>2c_*\rho W\log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma

2.6 and Theorem 2.2) gives $\max_{j\in\{A,B\}} P_j(d(\hat{\mu},\mu_j) \geq c_*\rho W \log(1/\rho)/\Lambda_N) \geq \frac{1}{4}e^{-1}$.

(ii) *Approximation.* Let $f$ be 1-Lipschitz on $[0,\rho]$ with $f(0) = 0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1,1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial $p$ of degree $L$ with $\sup|g-p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence $\sup|p| \leq 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \leq 2\sup|p| \leq 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates $f$ with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \leq \sum_{j\leq L} |b_j|\,\|T_j(2r/\rho-1)\|_{\mathrm{coef}} \leq 4\rho\sum_{j\leq L} B(\rho)^j \leq \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\mathrm{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\mathrm{coef}} \leq (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of $y^i$ by at most $(1+\alpha)^i$, and $B(\rho) - 1 \geq 4$. On the event $\max_{j\leq L}|\hat{k}_j - k_j| \leq \delta$ the true $\mu$ attains objective at most $\delta$, so the minimizer $\hat{\mu}$ satisfies $\max_{j\leq L}|\int r^j d\hat{\mu} - k_j| \leq 2\delta$ by the triangle inequality. *Moment comparison.* If $\max_{j\leq L}|\int r^j d\hat{\mu} - k_j| \leq 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\,d(\hat{\mu}-\mu) = \int (f-q)\,d(\hat{\mu}-\mu) + \sum_{j\leq L} q_j\Big(\int r^j d\hat{\mu} - k_j\Big) \leq \frac{\pi\rho(W+\delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

*Kernel recovery.* From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0,\ldots,k_L$ are the coefficients of $(1-az-\kappa/H)/(\kappa z)$, which depend on $h_0,\ldots,h_{L+1}$ only. Work in the algebra of power series modulo $z^{L+2}$ with the $\ell_1$ norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1\|E\|_1 \leq \frac{1}{2}$, then $1/\hat{H} = (1/H)(1+E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \leq 2\|1/H\|_1^2\|E\|_1$, so $\max_j|\hat{k}_j - k_j| \leq \delta \equiv 2((1+s)/\kappa)^2(L+2)\max_{t\leq L+1}|E_t|$. Each $E_t$ is Gaussian with variance $\sigma^2/N$, so with probability at least $1-\beta$, $\max_t|E_t| \leq \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true $\mu$ is feasible for the linear program with value at most $\delta$, so the minimizer satisfies the moment condition above. *Choice of $L$.* With $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, $B(\rho)^{L+1} \leq (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4}L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \leq 2\pi\rho W\log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For $N/\sigma^2$ larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1\|E\|_1 \leq \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W\log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

# S15 Near-colliding rates: a uniform rate for the merged measure

Theorem S4.2 needs the rates to stay $\gamma$-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights $w$ and $W-w$ at rates $r$ and $r+\Delta$ and two of weights $W-w$ and $w$ at the same rates have merged measures that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

**Theorem S15.1** (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most $K$ atoms in $[0,\bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From $N$ reset experiments as in Theorem S4.2, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is $C$ depending only on $(K,\bar{r},a,\kappa,B)$ such that, with probability at least*

$1-\beta$,

$$W_1(\hat{\bar{M}}, \bar{M}) \;\le\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \textit{uniformly over } \mathcal{M}_K.$$

*Hence $N = O(\sigma^2 m_{\min}^{-2}\epsilon^{-(4K-2)})$ experiments give accuracy $\epsilon$, and with the Wasserstein statement of Theorem S4.1 the minimax rate for the merged memory distribution is $N^{-1/(4K-2)}$, uniformly over near-colliding economies.*

*Proof.* As in the proof of Theorem S4.2, the kernel values $k_0, \dots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \le e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$ (the Lipschitz constant is for a bounded neighbourhood of the class; when the noise is too large for that, the right-hand side of the theorem exceeds $\bar{r} \ge W_1$ once $C$ is enlarged, so the bound holds trivially). The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \le 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \le k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy $|m_t(\hat{M}) - m_t(M)| \le 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \le 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries both normalized measures to distributions on $[-1,1]$ with at most $K$ atoms, multiplies $W_1$ by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1+2/\bar{r})^{2K-1}$. The moment comparison inequality for $K$-atomic distributions on $[-1,1]$ (Wu and Yang, 2020), $W_1 \le c_K\delta^{1/(2K-1)}$ when the first $2K-1$ moments differ by at most $\delta$, then gives the bound with $C = c_K(\bar{r}/2)(1+2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem S4.2 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes arbitrarily expensive as rates approach each other (Theorem S4.1) and impossible when they coincide (Theorem S2.3). A supervisor who needs the first can have it at the rate of Theorem S4.1; one who needs the second has to be told the number of groups and a separation, or has to obtain tagged data (Section S5).

# S16 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

**Theorem S16.1** (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance $V$ large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix}\lambda_1 & \theta \\ 0 & \lambda_2\end{smallmatrix}\right)$ with shock covariance $\mathrm{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in $\theta$. Here $s_1$ is the variance of the fresh price shock given the full prior state; a forecaster who sees prices only can face additional uncertainty about the hidden state, and the family is the one with positive coupling $\theta > 0$.*

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure S4).

Two clarifications prevent over-reading. First, the group economy of Section S1 is itself non-normal, yet its price output does not amplify (Lemma S8.1). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (full working paper). Second,

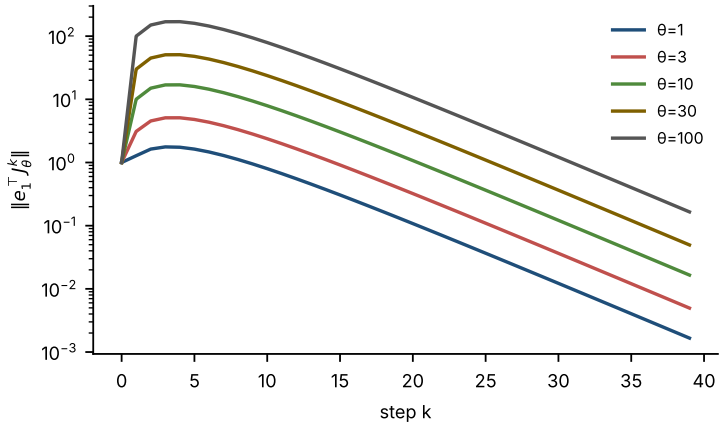



Figure S4: Price-row transient gain of the spectral-blindness family $J_\theta$: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every $\theta$.

in a network of $M$ identical group economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma S8.1; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator. A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial impact, which also depends on the input and output maps. The positive-group no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

**Proof of Theorem S16.1.** With $J_\theta$ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and $P_{11}$ is affine in $s_2$ with positive slope, so for each $\theta$ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when $V$ exceeds the value at $s_2 = 0$. The one-step price variance is $s_1$ throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

# S17 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (8) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\big(\sqrt{s^2 + 4s} - s\big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and $\eta$ is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

**Proposition S17.1** (Identification in economic primitives)**.** *Consider $K$ Kalman desks (Lemma S1.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$, and write $c_- = \eta'(\bar{s})$ and $c_+ = \eta'(\underline{s})$.*

*(i) Prices and aggregate order flow depend on the desks only through the measure*

$$\sum_i q_i\,\eta(s_i)\,\delta_{1-\eta(s_i)}.$$

*Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.*

*(ii) Fix a centre $s_c$ and let the ratios lie in $s_c + \Delta s\,[-1,1] \subset [\underline{s},\bar{s}]$. Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order $\Delta s$, or its weights $q_i\eta(s_i)$, from $N$ noisy reset experiments with worst-case error probability below $1/4$ requires $N \geq c\,\sigma^2(\Delta s)^{-(4K-2)}$; on separated configurations, Prony's method of Theorem S4.2 composed with $\eta^{-1}$ achieves this order. The constants are those of Theorems S4.1 and S4.2 multiplied by powers of $c_-$ and $c_+$.*

*Proof.* (i) By Lemma S1.1(i) desk $i$ is a group with memory rate $1-\eta(s_i)$ and weight $q_i$, so Theorem S2.3 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through $q_i$. (ii) On $[\underline{s},\bar{s}]$, $\eta$ is a $C^1$ bijection onto its image with $c_- \leq \eta' \leq c_+$. Lower bound: take the two economies of Theorem S4.1 with centre $r_c = 1-\eta(s_c)$ and rate spacing $\Delta\eta = c_-\Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1,1]$, so they are two populations of Kalman desks in the class, and the bound $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)}\sigma^2(\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because $\eta$ is $c_+$-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates, which Theorem S4.1 bounds below by a constant times $\Delta\eta = c_-\Delta s$. Upper bound: ratios in $s_c + \Delta s\,[-1,1]$ with spacing at least $\gamma\Delta s$ have gains in $\eta(s_c) + c_+\Delta s\,[-1,1]$ with spacing at least $c_-\gamma\Delta s$, a class of the form required by Theorem S4.2 with rescaled separation $\gamma c_-/c_+$; Prony's method recovers the rates and weights at that theorem's rate, and $\eta^{-1}$, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

**Proposition S17.2** (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio, $q = \varphi(s)$ with $\varphi$ positive and $C^2$ on $[\underline{s},\bar{s}]$ (for a mean–variance desk, $\varphi(s)$ increases with $s$), and let $n_i \geq 0$ denote the mass of desks with ratio $s_i$ (a divisible population, not an integer desk count). Write $s(\eta)$ for the inverse gain map and $w(\eta) = \varphi(s(\eta))\,\eta$ for the weight per desk as a function of its gain.*

*(i)* Unknown numbers of desks. *The merged measure is $\sum_i n_i\,w(\eta(s_i))\,\delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1-\eta(\bar{s}), 1-\eta(\underline{s})]$ is of this form, with $n_i = m_i/w(\eta_i)$ for an atom of mass $m_i$ at $1-\eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition S17.1(ii) holds unchanged.*

*(ii)* Known numbers of desks. *If the $n_i$ are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at*

*$\bar{\eta} \pm \delta$ with $n$ desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2[w(\eta)(1-\eta)^m]_{\bar{\eta}} + o(\delta^2)$, so separating them needs $N \geq c\,\sigma^2\delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

*(iii)* Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map $T$ from configurations $(s_1, \ldots, s_K)$ to perceived ratios that is continuous on a box $\prod_i [\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition S17.1 apply at it.*

*Proof.* (i) Desk type $i$ is a group with gain $\eta_i = \eta(s_i)$, memory rate $1 - \eta_i$ and total weight $n_i\varphi(s_i)$, so it contributes $n_i\varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1 - \eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i / w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1-\eta)^m$; the tap difference is $n[g_m(\bar{\eta} + \delta) + g_m(\bar{\eta} - \delta) - 2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta}) + o(\delta^2)$, with $\sum_m \sup |g_m''|^2 < \infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over $N$ experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how many desks of each kind there are, and even then the simplest separation problem keeps its $\delta^{-4}$ order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map $T$ from a model of the desks' learning.

# S18 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order $\delta^{-4}$ blocks (Online Supplement, Proposition S21.2), the same order as the passive Whittle bound of Theorem S4.4. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Niroula et al., 2026); group non-identification alone does not establish that the worst relevant cost exceeds the informed one.

# S19 Local information: the Cramér–Rao benchmark

**Proposition S19.1** (Cramér–Rao benchmark)**.** *Let $\theta \in \Theta \subset \mathbb{R}^{2K}$ collect the weights $w_1, \ldots, w_K$ and rates $\eta_1, \ldots, \eta_K$, with $\Theta$ open. Each of $N$ reset probes of size $V_{\text{probe}}$ records the taps $h_0, \ldots, h_T$ with independent $N(0, \sigma_p^2)$ errors, so the averaged, rescaled record is $Y = h(\theta) + \sigma_h Z$ with $Z \sim N(0, I_{T+1})$ and $\sigma_h = \sigma_p / (V_{\text{probe}}\sqrt{N})$. Let $J_{tj} = \partial h_t / \partial \theta_j$ and $F_{\text{unit}} = J^\top J$, and assume $T \geq 2K$ and that $J$ has full column rank at $\theta$, so that $F_{\text{unit}}$ is invertible.*

*(i) The Fisher information of $Y$ is $I(\theta) = \sigma_h^{-2} F_{\text{unit}}$. Every estimator $\hat{w}_k(Y)$ with finite variance at $\theta$ that is unbiased for $w_k$ on a neighbourhood of $\theta$ satisfies $\mathrm{Var}_\theta(\hat{w}_k) \geq \sigma_h^2 [F_{\text{unit}}^{-1}]_{kk}$. No further regularity is needed: the bound follows from the Hammersley–Chapman–Robbins inequality by letting the alternative approach $\theta$ along a line, without differentiating under the integral.*

Table S3: Cramér–Rao sample-size benchmark $N_{\mathrm{CR}}$ of (24) for multi-group weight recovery ($T = 120$ taps)

| Scenario | $K$ | $N_{\mathrm{CR}}$ probes | $\sigma_{\mathrm{crit}}$ | ML success ($f = 1$) | RMSE/CRB |
|---|---|---|---|---|---|
| $\eta = 0.30$ | 1 | 0.1 | $6.05 \times 10^{-3}$ | 94% | 1.02 |
| $\Delta\eta = 0.30$ | 2 | 47.8 | $2.31 \times 10^{-4}$ | 94% | 1.01 |
| $\Delta\eta = 0.20$ | 2 | 597 | $6.55 \times 10^{-5}$ | 96% | 0.98 |
| $\Delta\eta = 0.10$ ($\delta = 0.05$ pair) | 2 | 40,374 | $7.96 \times 10^{-6}$ | 93% | 1.03 |
| $\Delta\eta = 0.05$ | 2 | 2,620,320 | $9.88 \times 10^{-7}$ | 95% | 1.03 |
| spacing 0.10 | 3 | 17,349,791 | $3.84 \times 10^{-7}$ | 98% | 1.22 |
| spacing 0.20 | 3 | 27,038 | $9.73 \times 10^{-6}$ | 98% | 0.88 |

Tolerance $\pm 0.01$ at 95%. $\sigma_{\mathrm{crit}}$ is the single-probe noise at which one probe suffices; operational single-probe noise is $1.6 \times 10^{-3}$. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at $\sigma_{\mathrm{crit}}$; success means every weight within $\pm 0.01$. Fitted log–log slope of $N_{\mathrm{CR}}$ in $\Delta\eta$: $-6.04$. $N_{\mathrm{CR}}$ is a benchmark for an efficient, approximately normal estimator, not a lower bound for every procedure (Proposition S19.1(ii)).

*(ii)* Benchmark interpretation. *For an estimator that attains this variance and is approximately normal, a 95% interval of half-width* tol *needs*

$$N \geq N_{\mathrm{CR}} \equiv \Big(\frac{1.96}{\mathrm{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\mathrm{probe}}}\Big)^2 [F_{\mathrm{unit}}^{-1}]_{kk}. \tag{24}$$

*This is a sample-size benchmark, not a lower bound on the interval width of every procedure.*

*(iii) For $K = 2$, fix the horizon $T \geq 4$, known $a$ and $\kappa$ with $\kappa_{\min} \leq |\kappa| \leq \kappa_{\max}$, and weights $0 < w_i \leq w_{\max}$. For all rates with $0 \leq \eta_2 < \eta_1 \leq 1$, writing $\Delta\eta = \eta_1 - \eta_2$, $c\,(\Delta\eta)^{-6} \leq [F_{\mathrm{unit}}^{-1}]_{kk} \leq C\,(\Delta\eta)^{-6}$ for both weight coordinates $k$, where $c, C > 0$ depend only on $T$, $|a|$, $\kappa_{\min}$, $\kappa_{\max}$ and $w_{\max}$ (not on $\Delta\eta$ or on where the rates lie). Hence $N_{\mathrm{CR}} = \Theta((\Delta\eta)^{-6})$. The upper constant $C$ does not depend on $T$. The lower constant $c$ does in general. Under a small-gain margin it can be taken free of $T$: if $1 - \eta_2 \leq \rho < \theta < 1$ and $|a| + 2\kappa_{\max} w_{\max}\theta/(\theta - \rho) \leq \theta$, then the bound holds for every $T \geq 4$ with $c = (\theta - \rho)^8(1 - \theta)^3(3 + \theta)/(256\,\kappa_{\max}^4(1 + \theta)^2)$ (`p1_prop_S7_1_iii_uniform`, `p1_prop_S7_1_iii_two_uniform`). The margin is sufficient, not sharp. That an estimator attains this order is a separate statement, given by Theorem S4.2; the minimax lower bound of the same order is Theorem S11.1.*

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table S3), so the critical noise levels, not estimator failure, are what make close groups expensive. Under ordinary surveillance budgets ($N \leq 50$) unbiased estimators cannot resolve close weights.

**Proof of Proposition S19.1.** Part (i): for every alternative $\theta'$ the Gaussian likelihood ratio has second moment $\exp(\|h(\theta') - h(\theta)\|^2/\sigma_h^2)$, so the Hammersley–Chapman–Robbins inequality gives $(\mathbb{E}_{\theta'}\hat{w}_k - \mathbb{E}_\theta \hat{w}_k)^2 \leq \mathrm{Var}_\theta(\hat{w}_k)\,(e^{\|h(\theta') - h(\theta)\|^2/\sigma_h^2} - 1)$; put $\theta' = \theta + ta$, use local unbiasedness, divide by $t^2$ and let $t \to 0$ to get $a_k^2 \leq \mathrm{Var}_\theta(\hat{w}_k)\|Ja\|^2/\sigma_h^2$, and choose $a = F_{\mathrm{unit}}^{-1} e_k$. Part (iii): for $K = 2$ the columns of $J$ for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of $J$ is of order $(\Delta\eta)^3$, which gives the upper bound $[F^{-1}]_{kk} \leq \sigma_{\min}(J)^{-2} = O((\Delta\eta)^{-6})$. The matching lower bound for the weight coordinate does not follow from

the singular value alone; it uses the exact inverse row of the $4 \times 4$ confluent Vandermonde matrix in moment coordinates together with the bounded conditioning of the map from moments to the observed taps on the fixed-centre family.

# S20 Pricing, evasion and the choice of an audit trail

**Proposition S20.1** (Conditional endogenous-pricing invariance)**.** *Let $\kappa = \mathcal{K}(\Gamma_M(\kappa))$ on a common admissible interval, with identical innovation laws, initial-state conventions and other primitives across economies. Here $\Gamma_M(\kappa)$ is the specified covariance record at a fixed impact and $\mathcal{K}$ is a maintained dealer rule, not a derived general equilibrium.*

*(i) If equal merged measures imply the same map $\Gamma_M$ and the fixed point is unique (or a common selection rule depends only on that map), equal merged measures give equal selected impacts.*

*(ii) Suppose $\mathcal{K}$ is $L_K$-Lipschitz, each covariance map is $L_\Gamma$-Lipschitz in impact, $L_K L_\Gamma < 1$, and $\sup_\kappa \|\Gamma_A(\kappa) - \Gamma_B(\kappa)\| \le \epsilon$. For existing fixed points,*

$$|\kappa_A - \kappa_B| \le \frac{L_K \epsilon}{1 - L_K L_\Gamma}. \tag{25}$$

*Thus a covariance discrepancy of order $(\Delta\eta)^{2K-1}$ preserves that order only under these uniform conditions.*

**Limits of the pricing extension.** Anonymous flow alone implies neither a covariance-only dealer objective nor uniqueness or continuity of its equilibrium. Multiple equilibria, threshold quote changes, private information and inventory-dependent rules are outside this proposition; for the question of invariance, Theorem S2.1 below covers them. Preserving a price-record testing bound additionally requires a uniformly Lipschitz observation law with the same nondegenerate noise; adding anonymous flow is a different statistical experiment and requires its own divergence bound. We therefore make no unconditional claim that endogenous pricing or additional aggregate-flow observations preserve the price-only lower bound.

**Proof of Proposition S20.1.** Equal maps have the same fixed-point set; uniqueness or the declared common selection yields (i). For (ii), put $d = |\kappa_A - \kappa_B|$. The fixed-point equations and the two Lipschitz bounds give $d \le L_K(L_\Gamma d + \epsilon)$. Since $1 - L_K L_\Gamma > 0$, rearrangement proves the displayed bound. The scalar inequality is machine checked in `p1_endogenous_contraction_sensitivity`; the economic derivation and uniformity of the covariance map are assumptions, not consequences of that algebra. A discontinuous threshold pricing rule need not preserve a small discrepancy. □

**Proposition S20.2** (Conditional limits of three evasion arguments)**.** *Maintain the exogenous reporting-noise and valid-instrument assumptions of Proposition S5.2; fix the finite number of linked class labels and a uniformly stable model family.*

*(i) For a fixed split $u_{i,m} = \alpha_{i,m} u_i$ with known common ownership, independent mean-zero reporting errors and $\alpha_{i,m} \ge \underline{\alpha} > 0$, only the lagged-flow column of the population IV matrix scales. Thus $\Sigma_{i,m} = \Sigma_i \operatorname{diag}(\alpha_{i,m}, 1)$ and $\det \Sigma_{i,m} = \alpha_{i,m} \det \Sigma_i$. Subject to the remaining uniform moment and excitation conditions, fixed finitely many consistent sub-account estimates can be summed: the true identity is $w_i = \sum_m w_{i,m}$, and the estimated sum has sampling error. The share restriction implies $M_i \underline{\alpha} \le 1$. This is not robustness to arbitrarily many vanishing accounts; a common estimated rate does not identify ownership.*

*(ii) Assume separately an execution cost lower bound $\mathbb{E}C_\zeta \geq c_\zeta \sigma_\zeta^2$ with $c_\zeta > 0$ and a gross profit upper bound $\mathbb{E}\Pi \leq \overline{\Pi}$ uniform in the spacing. Camouflage is unprofitable when $c_\zeta \sigma_\zeta^2 > \overline{\Pi}$. A variance requirement of order $(\Delta\eta)^{-(4K-2)}$ requires an additional additive exogenous-noise estimation model with uniformly controlled instruments. Neither this variance requirement nor the execution bound is established for arbitrary strategic orders.*

*(iii) If intensity is scaled to $\gamma_i w_i$ while $1 - r_i \geq \eta_{\min} > 0$, then $q_i \leq \gamma_i w_i / \eta_{\min} \to 0$ as $\gamma_i \to 0$. The other groups remain in the aggregate budget; reducing one contribution does not itself establish that the ceiling is met.*

**Economic scope.** These are conditional calculations about fixed linked accounts, bounded execution costs and a persistence class. They do not solve an equilibrium game of evasion. In particular, endogenous orders can affect prices and invalidate the reporting-error instrument, and a persistence change can offset a reduction in weight. Adversarial account creation and response adaptation require further modelling.

**Proof of Proposition S20.2.** (i) The current-price regressor column is unchanged, whereas the lagged true flow column scales by $\alpha_{i,m}$; valid independent reporting noise has zero population covariance with the instruments. Multiplicativity of determinants gives the stated factor. Summing shares at least $\underline{\alpha}$ gives $M_i \underline{\alpha} \leq 1$, checked in `p1_sybil_share_count_bound`. For a fixed finite collection the summed estimation errors remain of order $T^{-1/2}$ under the maintained joint central limit assumptions; this asymptotic step is not newly formalized here. (ii) Subtract the cost lower bound from the uniform gross-profit upper bound. Once the former exceeds the latter, expected net profit is negative, as checked in `p1_camouflage_unprofitable`. The cost coercivity and variance-to-difficulty link are explicit hypotheses, not consequences of fairness alone. (iii) Divide the scaled weight by $1 - r_i \geq \eta_{\min}$; without that bound, taking $1 - r_i = \gamma_i$ can keep $q_i = w_i$ constant. □

Consider $M$ markets, a supervisory cycle of $\bar{T}$ periods, and in each market one algorithm whose conduct does harm $\ell > 0$ per period until it is attributed. With prices alone, attribution at the required error level takes $T_p(\Delta\eta) = c_p(\Delta\eta)^{-(4K-2)}$ periods; with class-tagged records it takes $T_t(\Delta\eta) = c_t(\Delta\eta)^{-2}$ periods, the orders of Theorems S4.4 and S5.3 with constants $c_p, c_t > 0$. We take the attribution times to be of exactly these orders, which is a modelling assumption: Theorem S4.4 gives only a lower bound for prices, and if price-only attribution is slower, $G$ below is larger and disclosure is chosen at least on the band the proposition describes. $T_t \propto (\Delta\eta)^{-2}$ is the rate target; for weights at fixed accuracy $T_t$ does not grow as $\Delta\eta$ shrinks, and the lower edge of the band disappears. Attribution that would take longer than the cycle does not happen within it. An audit trail costs $F$ per cycle to operate and $c$ per market to comply with, including any information leakage. The supervisor's expected loss is therefore $\mathcal{L}_{\text{price}} = M\ell \min\{\bar{T}, T_p\}$ without disclosure and $\mathcal{L}_{\text{disc}} = F + Mc + M\ell \min\{\bar{T}, T_t\}$ with it. Write $\Delta\eta_t = (c_t/\bar{T})^{1/2}$ and $\Delta\eta_p = (c_p/\bar{T})^{1/(4K-2)}$ for the spacings at which tagged and price-only attribution just fit within the cycle.

**Proposition S20.3** (When an audit trail is chosen)**.** *Let $K \geq 2$ and $\Delta\eta_t < \Delta\eta_p$, and let $G(\Delta\eta) = \min\{\bar{T}, T_p(\Delta\eta)\} - \min\{\bar{T}, T_t(\Delta\eta)\}$ be the attribution delay that disclosure saves.*

*(i) The supervisor strictly prefers disclosure if and only if $F/M + c < \ell\, G(\Delta\eta)$.*

*(ii) $G$ vanishes for $\Delta\eta \leq \Delta\eta_t$, increases on $(\Delta\eta_t, \Delta\eta_p]$ to its maximum $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$, decreases wherever it is positive beyond $\Delta\eta_p$, and is negative for $\Delta\eta > (c_p/c_t)^{1/(4K-4)}$, where it returns toward zero from below. Hence the spacings at which disclosure is chosen form an interval around $\Delta\eta_p$, possibly empty: disclosure is never chosen for algorithms so similar*

*that even tagged records cannot attribute them within the cycle, nor for algorithms so different that prices attribute them quickly.*

*(iii) The interval is non-empty if and only if the audit trail covers more than $M^* = F/(\ell G^* - c)$ markets (with $M^* = \infty$ if $\ell G^* \le c$), and it widens as $M$ grows.*

Part (i) restates the comparison of losses. Part (ii) says that disclosure is valuable in a middle band, where algorithms are close enough that prices cannot attribute them within the cycle but far enough apart that tagged records can; the polynomial gap between the two delays, $(\Delta\eta)^{-(4K-2)}$ against $(\Delta\eta)^{-2}$, is what makes the band wide. Part (iii) is an economies-of-scale result: an audit trail with a fixed operating cost pays only when it spans enough markets. This illustrates a possible economy of scale for market-wide reporting; the cost function is stylised, not estimated from CAT, and the result neither explains CAT's legal architecture nor shows that it is optimal. A supervisor that can trade has a second option, active probing, whose cost is the frontier of Table S9; because that cost grows at the same polynomial order, it shortens price-only attribution only for well-separated algorithms and leaves the comparison for close ones unchanged. The model takes the harm $\ell$ as given and holds the algorithms' behaviour fixed across regimes; neither is modelled here.

**Proof of Proposition S20.3.** (i) $\mathcal{L}_{\text{disc}} < \mathcal{L}_{\text{price}}$ is $F + Mc < M\ell G$. (ii) $T_p$ and $T_t$ are decreasing in $\Delta\eta$, with $T_t \ge \bar{T}$ exactly when $\Delta\eta \le \Delta\eta_t$ and $T_p \ge \bar{T}$ exactly when $\Delta\eta \le \Delta\eta_p$. On $\Delta\eta \le \Delta\eta_t$ both minima equal $\bar{T}$ and $G = 0$. On $(\Delta\eta_t, \Delta\eta_p]$, $G = \bar{T} - c_t(\Delta\eta)^{-2}$, which increases to $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$. On $\Delta\eta > \Delta\eta_p$, $G = c_p(\Delta\eta)^{-(4K-2)} - c_t(\Delta\eta)^{-2} = (\Delta\eta)^{-2}\big[c_p(\Delta\eta)^{-(4K-4)} - c_t\big]$; the bracket is positive exactly when $\Delta\eta < \Delta\eta_0 = (c_p/c_t)^{1/(4K-4)}$, and $G'(\Delta\eta) = (\Delta\eta)^{-3}\big[2c_t - (4K-2)c_p(\Delta\eta)^{-(4K-4)}\big] < 0$ whenever $c_p(\Delta\eta)^{-(4K-4)} > c_t$, because $4K - 2 > 2$. So $G$ decreases wherever it is positive beyond $\Delta\eta_p$, and is non-positive beyond $\Delta\eta_0$. Every superlevel set of $G$ at a positive level is therefore an interval (the negative tail, where $G$ rises back toward zero from below, never reaches a positive level), and the level $(F/M + c)/\ell > 0$ lies above $G = 0$, so the set where (i) holds is an interval contained in $(\Delta\eta_t, \Delta\eta_0)$ that contains $\Delta\eta_p$ whenever it is non-empty. (iii) The interval is non-empty if and only if the level is below the maximum, $F/M + c < \ell G^*$, that is $M > M^*$; the level decreases in $M$, so the superlevel set grows. □

# S21 Finite-sample surveillance bounds

## S21.1 A finite-window price-surveillance bound

**Theorem S21.1** (Finite-window abstention)**.** *Observe $x_t = sx_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \le T$, with $s \in (0, 2)$. Let a possibly randomized rule $\Psi$ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ and $\sup_{s\ge1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$ with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. On $B_T = (1 - c_0/T, 1 + c_0/T)$,*

$$\inf_{s \in B_T} P_s(\Psi = \text{Abstain}) \;\ge\; 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

*and for every finite $T$ the minimax worst-case error of binary rules over $s \in (0, 2)$ is at least $1/2$.*

*Proof.* Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\text{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval $B_T$ is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

## S21.2 Passive distinction of a known pair in finite samples

**Proposition S21.2** (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma^2)$, use independent three-step reset blocks and no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q \eta$, $b_{2,0} = b_1^2 + \kappa q \eta(1-\eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q \delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With*

$$R = \max_{r \in \{0,\delta\}} \Big\{ (s_w^2 + \sigma^2)\big[s_w^2(1 + b_1^2 + b_{2,r}^2) + \sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

*the midpoint test of the average product has worst-case error at most $\alpha$ whenever $n \geq 4R/(\alpha\Delta^2)$ blocks are observed: zero controlled quantity and order $\delta^{-4}$ blocks.*

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for $3.9 \times 10^7$ blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with $2 \times 10^5$ blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

*Proof.* In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \leq 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \leq 0$ for uniform innovations, so it is at most $R$. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

# S22 Account-level capital

Lemma S10.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

**Numerical values.** Decimal optima and brackets below are numerical evaluations of the stated bounds.

## S22.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates $\theta_0$ and $\theta_1 = \theta_0 + \delta$ over $N$ periods, submitting probes $V = (V_1, \ldots, V_N)^\top$ that generate inventory $I_t = \sum_{s \leq t} V_s$. Under contract $I_0$ (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{26}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi|V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \geq \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \leq V_{\mathrm{eff}}$ and $|I_t| \leq I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \leq B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

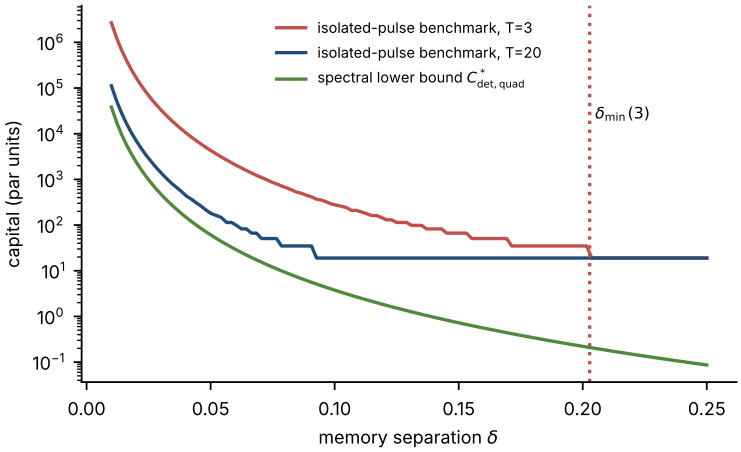



Figure S5: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem S22.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

**Proposition S22.1** (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon $T$ per pulse, $(\delta, B)$ space splits into three regions.*

*(i)* Feasible single pulse. *For $\delta \geq \delta_{\min}(T)$ and $B \geq C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

*(ii)* Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds $V_{\text{eff}}$. Capped pulses each supply divergence $V_{\text{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\text{pulses}}(T, \delta) = \left\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\text{eff}}^2 \|d_T\|_2^2} \right\rceil. \tag{27}$$

*(iii)* Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\text{session}})$ is the root of $V_{\text{eff}} \|d_{T_{\text{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha / \sqrt{\lfloor T_{\text{session}}/T_{\text{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\text{session}}^{-1/4})$.*

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\text{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\text{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\text{eff}}^2 + \phi P_* V_{\text{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure S5).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

Table S4: Account-level initial capital at $\delta = 0.05$ (par units)

| Design | $C_0$ | Error | End inv. | Peak margin | Max $\lvert v_t \rvert$ | Trades |
|---|---|---|---|---|---|---|
| 12 buys, $T = 20$ apart, never unwound | 229.13 | 0.0578 | 75.00 | 39.38 | 6.25 | 12 |
| 6 buy/sell round trips | 189.45 | 0.0308 | 0.00 | 3.28 | 6.25 | 12 |
| 5 buy/sell round trips | 157.89 | 0.0442 | 0.00 | 3.28 | 6.25 | 10 |
| 12 buys, each unwound after 20 steps | 378.82 | 0.0040 | 0.00 | 3.28 | 6.25 | 24 |
| Generalized-eigenvector probe ($L = 120$) | 65.06 | 0.0500 | 0.00 | 2.74 | 1.24 | 120 |
| Generalized-eigenvector probe ($L = 160$) | 64.86 | 0.0500 | 0.00 | 2.17 | 1.09 | 160 |
| Fee-refined probe ($L = 240$) | 64.48 | 0.0500 | 0.00 | — | 1.30 | 240 |

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The numerically evaluated lower bound of Theorem S22.2 is 61.41; the explicit constant of Lemma S10.5 is $4.9 \times 10^{-4}$.

## S22.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table S4 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.24 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{28}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon $L$ into $L+1$ at identical cost and information, so the minimum is non-increasing in $L$.

**Theorem S22.2** (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{29}$$

*(i)* Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

*(ii)* Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most $\alpha$ has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\text{det,quad}}(\delta). \tag{30}$$

*(iii)* Bracket. *At $\delta = 0.05$, $C^*_{\text{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the numerical lower estimate $C_{\text{det}} \geq 61.41$. A feasible 240-step sequence attains 64.48. So, up to the accuracy of the numerical minimization over $\omega$, the fee-paying*

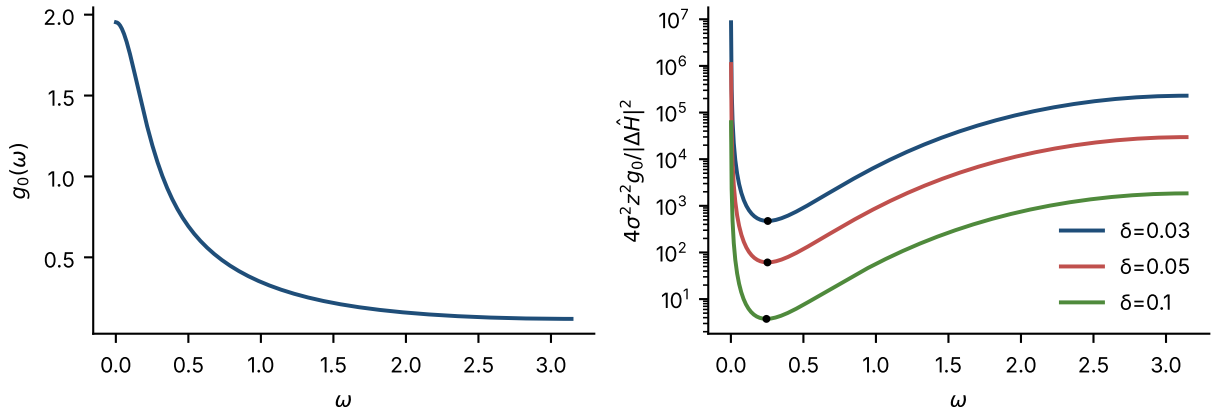



Figure S6: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (30); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

*optimum lies in* $[61.41, 64.48]$. *The upper endpoint is attained by a feasible probe. The lower endpoint is a numerical estimate of an infimum, found by a frequency grid and local refinement, and is not certified: a grid minimum can lie above the true infimum.*

*(iv)* Process noise. *Let the price also receive the innovations* $w_t \sim N(0, \sigma_w^2)$ *of* (2)*, independent of observation noise* $\nu_t \sim N(0, \sigma_\nu^2)$*, so that from the reset state the observer sees* $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$*,* $t < T$*. For the known pair of Theorem S10.1, put* $\sigma_{\text{eff}}^2 = \sigma_\nu^2 + \sigma_w^2/4$*,* $\bar{M} = 1/(1-S)$ *and* $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2(\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma_{\text{eff}}^4)$*. Every deterministic round trip whose test has errors* $\alpha$ *and* $\beta$ *under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma_{\text{eff}}^2 \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_{\omega} \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{31}$$

*with* kl *as in Theorem S22.3. In-loop noise therefore leaves the* $\delta^{-4}$ *capital order intact for every horizon* $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$*; at horizons of order* $\delta^{-4}$ *and longer, the noise itself separates the pair without a probe (Theorem S4.4).*

Positivity of $g_i$ follows from Lemma S8.1: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\text{det,quad}}$ across separations (Table S5): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\text{det,quad}}$ between $3.67 \times 10^{-4}$ and $3.84 \times 10^{-4}$, confirming the $\delta^{-4}$ law. The fee term, the feasible upper bounds and the adaptive bound of Theorem S22.3 are reported in the same table.

The infimum over $\omega$ is evaluated on a fine frequency grid; a local refinement agrees to four digits (61.1377 at $\omega^* = 0.2526$), and the reported constant is the larger of the two hypotheses' bounds. "Certified" here means a numerical evaluation of the stated bound, not an interval-arithmetic enclosure.

**Theorem S22.3** (Adaptive and sequential probing)**.** *Maintain a common-variance conditional Gaussian likelihood for the causal reset observations. Require finite expected total squared conditional mean separation, finite expected execution loss, and the stopped likelihood/Kullback–Leibler identity used below. Let* $(v_t)_{t \leq \tau}$ *be a non-anticipating policy adapted to that filtration, with an*

Table S5: Capital lower bounds and feasible upper bounds across separations (par units)

| $\delta$ | $\omega^*$ | $C^*_{\mathrm{det,quad}}$ | $\delta^4 C^*_{\mathrm{det,quad}}$ | Certified $C_{\mathrm{det}}$ | Feasible UB | Adaptive LB |
|---|---|---|---|---|---|---|
| 0.02 | 0.2546 | 2397.85 | $3.84 \times 10^{-4}$ | 2399.58 | 2428.25 | 1176.04 |
| 0.03 | 0.2541 | 473.19 | $3.83 \times 10^{-4}$ | 473.97 | 484.50 | 232.51 |
| 0.05 | 0.2526 | 61.14 | $3.82 \times 10^{-4}$ | 61.41 | 64.48 | 30.22 |
| 0.08 | 0.2488 | 9.26 | $3.79 \times 10^{-4}$ | 9.37 | 10.38 | 4.64 |
| 0.10 | 0.2453 | 3.77 | $3.77 \times 10^{-4}$ | 3.83 | 4.44 | 1.91 |
| 0.15 | 0.2331 | 0.73 | $3.67 \times 10^{-4}$ | 0.76 | 1.01 | 0.38 |

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. the numerical-bound column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \le 240$; the adaptive column adds the fee lower bound to the adaptive bound.

*almost surely finite stopping time and* $\sum_{t\le\tau} v_t = 0$*, followed by a terminal test* $\psi_\tau \in \{0,1\}$ *with* $P_0(\psi_\tau = 1) \le \alpha$ *and* $P_1(\psi_\tau = 0) \le \beta$*. Then*

$$\mathbb{E}_0[L_0(v)] \;\ge\; 2\sigma^2\,\mathrm{kl}(\alpha, 1-\beta)\inf_{\omega}\frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{32}$$

*where* $\mathrm{kl}(\alpha,1-\beta) = \alpha\log\frac{\alpha}{1-\beta} + (1-\alpha)\log\frac{1-\alpha}{\beta}$*. The adaptive lower-bound constant divided by the deterministic lower-bound constant is the factor* $\mathrm{kl}(\alpha,1-\beta)/(2z^2_{1-\alpha}) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, under the stated likelihood and execution assumptions. Comparing two lower bounds alone does not bound the achievable savings relative to an optimal deterministic policy.*

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a numerical ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under $H_0$, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

**Theorem S22.4** (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of* $L \ge 4$ *trading periods, let* $E \in \mathbb{R}^{L\times(L-1)}$ *be the first-difference basis of flat probes (*$\sum_t v_t = 0$*),* $Q_i = \frac{1}{2}T_{h_i}$ *(so that* $L_i(v) = v^\top Q_i v$*),* $A_{\mathrm{diff}} = (H_1 - H_0)^\top(H_1 - H_0)$*, and let* $G_i = E^\top Q_i E \succ 0$ *and* $M = E^\top A_{\mathrm{diff}} E \succeq 0$ *be the reduced execution-loss and information matrices, with* $G_i$*-orthonormal generalized eigenvectors* $e_j$ *and eigenvalues* $0 \le \lambda_1 \le \cdots \le \lambda_{L-3} < \lambda_{L-2} \le \lambda_{L-1}$*; put* $c_{\mathrm{req}} = 4\sigma^2 z^2_{1-\alpha}$ *and* $c_{\mathrm{fee}} = \phi P_*$*. For* $\theta \in [0,\pi)$ *let* $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$*,* $\lambda(\theta) = \lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$*,* $s_\theta = \mathrm{sign}(Ed(\theta))$*, and*

$$\begin{aligned} F_{\mathrm{dom}}(\theta) &= c_{\mathrm{fee}}\sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c^2_{\mathrm{fee}}\sum_{j\le L-3}\frac{(s_\theta^\top E e_j)^2}{1-\lambda_j/\lambda(\theta)}, \\ \gamma(\theta) &= \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta)-\lambda_{L-3})\,c_{\mathrm{req}}}. \end{aligned} \tag{33}$$

*Then every flat probe meeting the separation constraint of* (28) *satisfies*

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; \min_\theta\Big\{\frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta)\Big\} \;\equiv\; C_{\mathrm{RH}}(L) \;\ge\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{34}$$

Table S6: Finite-horizon capital at $\delta = 0.05$ (par units)

| Horizon $L$ | Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$ | Full bound $C_{\mathrm{RH}}(L)$ | Best feasible probe | Gap to $C_{\mathrm{det}} = 61.41$ |
|---|---|---|---|---|
| 80 | 62.55 | 64.72 | 65.02 | 5.9% |
| 120 | 61.84 | 63.39 | 64.59 | 5.2% |
| 160 | 61.54 | 62.46 | 64.50 | 5.0% |
| 240 | 61.32 | 61.76 | 64.48 | 5.0% |
| 320 | 61.25 | — | $\leq 64.48$ | $\leq 5.0\%$ |
| 480 | 61.19 | — | $\leq 64.48$ | $\leq 5.0\%$ |

Quadratic bound $c_{\mathrm{req}}/\lambda_{L-1}$ from the generalized eigenproblem of Theorem S22.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with $\lambda'$ computed against $T_{h_0}$ rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (34) over 3,600 values of $\theta$; at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

$$\Psi(\theta) = \min_{x \geq 0} \Big\{ x^2 + \Big[ F_{\mathrm{dom}}(\theta)\big(1 - \gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\, x \Big]_+ \Big\} \geq 0,$$

*a piecewise-quadratic minimization in closed form; when $\gamma F_{\mathrm{dom}} < 1$ and the minimizer has $\gamma x^2 \leq 1$, $\Psi \geq \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1 - \gamma F_{\mathrm{dom}}))\big)$, which is what the bound uses.*

The bound is computable for any horizon. Table S6 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective (Table S6 reports 65.02 for ledger capital, which also includes margin), so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or $L_1$-localized probes are needed at long horizons.

## S22.3 Robustness: heteroskedastic impact and probing protocols

**Corollary S22.5** (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as $\sigma^2_{i,t}(v) = \sigma^2(1 + \xi[(h_i * v)_t]^2)$ and slippage scales as $\kappa_{\mathrm{eff}} = \kappa(1 + \xi_0|v_t|)$. For capacity-bounded probes, $D_{\mathrm{KL}}(P_0^v \| P_1^v) \leq \frac{1+O(\xi^2 V_{\mathrm{eff}}^2)}{2\sigma^2} \|(H_1 - H_0)v\|_2^2 = O(\|v\|_2^2 \delta^4)$, and convex impact only raises dissipation. Required capital remains $\Theta(\delta^{-4})$.*

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near $-4$ in $\log \delta$: $-3.90$, $-3.84$, $-3.47$ and $-3.81$ over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (full working paper). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

Table S7: Capital lower bound under alternative observation-noise calibrations (USDT)

| Noise calibration | $\delta = 0.02$ | $\delta = 0.05$ | $\delta = 0.10$ |
|---|---|---|---|
| Roll estimator, median (0.40 USDT) | 3,874,696 | 99,192 | 6,200 |
| One-second price-change s.d., median (1.98 USDT) | 93,688,369 | 2,398,422 | 149,901 |
| Earlier assumption (4 USDT, upper scenario) | 383,812,500 | 9,825,600 | 614,100 |

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with $\sigma$ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Table S8: Daily observation-noise estimates, Binance BTCUSDT one-second closes

| Date | Roll $\hat{\sigma}$ (USDT) | One-second change s.d. (USDT) | Mean price (USDT) |
|---|---|---|---|
| 2023-01-10 | 0.152 | 0.679 | 17,288 |
| 2023-01-11 | 0.092 | 0.891 | 17,456 |
| 2023-01-12 | 0.232 | 2.292 | 18,343 |
| 2023-01-13 | 0.348 | 1.762 | 19,075 |
| 2023-01-14 | 0.773 | 3.511 | 20,818 |
| 2023-01-15 | 0.482 | 1.790 | 20,778 |
| 2023-01-16 | 0.629 | 2.357 | 21,036 |
| 2023-01-17 | 0.457 | 1.976 | 21,194 |
| 2023-01-18 | 0.695 | 2.541 | 21,122 |
| 2023-01-19 | 0.402 | 1.316 | 20,850 |
| 2023-01-20 | 0.316 | 2.038 | 21,306 |
| Median | 0.402 | 1.976 | |

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

## S22.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2\delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\mathrm{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about $9.9 \times 10^4$ USDT, which is not prohibitive for a well-capitalized desk (Table S7). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem S4.1, when several groups are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

**Proofs.**

**Proposition S22.1.** Parts (i) and (ii) follow from (26) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii)

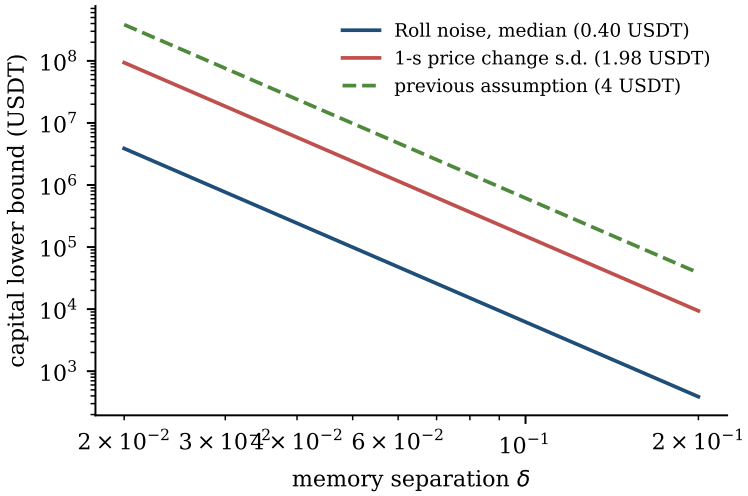



Figure S7: Capital lower bound against memory separation for three observation-noise calibrations.

solves the session constraint for $\delta$; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

**Theorems S22.2 and S22.3.** For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2} v^\top T_{h_i} v = \frac{1}{2\pi} \int g_i(\omega) |V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma S8.1. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi} \int |\Delta \hat{H}|^2 |V|^2 \le \sup_\omega (|\Delta \hat{H}|^2 / g_i) L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \ge 2\sigma z_{1-\alpha}$ gives (30). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \ge \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0 \sum_t (\Delta\mu_t)^2/(2\sigma^2) \le \sup(|\Delta \hat{H}|^2/g_0) \mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (32).

*Part (iv).* Let $G_\theta$ be the $T \times T$ lower-triangular Toeplitz section of $H_\theta/\kappa$, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \le \sup_\omega |H_\theta|/\kappa \le \bar{M}$ because $|H_\theta(e^{i\omega})| \le \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; $\kappa/H_\theta$ is analytic on and outside the unit circle because $H_\theta$ is minimum phase (Theorem S4.3), and Lemma S8.1 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell (1 - \lambda_\ell \cos\omega)/(1 - 2\lambda_\ell \cos\omega + \lambda_\ell^2) \ge \sum_\ell \omega_\ell/(1+\lambda_\ell) \ge \kappa/2$, so $\|G_\theta^{-1}\| \le 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$ (the smallest singular value of $G_\theta$ is at least 0.522 in 762 random positive economies and sections). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top \Sigma_1^{-1} \Delta\mu \le \|(H_1 - H_0)v\|_2^2/(2\sigma_{\mathrm{eff}}^2)$. For the covariance part, the eigenvalues $\lambda_k$ of $A = \Sigma_1^{-1/2} \Sigma_0 \Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)$, and $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \le \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\mathrm{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1 \Delta G^\top)$ and $\|\Delta G\|_F^2 \le T\|h^\delta - h^0\|_2^2/\kappa^2 \le T K_*^2 \delta^4/\kappa^2$ (proof of Theorem S10.1) gives a covariance part of at most $C_w T \delta^4$. The data-processing inequality turns errors $(\alpha, \beta)$ into $D_{\mathrm{KL}} \ge \mathrm{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega (|\Delta \hat{H}|^2/g_i) L_i(v)$, which gives (31).

**Theorem S22.4.** Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the $G_i$-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \ge c_{\mathrm{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \ge 0$; replacing $\theta$ by $\theta + \pi$ flips $d$ and $s_\theta$ and leaves (33) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$ and $D = L_i(v) - L_0(\theta)$.

The constraint gives $r^2\lambda(\theta) \ge c_{\mathrm{req}} - \sum_{j\le L-3}\lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j\le L-3} c_j^2 - L_0(\theta) \;\ge\; \sum_{j\le L-3}\Big(1-\frac{\lambda_j}{\lambda(\theta)}\Big)c_j^2 \;\ge\; \Big(1-\frac{\lambda_{L-3}}{\lambda(\theta)}\Big)\sum_{j\le L-3} c_j^2 \;\ge 0,$$

and $r^2 \ge L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)}\sum_{j\le L-3} c_j^2 \ge L_0(\theta)(1-\gamma(\theta)D)$. By Hölder's inequality with the sign vector $s_\theta$, $c_{\mathrm{fee}}\|Eu\|_1 \ge c_{\mathrm{fee}} r\|Ed(\theta)\|_1 + c_{\mathrm{fee}}\sum_{j\le L-3} c_j\, s_\theta^\top E e_j$, and Cauchy–Schwarz with weights $1-\lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\,W_\perp(\theta)}$. Using $r \ge \sqrt{L_0(1-\gamma D)_+} \ge \sqrt{L_0}\,(1-\gamma D)_+$ and the fact that the fee is nonnegative, with $x = \sqrt{D}$,

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; L_0(\theta) + x^2 + \Big[F_{\mathrm{dom}}(\theta)(1-\gamma(\theta)x^2)_+ - \sqrt{W_\perp(\theta)}\,x\Big]_+ \;\ge\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown $\theta$ gives (34), and $\Psi \ge 0$, $L_0(\theta) \ge c_{\mathrm{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \le 1$ with $\gamma F_{\mathrm{dom}} < 1$ the bracket is a convex quadratic in $x$ with minimum $F_{\mathrm{dom}} - W_\perp/(4(1-\gamma F_{\mathrm{dom}}))$, which gives the closed form.

**Corollary S22.5.** Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma_{1,t}^2 - \sigma_{0,t}^2 = \sigma^2\xi[(h_1+h_0)*v]_t[\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x-1-\log x)$ with $x = \sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

# S23 A hypothetical scale illustration: disclosure in dollars

*This subsection is a hypothetical scale illustration of an outside prober's shadow loss, not a public cost–benefit calculation.*

Theorem S5.3 states the value of class-tagged disclosure in samples. Theorems S22.2 and S4.1 let us translate that value into a flow of round-trip trading losses for an outside participant. Two adjustments make the figures exact:

(i) *Exact $K$-group moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem S22.2,

$$L_{\mathrm{rt}}^*(K,\Delta\eta) \;=\; \frac{4\sigma^2 z_{1-\alpha}^2}{\sup_\omega\big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2/g_0(\omega)\big)},$$

directly for the $K$-point Gauss versus Gauss–Radau economies of Theorem S4.1, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss $L_{\mathrm{rt}}^*$ is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over, we report the annualised round-trip trading loss $52 \times L_{\mathrm{rt}}^*$ under weekly renewal (matching the CFTC's weekly reporting cycle). For orientation only, we also record official audit-trail figures of two different kinds; we do not divide one by the other, because a prober's trading loss is a private transfer to other traders calibrated to a Bitcoin perpetual, while an audit trail's budget is a resource cost for equities and options. The first is CAT LLC's own operating

Table S9: Price-only round-trip probing loss per market and annualised weekly-renewal cost (hypothetical outside prober)

| | Loss / probe (USDT) | | Annual (52×, USDT) | |
|---|---|---|---|---|
| Economy ($\Delta\eta$) | Roll | 1-sec | Roll | 1-sec |
| $K = 2$, known (0.20) | 6,200 | 149,901 | 322,400 | 7,794,852 |
| $K = 2$, known (0.10) | 99,192 | 2,398,422 | 5,157,984 | 124,717,944 |
| $K = 2$, known (0.04) | 3,874,696 | 93,688,369 | 201,484,192 | 4,871,795,188 |
| $K = 2$, Radau (0.20) | 4,292 | 103,788 | 223,184 | 5,396,976 |
| $K = 2$, Radau (0.10) | 283,488 | 6,854,612 | 14,741,376 | 356,439,824 |
| $K = 3$, Radau (0.20) | 804,403 | 19,450,107 | 41,828,956 | 1,011,405,564 |
| $K = 3$, Radau (0.10) | 1,137,325,880 | 27,500,017,407 | 59,140,945,760 | 1,430,000,905,164 |
| $K = 4$, Radau (0.20) | 156,632,086 | 3,787,291,895 | 8,144,868,472 | 196,939,178,540 |

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact $K$-group quadrature pair of Theorem S4.1 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem S22.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning; it is a private transfer to other traders, not a social or resource cost, and it is not comparable with audit-trail budgets.

budget: $248.8 million for 2025 in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to $228.3 million in May 2025 and reduced further later in 2025, and the revised 2026 Financial and Operating Budget of 31 March 2026, with total expenses of $147,044,869, which we use (CAT LLC, 2026). The second is historical, ex ante estimates of industry-wide compliance costs, which are not operating budgets and cover a different scope: the SEC's 2016 CAT NMS Plan Approval Order estimate of $2.4 billion in initial industry build costs and $1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating $4.0 billion initial and $2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

The figures are a shadow cost of learning by trading, the price an outside participant would pay; a supervisor without an audit trail does not probe, and its cost is detection delay, which Proposition S20.3 models. The cost results require a positive economy: with signed weights the execution form $\frac{1}{2}v^\top T_h v$ can be indefinite and a round trip can earn money, so none of the dollar figures applies there. Probing cost is a lower bound on the cost of learning from prices by trading, not on passive learning and not a social cost. It is also a lower bound for deterministic probes only. For sequential probes, Theorem S22.3 gives a lower bound whose constant is about 0.49 times the deterministic one. That ratio compares two lower bounds. It does not show that any sequential policy achieves a saving. Only the observation noise is estimated from data; the structural parameters are illustrative, and the constant is sensitive to them. At $\delta = 0.05$ the numerically evaluated lower bound of Theorem S22.2 is 358.8, 179.0, 61.1 and 13.9 par units for $a = 0$, 0.25, 0.5 and 0.7, and 3.9, 19.6, 61.1 and 305.2 for $\bar{\eta} = 0.2$, 0.3, 0.4 and 0.6; at $\Delta\eta = 0.20$ instead

of 0.10 the unknown-pair figures fall by a factor of about 64. The Gauss–Radau pair is used in the orientation that gives the larger, still valid, lower bound. The known-pair rows span a rate width of $2\delta = \Delta\eta$, whereas the $K = 2$ Gauss–Radau rows on $r_c + \Delta\eta[-1, 1]$ span $\frac{4}{3}\Delta\eta$. Within that interpretation, Table S9 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant $\Delta_K$ turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly renewal of price-only probing costs \$0.32 million per market per year at Roll noise and \$7.79 million at one-second noise. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs \$14.74 million (Roll) or \$356.44 million (one-second) per market per year; with $K = 3$ unknown groups at $\Delta\eta = 0.20$ the figures rise to \$41.83 million and \$1.01 billion, and at $K = 3$, $\Delta\eta = 0.10$ or $K = 4$, $\Delta\eta = 0.20$ they run to billions. For orientation only, CAT LLC's revised 2026 budget is \$147.0 million; as explained above, the two numbers measure different things. The renewal frequency is an assumption: annual figures scale with it, so monthly renewal divides them by 4.3 and renewal on each of 252 trading days multiplies them by 4.8.

# S24 Robustness to nonlinear and adaptive trading rules

The theorems of Sections S2–S4 use the linear exponential-memory recursion (3). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size $A$ moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the group's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1 - \eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size $\eta_i$; and (5) a *tabular value rule* that updates a 21-bin table at rate $\eta_i$, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate $\eta_i$, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one group with weight $q$ and position limit 1 against two groups with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B ($\delta^{-4}$ law):* one group at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as $\delta^2$ and the capital lower bound of Theorem S22.2 as $\delta^{-4}$. The saturation threshold $A_{\text{sat}}$ is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-specified a second experiment in which online learners are trained from scratch inside the closed-loop market for

Table S10: Merged-measure equivalence and the $\delta^{-4}$ law under stylized nonlinear rules and closed-loop trained online learners

| Rule | Test A raw ($O(A^2)$) | Test A norm. ($O(A)$) | Test B slope ($\|\Delta h\|_2$) | Test B slope (capital) | Capital at $\delta = 0.05$ | $A_{\text{sat}}$ / $R^2_{\text{exp}}$ |
|---|---|---|---|---|---|---|
| *Panel A. Stylized nonlinear and adaptive rules (fixed operating point)* | | | | | | |
| Linear benchmark | — | — | 2.007 | −4.001 | 61.42 | — |
| 1. Momentum (tanh limit) | 2.015 | 1.015 | 2.007 | −4.001 | 61.47 | 0.834 |
| 2. Threshold rule, averaged | 2.043 | 1.043 | 2.007 | −4.001 | 61.42 | 0.049 |
| 3. Adaptive gain (RLS) | 2.064 | 1.064 | 2.007 | −4.001 | 61.46 | 0.751 |
| 4. Online neural readout | 2.011 | 1.011 | 2.007 | −4.001 | 61.48 | 0.751 |
| 5. Tabular value, smoothed | 2.039 | 1.039 | 2.007 | −4.001 | 61.44 | 1.271 |
| *Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)* | | | | | | |
| 6. Recursive least squares | 3.000 | 2.000 | 2.008 | −4.014 | 54.58 | $R^2 = 0.993$ |
| 7. Two-layer network (SGD) | 1.878 | 0.878 | 2.007 | −4.012 | 65.19 | $R^2 = 0.994$ |
| 8. Policy gradient on profit | 3.000 | 2.000 | 2.004 | −4.006 | 58.05 | $R^2 = 0.995$ |
| 9. Tabular Q-learner (TD(0)) | 2.108 | 1.108 | 1.994 | −3.973 | 186,821 | $R^2 = 1.000$ |
| *Panel C. Where the trained-learner results change (range over the four learners)* | | | | | | |
| Independent training streams | rel. gap $\leq 10^{-5}$ | | 0.13 to 0.92 | −1.75 to −0.38 | — | (c) fails |
| Learning during the probe | rel. gap $\leq 1.1 \times 10^{-3}$ | | 1.99 to 2.01 | −4.02 to −3.98 | — | (c) holds |

Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past $\tau$ only for $A > 0.139$.
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure S8). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most $10^{-5}$. A raw Test A slope above 2 means the gap is of higher order than $A^2$. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, lower bound of Theorem S22.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital.

$T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate $\eta_i$: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and $\delta^{-4}$ predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative $L_2$ response at $A = 0.005$; (c) the capital lower bound to separate trained populations whose memory rates differ by $2\delta$ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the $\delta^{-4}$ law fail when learning continues during the probe.

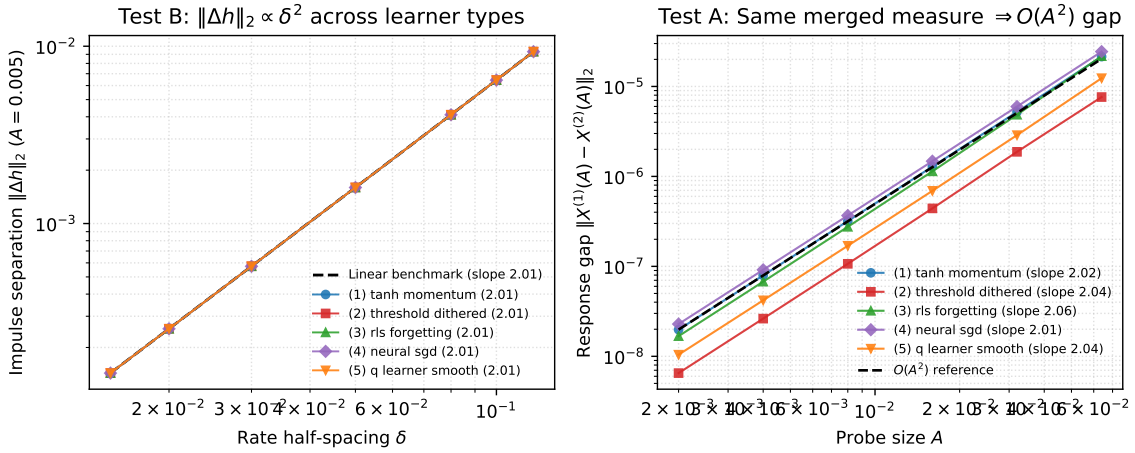



Figure S8: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing $\delta$ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size $A$, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

Table S10 and Figure S8 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and capital lower bound scale as $\delta^2$ (slope 2.007) and $\delta^{-4}$ (slope $-4.001$), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most $10^{-5}$ in relative response at $A = 0.005$, and capital lower bound scales with slopes $-4.014$ to $-3.973$, inside the registered band. The level of capital differs across learners because training sets the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are $-1.75$, $-0.97$, $-1.62$ and $-0.38$. Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The $\delta^{-4}$ law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes $-4.02$ to $-3.98$) and the merged-measure gap rises only to $1.1 \times 10^{-3}$, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those groups from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights, not their rates.

# S25 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma S1.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar\eta = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem S10.1 around $\bar\eta$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q\delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar\eta$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: the spectral lower bound of Theorem S22.2, evaluated at this pair ($\bar\eta = 0.2931$, $\delta = 0.0348$), gives about 77 par units for the optimal flat probe (rescaling the $\bar\eta = 0.4$ constant by $\delta^{-4}$ alone would give 262, because the constant also depends on $\bar\eta$). Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly $1.2\times10^5$ USDT and the naive pulse about $3 \times 10^7$ USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem S4.1).

# S26 Do the rates survive real market noise?

Theorems S4.1 and S4.2 are proved for observation errors that are independent and Gaussian, and the full working paper checks them with Gaussian noise. Real price changes are different: one-minute returns have heavy tails, although they are almost uncorrelated from one minute to the next. To see whether the rates depend on the Gaussian assumption, we ran a pre-specified test (opened once) in which the trader groups are synthetic, with known memory spacing, and only the noise is real. Each reset experiment returns the price response to a unit impulse plus a window of consecutive one-minute log-price changes of Binance BTCUSDT, demeaned and scaled to unit variance. The changes come from 21 days, the 6th of each month from January 2025 to September 2026, that are used nowhere else in this project; their kurtosis is 19.4, against 3 for Gaussian noise. With class tagging, the experiment also returns each group's flow response, observed through independent windows of the same noise. A control arm uses i.i.d. Gaussian noise of the same variance. For each spacing, $N^*$ is the smallest number of experiments at which every recovered weight lies within 25% of the smallest true weight in at least 75% of 200 replications, using Prony's method on prices alone or the per-group regression of Proposition S5.2 with tagging. The noise level and the spacing grids were fixed on Gaussian noise alone, before any real noise was downloaded.

Both pre-specified hypotheses hold, so the test passes (Table S11). With real noise, the price-only requirement for two groups rises from 64 experiments at $\Delta\eta = 0.5$ to 370,728 at $\Delta\eta = 0.125$, a factor of about 5,800, and the fitted slope is $-6.01$; for three groups it is $-9.81$. Over the same grid the tagged requirement stays between 256 and 724 experiments, and it falls slightly rather than rises as the spacing shrinks. At every spacing the real-noise requirement is within one step of the grid (a factor $\sqrt{2}$) of the Gaussian control. The slopes are least-squares fits of $\log N^*$ on $\log\Delta\eta$ over the stated grids (seven spacings from 0.5 to 0.125 for two groups, five from 0.6 to 0.25 for three). No uncertainty interval was pre-specified; as a post hoc description, the ordinary least-squares standard errors are 0.17 and 0.47 with real noise (0.12 and 0.31 for the Gaussian control, slopes $-6.22$ and $-9.96$), and each $N^*$ is itself read off a grid with steps of $\sqrt{2}$. For two groups, the last four price-only points lie above 4,096 and use the normal draw, so the heavy tails enter the

Table S11: Sample requirement with real Bitcoin price noise: fitted slope of $\log N^*$ in $\log \Delta\eta$

| | Theory | Real noise | Gaussian control | Pre-specified criterion | Result |
|---|---|---|---|---|---|
| Price only, $K = 2$ (H1) | $-6$ | $-6.01$ | $-6.22$ | slope in $[-7.5, -4.5]$ | pass |
| Tagged, $K = 2$ (H2) | bounded | 256–724 | 256–1,024 | max/min $\leq 8$ | pass (2.8) |
| Price only, $K = 3$ (H3) | $-10$ | $-9.81$ | $-9.96$ | slope in $[-12.5, -7.5]$ | holds |

Synthetic groups at rates $r_c + \Delta\eta\, s$ ($s$ the $K$-point Gauss nodes; $r_c = 0.6$ and $\Delta\eta$ from 0.5 to 0.125 for $K = 2$, $r_c = 0.5$ and $\Delta\eta$ from 0.6 to 0.25 for $K = 3$), $a = 0.5$, $\kappa = 0.8$, total weight 0.08, $\sigma = 10^{-3}$, $\sigma_f = 0.02$. The test passes if H1 and H2 hold; H3 is reported either way. The tagged row gives the range of $N^*$ over the spacing grid. Requirements above 4,096 use the central-limit draw described in the text. Requirements at each spacing are in the full working paper. Test opened once.

fit only through the first three.

The test has a limit, stated in its specification. Averages of $N$ real windows are resampled exactly up to $N = 4{,}096$; beyond that they are drawn as Gaussian vectors with the real windows' covariance, as the central limit theorem suggests. Real one-minute changes are almost uncorrelated (lag-one autocorrelation between $-0.007$ and 0.009 within the windows), so beyond 4,096 the real and Gaussian arms differ little by construction. The direct evidence about heavy tails therefore comes from the points in the exact range: every tagged requirement, and the price-only requirements for two groups at the three widest spacings (64, 362 and 1,448 experiments, against 64, 362 and 1,024 with Gaussian noise). The other four price-only points for two groups, and all points for three groups, check the rate with the real noise's variance and covariance but not its tails. Averaging should make heavy tails matter less as $N$ grows, but the test does not show this.

# S27 Machine checks in detail

**Theorem ~S3.1.** `Paper1InventoryEquilibrium.lean` checks the algebraic core of part (i):

- the dealers' first-order condition holds in every state;
- the next price loads on the next flow shock with coefficient $\kappa/D$;
- the no-feedback existence condition.

Part (iii) is proved on paper. For part (iv), `Paper1InventoryIFT.lean` applies Mathlib's inverse function theorem along a one-parameter family of feedback $G = \epsilon g$, with $g$ strictly differentiable at $D = 1$. For all small $\epsilon$ it gives a zero near $(\kappa_-, 1)$, its uniqueness in a fixed neighbourhood, and the two first-order derivatives in the statement (`p1_inventory_small_feedback`). The version that is uniform over all small group measures, in the norm of the statement, is proved on paper.

**Theorem S4.1.** Lean proves the bound for every $K$ and every horizon $T$, with a constant that does not depend on $T$. It uses the theorem's stability hypothesis $|a| + |\kappa|W/(1 - r_c - \Delta\eta) \leq \varsigma < 1$, and holds for all error probabilities with $\alpha_0 + \beta_0 \leq 1$ (`p1_thm_kcohort_lower_bound_uniform`): $4\sigma^2(1-\alpha_0-\beta_0)^2 \leq N\, C_U^2 (\Delta\eta)^{4K-2}$. The checked pair is a different moment-matched pair with the same order: binomial weights on $2K$ equally spaced nodes, split into odd and even halves. For the Gauss and Gauss–Radau rules themselves, Lean checks the case $K = 3$ (positive weights, matched moments and the fifth-moment gap). Lean also checks both estimator statements for this pair, for every horizon (`p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1`). The first covers an estimator of the weights, listed in order of memory rate, that is accurate to $LW/4^L$ in each coordinate with probability at least 3/4 under both economies. The second covers

an estimator of the merged measure that integrates every 1-Lipschitz function vanishing at zero to within $2\Delta\eta W/(\pi(2K-1))$. For measures of equal mass, a Wasserstein error below that accuracy implies this. Both force $\sigma^2 \le N\,C_U^2(\Delta\eta)^{4K-2}$, here with $L = K-1$. The matching lower order $\|h^A - h^B\|_2 \gtrsim (\Delta\eta)^{2K-1}$ is proved on paper.

**Theorem S4.2.** Lean verifies this theorem for every $K \ge 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \le 1$, so that every memory rate has modulus at most one.

**Theorem S4.3.** For every $K$, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses $H$ in the rational form $\kappa z D(z)/((z-a)D(z) - \kappa N(z))$ of Theorem S2.3, and *stable* in the sense that the poles of $H$ lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \dots, y_{K,t})$ are exactly the zeros of the denominator of $H$, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response $h_t$ to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1} B(1/x) = x^{K+1}\kappa x^{-1} D(1/x)$ as formal power series, where $B$ is the denominator of $H$; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

**Theorem S4.4.** Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest and runs for any number $M$ of warm-up periods before the $T$ recorded prices. The bound does not depend on $M$, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most $W$, normalized rates in $[-1, 1]$ and equal moments below order $2K-1$ (`p1_thm_passive_cost_matched`). This covers the $K$-group Gauss and Gauss–Radau economies of Theorem S4.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K-2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`, `p1_thm_passive_cost_gauss_radau_three`), so for these $K$ the check covers the paper's economies with no extra hypothesis. For general $K$, the exactness of the two quadrature rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

# S28 Machine-checked declarations

The table lists, for each main result, the Lean 4 declarations that check it and what is proved only on paper. The project builds against a pinned Mathlib and has 1,141 named declarations in 127 modules, with no `sorry`. An axiom audit of every named declaration finds only Lean's three standard axioms (`propext`, `Classical.choice` and `Quot.sound`). A machine check confirms the

formal statement, not that it matches the prose. The prose statements above say where the two differ.

| Result | Declarations (module) | On paper only |
|---|---|---|
| Lemma S1.1 | `tracking_gain_explicit_root`, `tracking_prior_riccati_equation` (`Microfoundations.lean`); the third example's recursion, `p1_proportional_trade_recursion` (`Paper1StressIdentification.lean`) | Convergence of sample covariances. The third example's first-order condition is assumed in the checked statement, not derived |
| Theorem S2.1 (sufficiency) | `p1_eqAggFlow_kernel`, `p1_eqAggFlow_of_kernel`, `p1_kernel_of_eqAggFlow`, `p1_eq_market_invariance`, `p1_eq_insider_profit_invariance`, `p1_eq_group_level_differs` (`Paper1EquilibriumInvariance.lean`) | The unbounded-horizon moment statement; the passage from pathwise equality to equal equilibria |
| Proposition S2.2 | `p1_anylinear_flow_kernel`, `p1_anylinear_iff`, `p1_anylinear_invariance` (`Paper1CoreExtensions.lean`) | The passage from pathwise equality to equal equilibria |
| Theorem S3.1 | `p1_inventory_foc`, `p1_inventory_innovation`, `p1_inventory_benchmark` (`Paper1InventoryEquilibrium.lean`); `p1_inventory_small_feedback` (`Paper1InventoryIFT.lean`) | Part (iii); the version of part (iv) uniform over small group measures |
| Theorem S3.2 | `p1_eq_realize_D`, `p1_eq_realize_dealer`, `p1_eq_scaled_cohort` (`Paper1EquilibriumReducedForm.lean`) | Parts (ii)–(iii), and part (iv) beyond the realization |
| Theorem S4.1 | `p1_thm_kcohort_lower_bound_uniform` (`Paper1KCohortUniform.lean`); `p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1` (`Paper1KCohortEstimators.lean`) | The matching lower order of the response gap; the checked pair is a binomial moment-matched pair of the same order |
| Theorem S4.2 | `p1_thm_prony_upper_bound` (`Paper1PronyUpper.lean`) | — |

| Result | Declarations (module) | On paper only |
|---|---|---|
| Theorem S4.3 | `p1_thm_minphase_zeros` (`Paper1MinPhase.lean`); `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged` (`Paper1SpectralH.lean`); model link `p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`, `p1_impulse_generating` (`Paper1ModelLink.lean`) | — |
| Theorem S4.4 | `p1_thm_passive_cost_matched`, `p1_thm_passive_cost_gauss_radau_two` (`Paper1PassiveMatched.lean`); `p1_thm_passive_cost_gauss_radau_three` (`Paper1GaussRadau3.lean`); `p1_thm_passive_cost_every_K` (`Paper1PassiveKCohort.lean`) | The divergence-rate statements; the stationary limit; Gauss–Radau exactness for general $K$ enters as a hypothesis |
| Theorem S9.3 | `p1_insider_reduced_form`, `p1_insider_best_response`, `p1_insider_kyle_limit`, `p1_insider_posterior_variance` (`Paper1EquilibriumInsider.lean`) | The fixed-point characterization in part (i), part (iii), and the numerical comparative statics |
| Proposition S5.1 | `p1_prop_target_two_cohorts`, `p1_prop_target_every_K` (`Paper1SupervisorySeparation.lean`) | The checked two-group pair is a different moment-matched pair from the Gauss and Gauss–Radau pair, with the same orders; the every-$K$ version assumes a dominant-group margin; for the quadrature pairs the largest-weight comparison is checked numerically for $K \leq 200$, not proved |
| Proposition S12.4 and Lemma S12.3 | The counting step of Lemma S12.3(iii) only: `p1_largest_value_not_constant`, `p1_largest_weight_not_constant` (`Paper1EveryKHalt.lean`) | Lemma S12.3(i)–(ii), the finiteness of the level sets, and all of Proposition S12.4 |
| Proposition S4.5 | none | All of it; checked numerically |
| Proposition S5.4 | none; it rests on Theorem S2.1 | All of it |
| Proposition S5.2 | `p1_tagged_iv_determinant_bound` (`Paper1TaggedIV.lean`); `p1_iv_boundary_decomposition` (`Paper1IVBoundary.lean`) | The stationary IV central limit theorem |
| Theorem S6.1 | `p1_sat_sufficiency`, `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`) | The passage from pathwise equality to equal equilibria |

| Result | Declarations (module) | On paper only |
|---|---|---|
| Theorem S6.2 | `p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`, `p1_sat_capacity_differs` (`Paper1SaturatingSufficiency.lean`) | Parts (ii)–(iv): the total-variation, Gaussian and probe bounds |
| Proposition S6.3 | For a cubic shape $u - \gamma u^3$ only: `p1_cubic_pulse_tap1_kernel_cancel`, `p1_cubic_pulse_tap1_price_gap` (`Paper1StressIdentification.lean`) | All of the proposition as stated for tanh: part (i), and the tanh expansion and test in part (ii). The checked statements show the linear cancellation and the exact $A^3$ gap for the cubic shape |
| Theorem S6.4 | General statement, any number of traders: `p1_thm_stress_identification`; hard limit: `p1_clip_indep_dilations`, `p1_thm_stress_identification_hard_limit`; cubic counterexample: `p1_cubic_not_indep_dilations` (`Paper1StressIdentificationGeneral.lean`); tanh: `p1_tanh_indep_dilations`, `p1_thm_stress_identification_tanh` (`Paper1StressIdentificationTanh.lean`); steps: `p1_probe_steer_target`, `p1_probe_flow_recovery_eq`, `p1_sat_signed_capacity_sufficiency` (`Paper1StressIdentification.lean`) | The small-probe statement in part (ii); the last sentence of part (iii); the statement in part (i) on probes inside the bands, which is Theorem S6.2(iv) |
| Theorem S6.7 | Parts (i)–(ii) for any flow map: `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`); `p1_general_identification`, `p1_general_identification_iff`, `p1_responses_of_flow_eq`, `p1_identified_functional` (`Paper1GeneralIdentification.lean`); band-linear embedding: `p1_sat_follows_linear` | The transfer of the lower bounds in part (iii); the passage to equilibria in part (i) |
| Proposition S12.5 | none; it rests on Theorem S6.4 | All of it (a standard extremum-estimator argument) |

| Result | Declarations (module) | On paper only |
|---|---|---|
| Theorem S14.2 (general memory) | Lower bound: `p1_thm_general_memory_lower` (`Paper1GeneralMemoryLower.lean`); upper bound: `p1_thm_general_memory_upper` (`Paper1GeneralMemoryUpper.lean`), with a Jackson theorem built in `Paper1Jackson.lean`; rate: `p1_thm_general_memory_rate` (`Paper1GeneralMemoryRate.lean`) | The constants of the paper statement (the formal constants are larger; the order is the same) and Lemma S14.1 via Bernstein's theorem (the formal proof uses an elementary pair) |
| Theorem S15.1 | The arithmetic step of the lower bound: `p1_thm_uniformw1_minimax_lower_bound` (`Paper1FullFrontiers.lean`) | The probabilistic content of the lower bound, and the upper bound |
| Proposition S19.1 | `p1_prop_S7_1_iii_uniform`, `p1_prop_S7_1_iii_two_uniform` (`Paper1CRUniform.lean`, `Paper1CRUniformW2.lean`) | All but the two-sided Fisher-information bound for $K = 2$, uniform in the horizon |
| Proposition S20.1 | The displayed bound of part (ii), as an arithmetic step: `p1_endogenous_contraction_sensitivity` (`Paper1ClaimRepairs.lean`) | Part (i), and the fixed-point argument of part (ii) |
| Proposition S20.2 | Two arithmetic steps: `p1_camouflage_unprofitable`, `p1_sybil_share_count_bound` (`Paper1ClaimRepairs.lean`) | The determinant factor of part (i) and the economic arguments |
| Theorem S16.1, Propositions S17.1, S17.2, S22.1, S20.3, S21.2, Theorems S21.1, S22.2, S22.3, S22.4 and Corollary S22.5 | none in this table | All of them (proofs in the supplement) |
| Proposition S6.5 | `p1_delever_stress_invariance` (`Paper1CoreExtensions.lean`) | Parts (ii)–(iv), as for Theorem S6.2 |
| Proposition S6.6 | none | All of it |

# S29 Notation

| Symbol | Meaning |
|---|---|
| $x_t$ | pricing error (price minus fundamental) |
| $a$ | price persistence (anchor), $0 \le a < 1$ |
| $\kappa$ | price impact per unit of executed quantity |

| Symbol | Meaning |
|---|---|
| $q_i,\ \eta_i,\ r_i$ | group weight, adaptation rate, memory rate $r_i = 1 - \eta_i$ |
| $w_i = q_i\eta_i$ | merged weight |
| $M = \sum_i w_i \delta_{r_i}$ | merged rate–weight measure |
| $k_m,\ G(z)$ | feedback kernel and its transform |
| $H(z),\ h_j$ | closed-loop transfer function and impulse response, $h_0 = \kappa$ |
| $J,\ S$ | state transition matrix and $S = a + \kappa \sum q_i$ |
| $v_t,\ V$ | executed probe quantity and total quantity $\sum \lvert v_t \rvert$ |
| $d_T$ | impulse-response difference between hypotheses, first $T$ taps |
| $\sigma$ | observation-noise standard deviation |
| $\delta,\ \Delta\eta$ | half-separation of a known pair; spacing of unknown groups |
| $K_*$ | transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$ |
| $T_h$ | Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost |
| $g_i(\omega)$ | dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$ |
| $B,\ \mu,\ c,\ \phi$ | capital, maintenance-margin fraction, proportional cost, taker fee |
| $V_{\text{eff}}$ | per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$ |
| $\hat{Q},\ \hat{S}$ | reduced-form loop gain and persistence indices |